

Electronics & Physical Computing (SCS2315/IS2207)

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Electronic Components

Dr. Hiran Ekanayake

Electronic Components

- Passive Components
- Active Components
- Electromechanical Components
- Semiconductor Devices
- Power Components
- Sensors & Transducers
- Communication Components
- Protection Components
- Display & Output Devices
- Embedded & Digital Components
- Interconnection Components
- Etc.

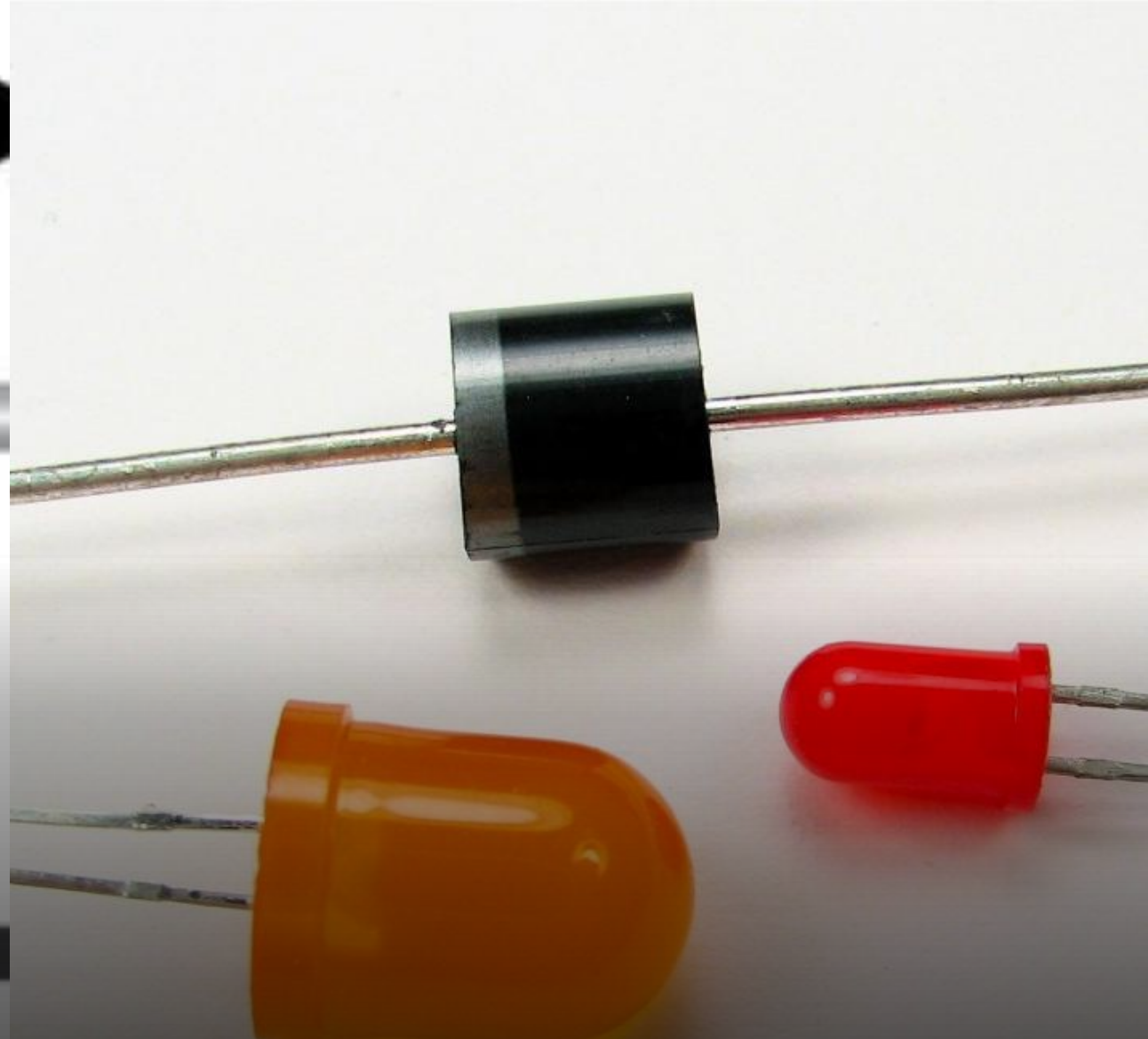
Active Components

- Components that require an external source of energy (power supply) to operate and can **control, amplify, or generate electrical signals**.
- Capable of providing **power gain** – the output signal can be stronger than the input signal.
- E.g., Diode, Transistor, IC, LED, Microcontroller, Op-Amp

Current Flow



Diode



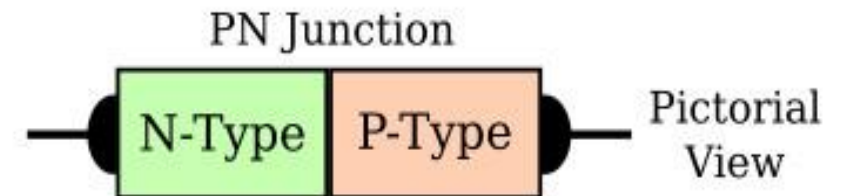
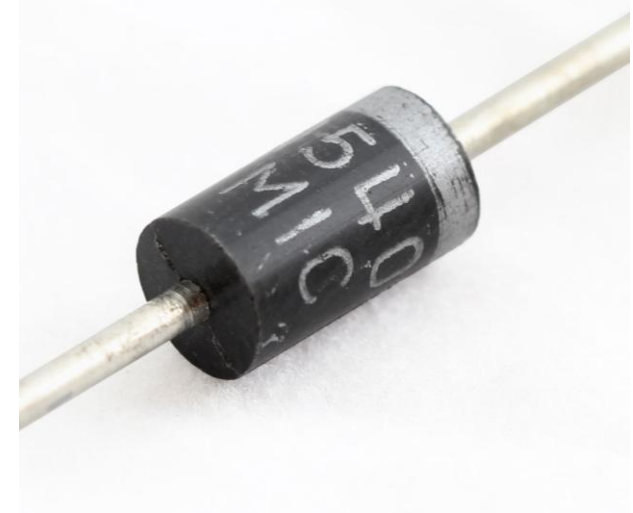
Online Tutorial on Diode:

https://www.electronics-tutorials.ws/diode/diode_1.html

- ▶ 1. Semiconductor Basics
- 2. PN Junction Theory
- 3. PN Junction Diode
- 4. The Signal Diode
- 5. Power Diodes and Rectifiers
- 6. Full Wave Rectifier
- 7. The Zener Diode
- 8. The Light Emitting Diode
- 9. Bypass Diodes in Solar Panels
- 10. Diode Clipping Circuits
- 11. The Schottky Diode
- 12. Photodiode Basics
- 13. Diode Clamping Circuit
- 14. Biased Clamper Circuit

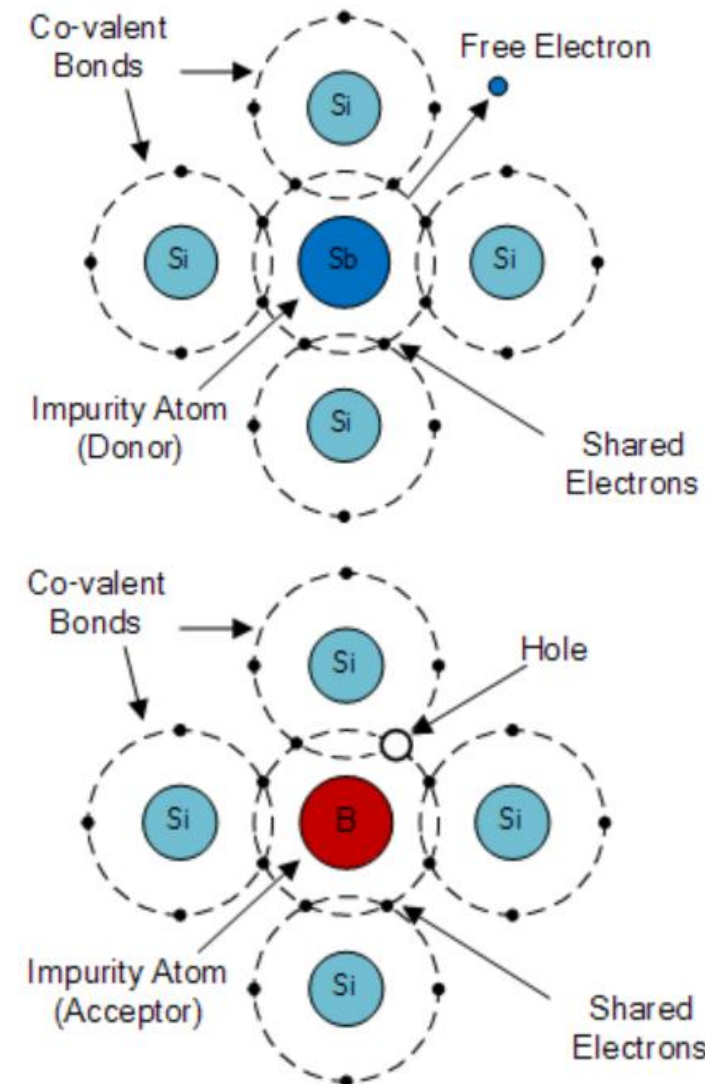
Introduction to Semiconductor Diodes

- A **diode** is a two-terminal semiconductor device that allows current to flow mainly in **one direction**.
- Acts like a **one-way electrical valve**.
- Formed using a **PN junction**:
 - **P-type semiconductor** → majority carriers are **holes**
 - **N-type semiconductor** → majority carriers are **electrons**
- Diode Symbol:
 - **Anode (P-side)** → Current enters
 - **Cathode (N-side)** → Current exits



Semiconductor Materials & Doping Process

- Common semiconductor materials: Silicon (Si), Germanium (Ge), Gallium Arsenide (GaAs)...
- N-Type Semiconductor:
 - Doped with **pentavalent impurities** (e.g., Phosphorus, Antimony)
 - Produces extra **free electrons**
- P-Type Semiconductor:
 - Doped with **trivalent impurities** (e.g., Boron, Aluminium)
 - Produces extra **holes**
- When P-type and N-type materials are joined together, a **PN junction** is formed, creating the basis of diode operation.



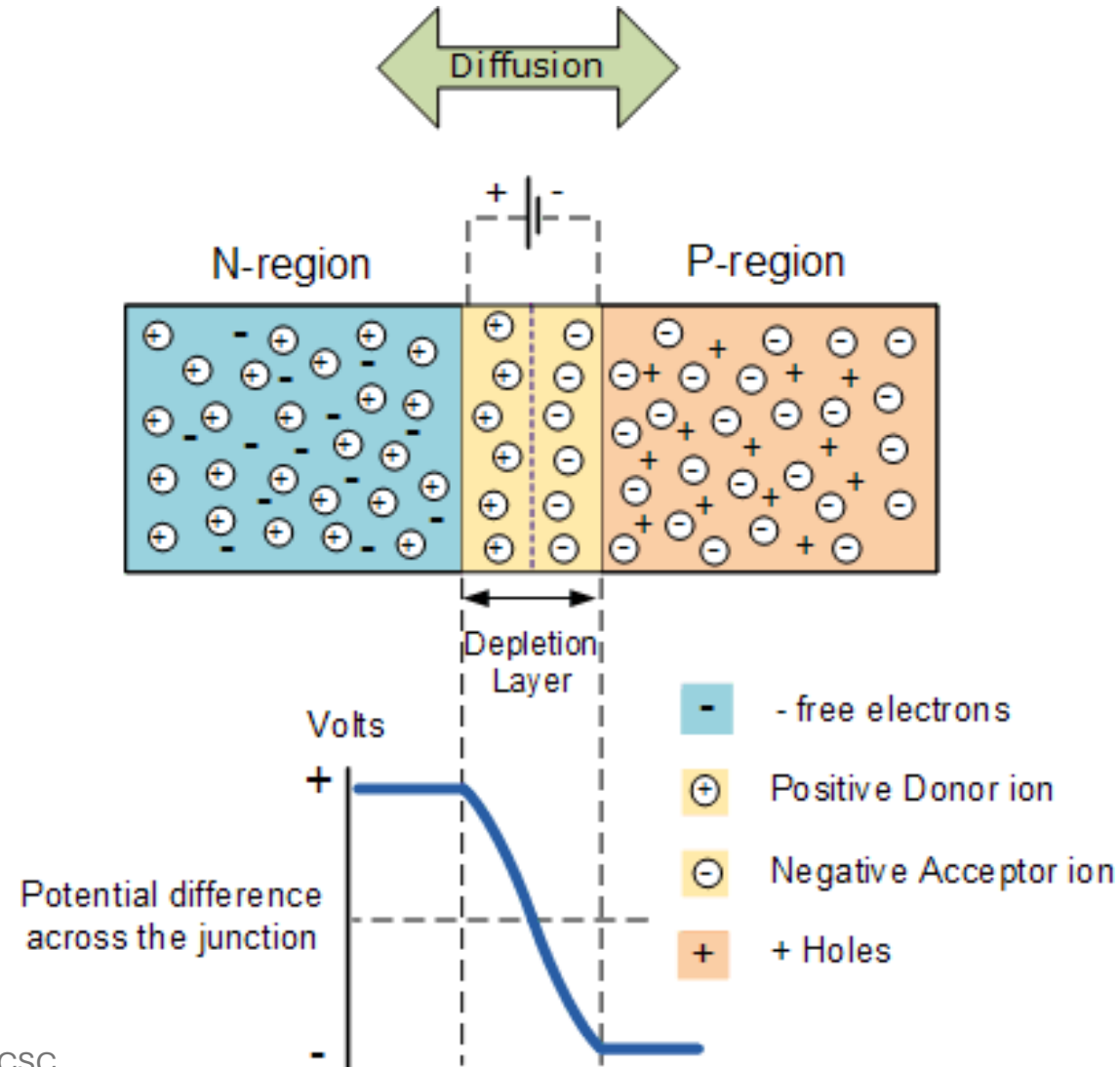
PN Junction and Diode Operation

- **Depletion Region:**

- At the junction electrons diffuse from N-side to P-side and holes diffuse from P-side to N-side
- This creates a region with very few charge carriers called the **Depletion Layer**

- **Potential Barrier:**

- An internal electric field forms across the junction.
- Typical barrier voltage: Silicon diode $\approx 0.7\text{ V}$, Germanium diode $\approx 0.3\text{ V}$



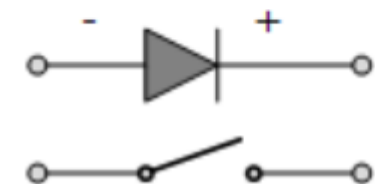
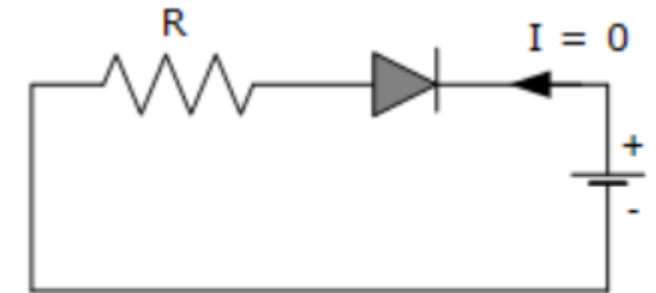
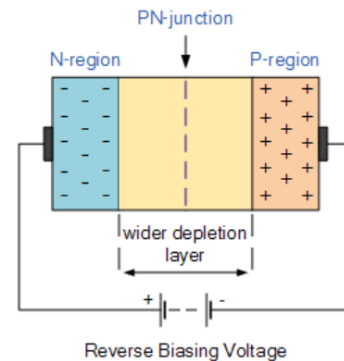
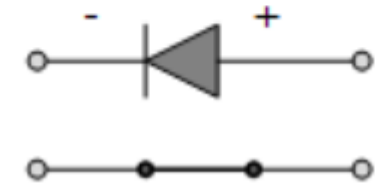
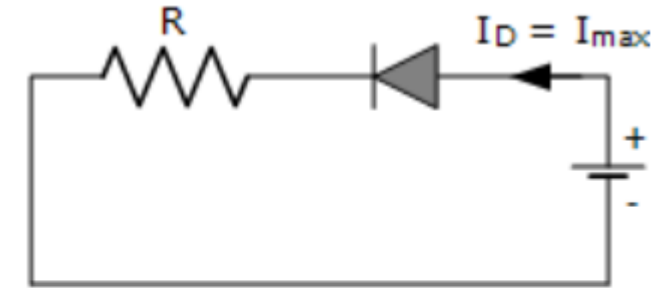
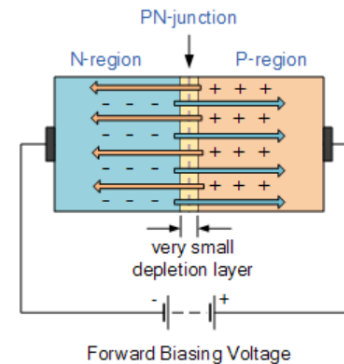
Biasing Conditions

- **Forward Bias:**

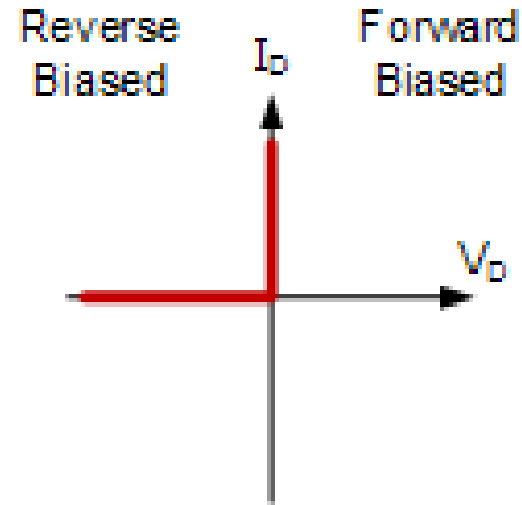
- P-side connected to positive terminal
- N-side connected to negative terminal
- Depletion layer becomes thinner
- Current flows easily

- **Reverse Bias:**

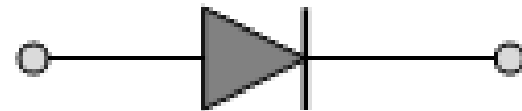
- P-side connected to negative terminal
- N-side connected to positive terminal
- Depletion layer widens
- Current flow is blocked



Ideal and Real Diode Characteristics



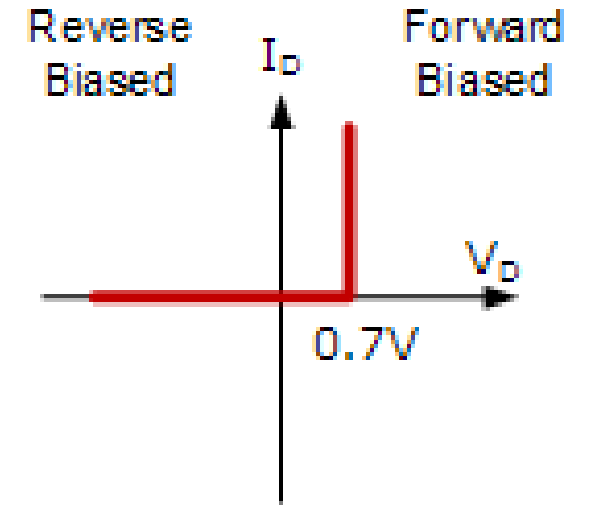
Ideal Diode



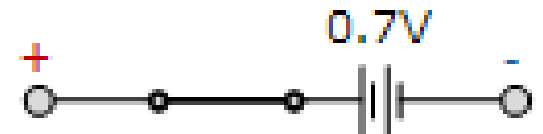
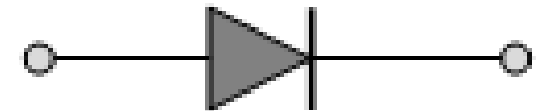
Forward Biased



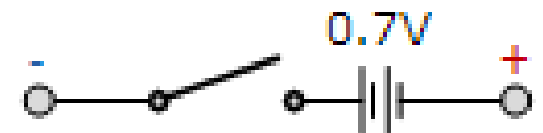
Reverse Biased



Real Diode

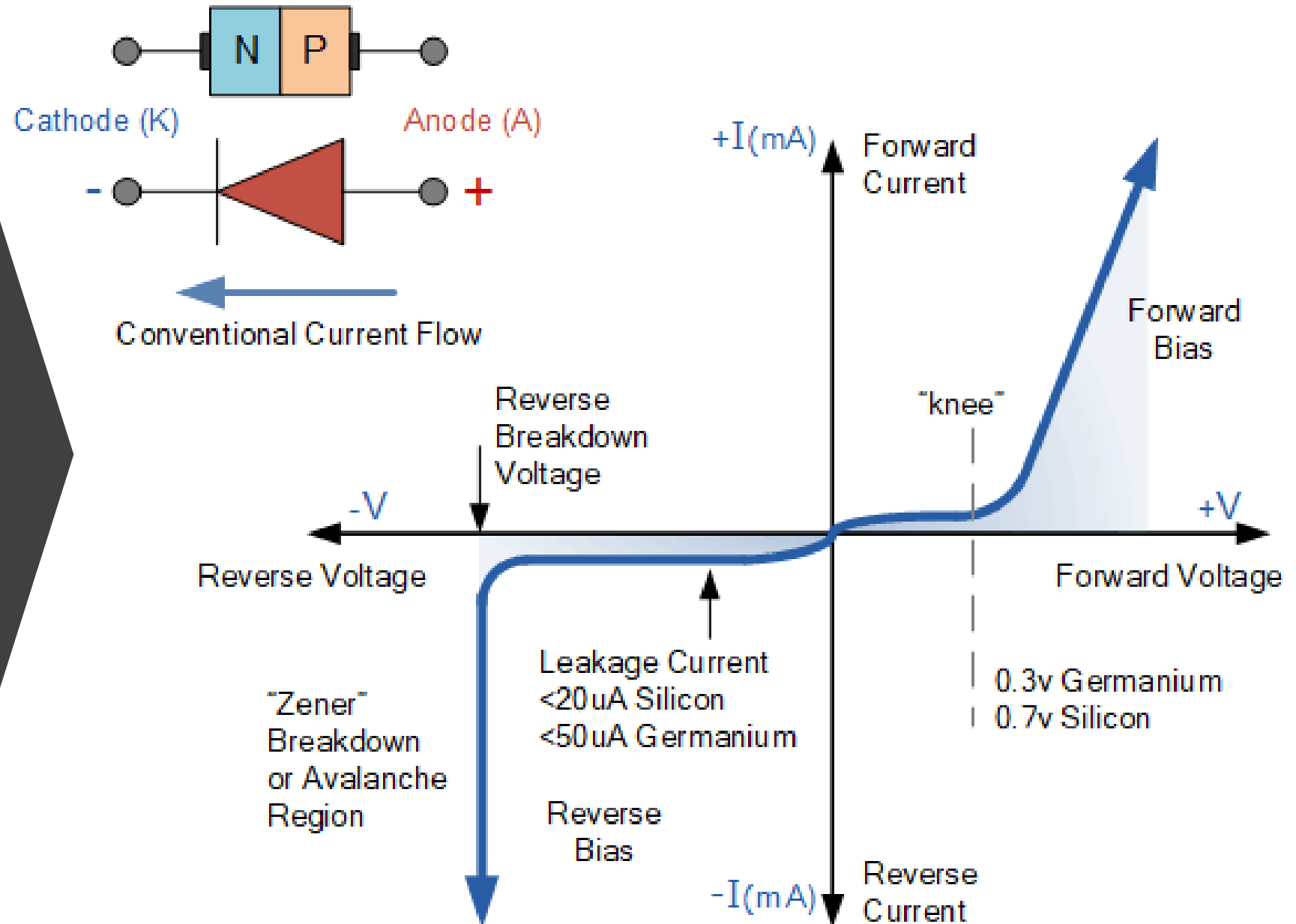


Forward Biased



Reverse Biased

Junction Diode Static I-V Characteristics



Diode Applications

- Rectifiers (AC to DC conversion)
- Voltage regulation
- Signal clipping/clamping
- Protection circuits
- LEDs and switching circuits

TYPES OF DIODES



Junction Diode



Zener Diode



Tunnel Diode



Schottky Diode



Varactor Diode



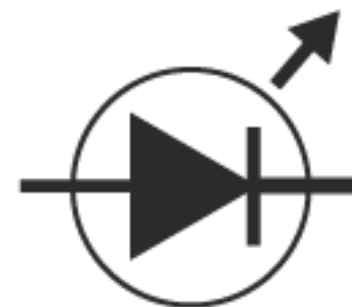
Diac



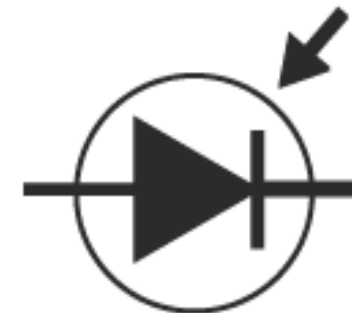
Triac



SCR



Light Emitting Diode (LED)



Photodiode

Signal Diodes

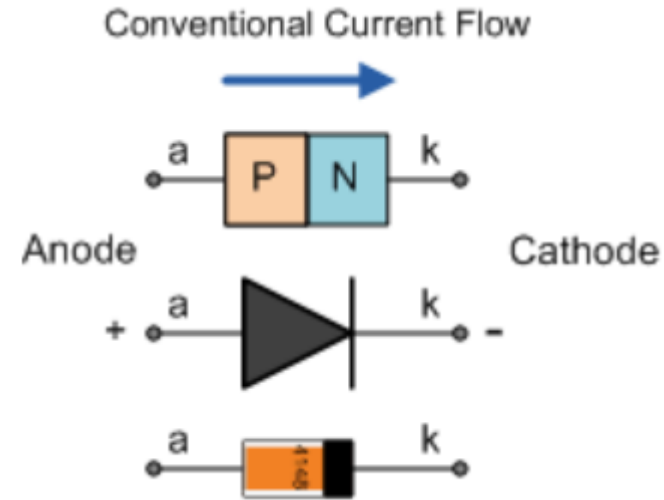
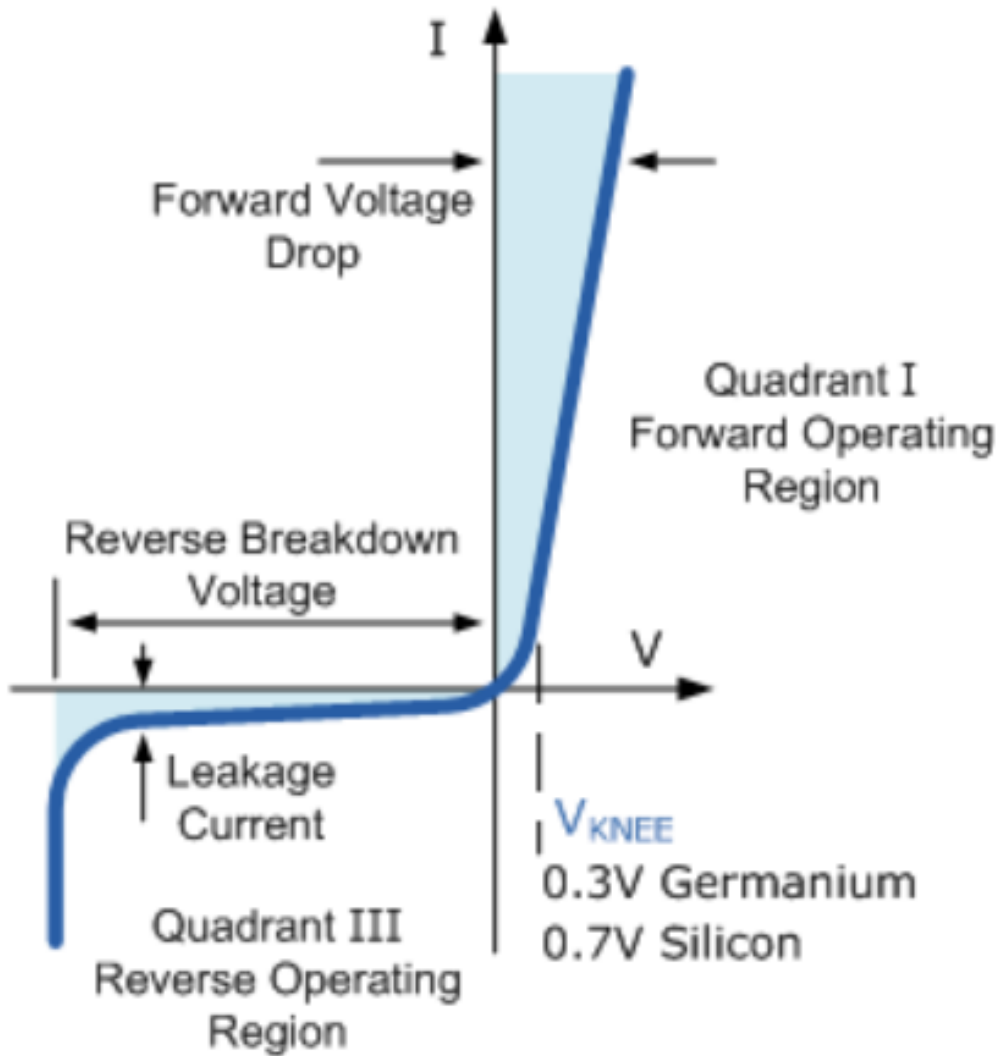


- A **signal diode** is a small semiconductor diode used for low-current applications, high-frequency switching, and signal processing circuits.
- Common Applications:
 - Radio and TV circuits, digital logic circuits, wave shaping, clipping and clamping circuits, high-speed switching, etc.
- Physical Features:
 - Usually packaged in **glass encapsulation**. Cathode terminal identified by a black or red band.
 - E.g., 1N4148 and 1N914

Germanium vs Silicon Signal Diodes

Feature	Germanium Diode	Silicon Diode
Forward Voltage Drop	~0.2–0.3 V	~0.6–0.7 V
Reverse Resistance	Lower	Higher
Leakage Current	Higher	Lower
Temperature Stability	Lower	Better
Common Applications	Detection circuits	General electronics

Silicon Diode V-I Characteristic Curve

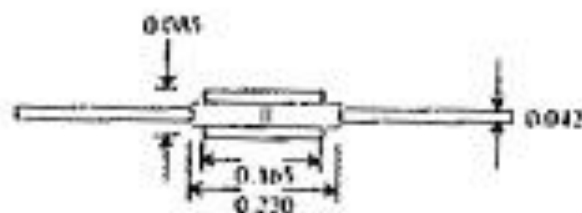


Signal Diode Parameters

Parameter	Description	Unit	Importance	Example: 1N4148
Maximum Forward Current (IF max)	Maximum continuous current the diode can safely conduct in forward bias	mA or A	Exceeding this may overheat and damage the diode	150 mA
Peak Inverse Voltage (PIV) / Reverse Voltage (VR max)	Maximum reverse voltage the diode can withstand before breakdown	Volts (V)	Protects diode from reverse-bias failure	100 V
Forward Voltage Drop (VF)	Voltage across diode during forward conduction	Volts (V)	Determines power loss and turn-on voltage	0.7 V (Silicon)
Reverse Leakage Current (IR)	Small current flowing during reverse bias	μA or nA	Lower leakage improves efficiency and accuracy	25 nA typical
Power Dissipation (PD max)	Maximum power the diode can safely dissipate as heat	mW or W	Important for thermal design	500 mW
Reverse Recovery Time (trr)	Time required to switch from ON to OFF state	ns or μs	Critical in high-speed switching circuits	4 ns

Type 1N914 Through 1N917 Silicon Switching Diodes

mechanical data



dimensions in inches

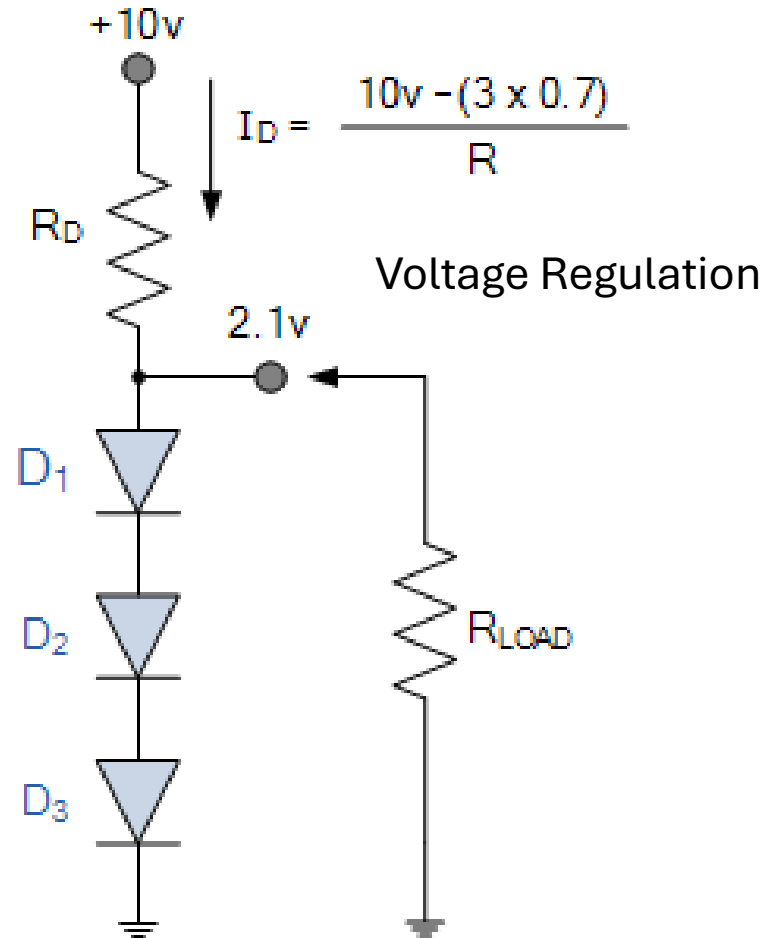
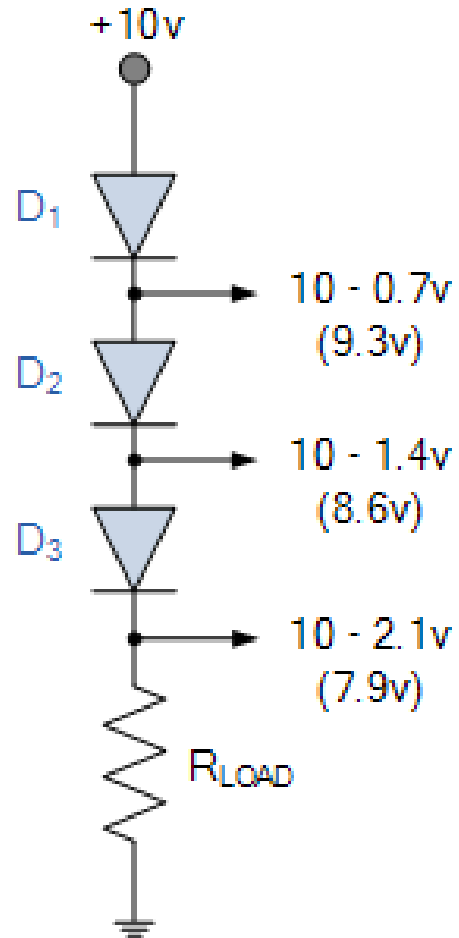


absolute maximum ratings at 25°C ambient temperature

	1N914	1N914A	1N914B	1N915	1N916	1N916A	1N916B	1N917	Unit
V_R Reverse voltage	75	75	75	50	75	75	75	30	V
I_O Average Rectified Forward Current	75	75	75	75	75	75	75	50	mA
I_{FRM} Repetitive Peak Forward Current	225	225	225	225	225	225	225	150	mA
P Power Dissipation	250	250	250	250	250	250	250	250	mW

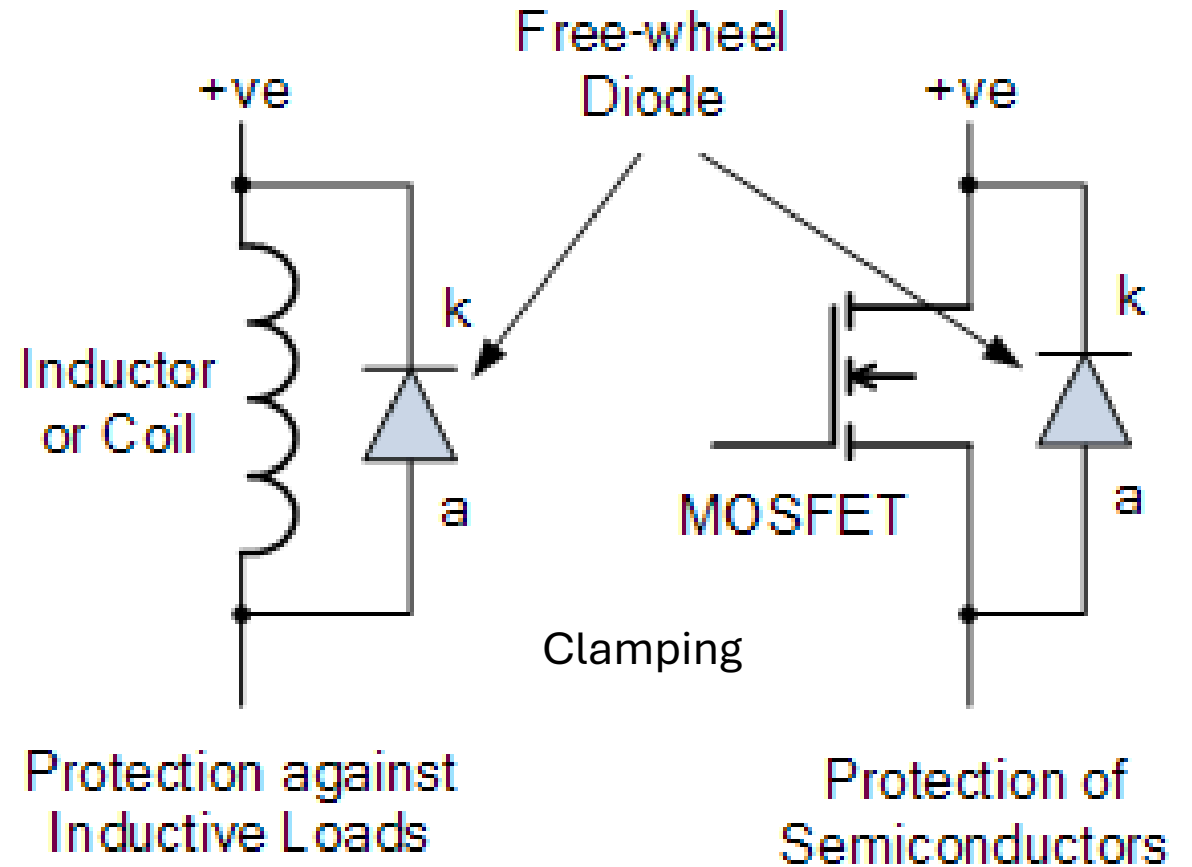
Figure 2-21
Portion of data sheet
for 1N914 through
1N917 diodes.

Application: Signal Diode in Series/Parallel



Application: Freewheel Diodes

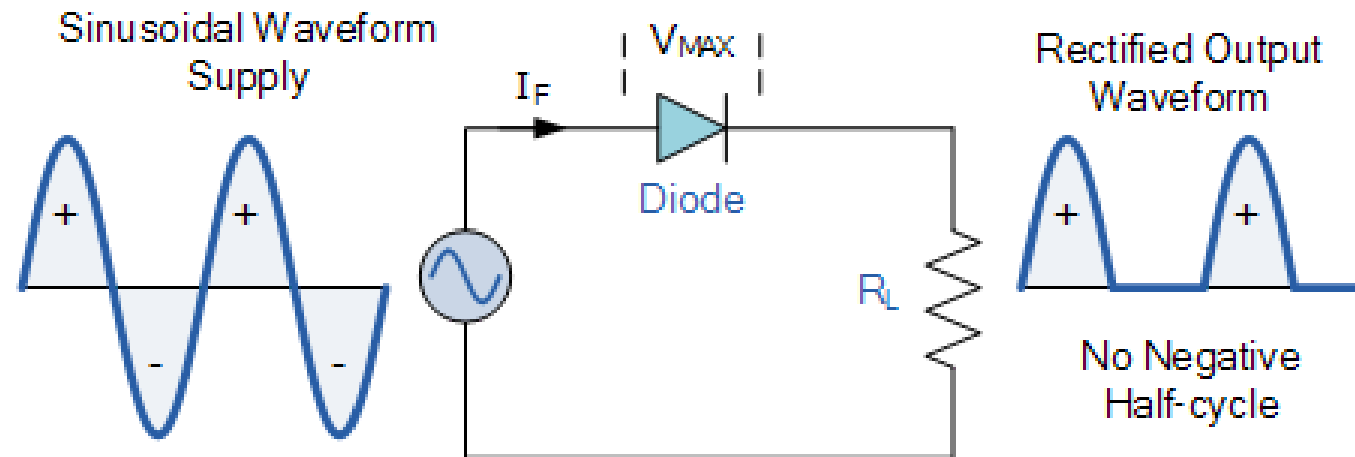
- The **Freewheel (Flywheel) diode** is used to protect solid state switches such as power transistors and MOSFET's from damage by reverse battery protection as well as protection from highly inductive loads such as relay coils or motors.



Power Diodes & Rectifiers



- A **power diode** is a PN-junction semiconductor device designed to conduct large forward currents and to withstand high reverse voltages.
- Mainly used to convert **Alternating Current (AC)** into **Direct Current (DC)** – called **Rectification**.



Power Diode Parameters

Parameter	Description	Unit	Importance	E.g.: 1N4001
Maximum Forward Current (IF(AV))	Maximum average current the diode can conduct continuously in forward bias	A	Determines load current capability	1 A
Peak Inverse Voltage (PIV) / Maximum Reverse Voltage (VRRM)	Highest reverse voltage the diode can withstand without breakdown	V	Prevents reverse-bias damage	50 V
Forward Voltage Drop (VF)	Voltage drop across the diode during conduction	V	Affects power loss and efficiency	≈ 1.1 V
Surge Current (IFSM)	Maximum short-duration surge current the diode can tolerate	A	Important during power-on surges	30 A
Power Dissipation (PD)	Maximum heat power the diode can safely dissipate	W	Determines cooling requirements	≈ 3 W
Reverse Leakage Current (IR)	Small current flowing under reverse bias	μA	Lower leakage improves efficiency	≈ 5 μA
Reverse Recovery Time (trr)	Time required to switch from conducting to blocking state	μs	Important in switching circuits	≈ 30 μs
Maximum Junction Temperature (TJ max)	Highest safe internal operating temperature	°C	Affects reliability and lifetime	150°C

1N4001 - 1N4007

General Purpose Rectifiers

Features

- Low forward voltage drop.
- High surge current capability.



DO-41

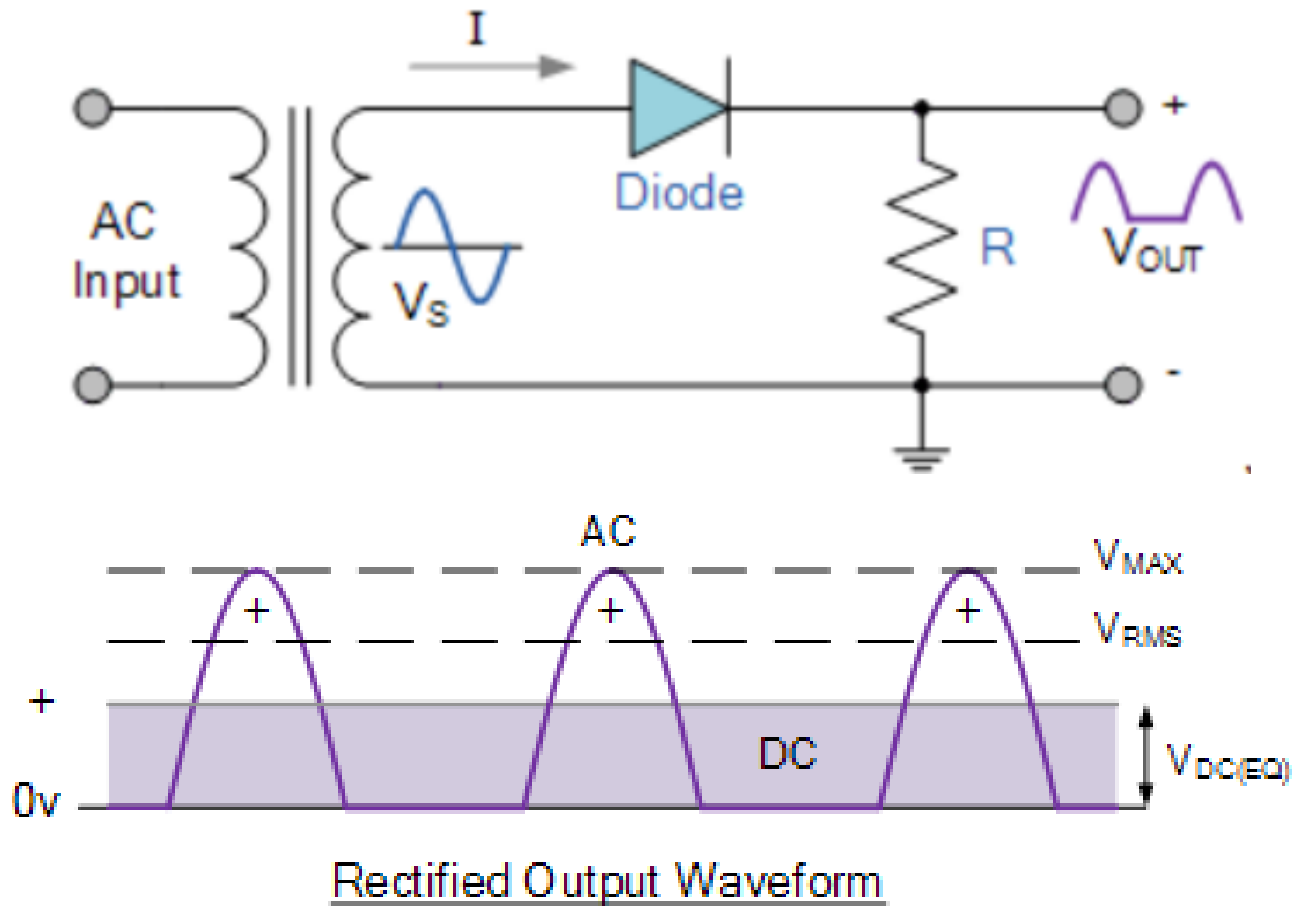
COLOR BAND DENOTES CATHODE

Absolute Maximum Ratings * $T_A = 25^\circ\text{C}$ unless otherwise noted

Symbol	Parameter	Value							Units
		4001	4002	4003	4004	4005	4006	4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " lead length @ $T_A = 75^\circ\text{C}$	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing ($t < 8.3\text{ms}$)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							$^\circ\text{C}$
T_J	Operating Junction Temperature	-55 to +175							$^\circ\text{C}$

Half-Wave Rectifier

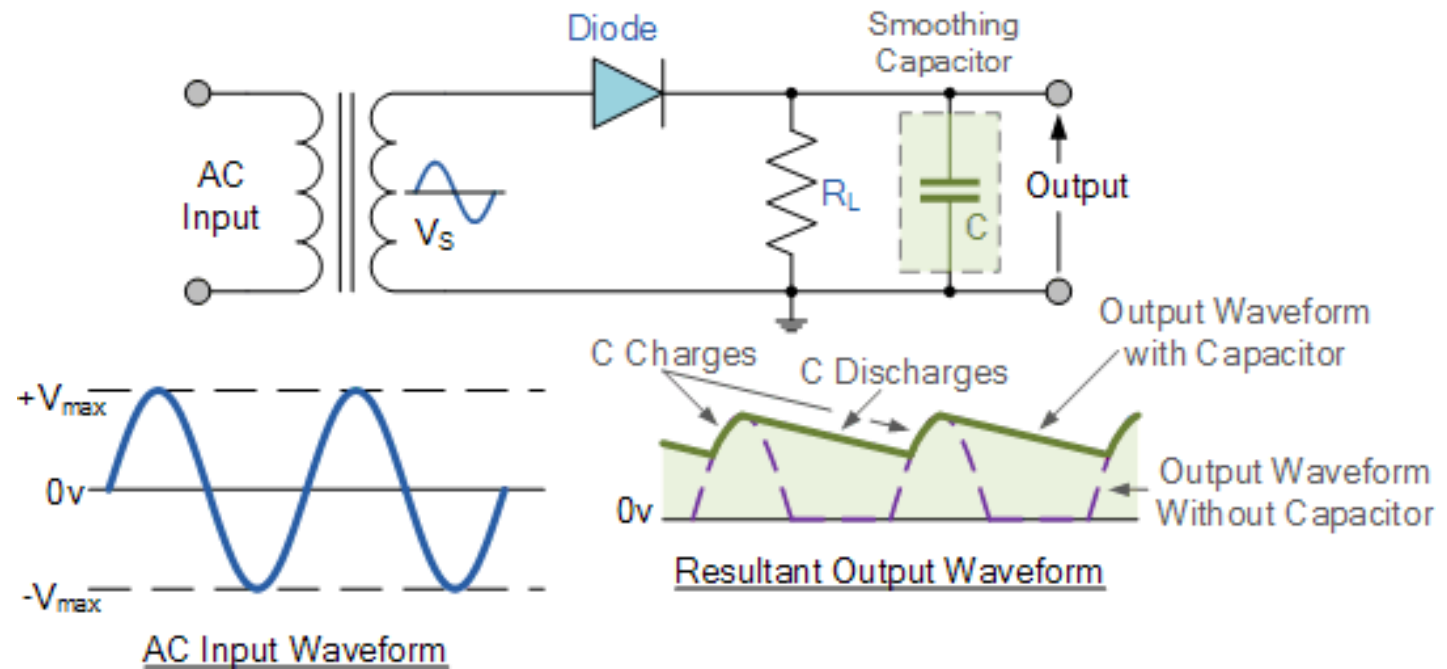
- The power diode in a half wave rectifier circuit passes just one half of each complete sine wave of the AC supply in order to convert it into a DC supply.
- Advantages: Simple design and low cost
- Disadvantages: Low efficiency and high ripple content



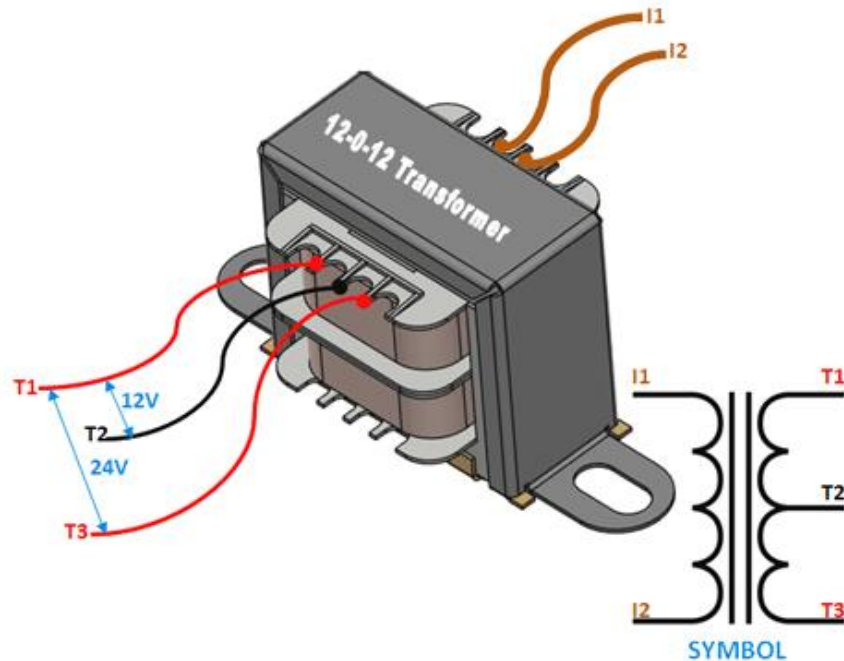
$$V_{d.c.} = \frac{V_{MAX}}{\pi} = 0.318V_{MAX} = 0.45V_{RMS}$$

Half-wave Rectifier with Smoothing Capacitor

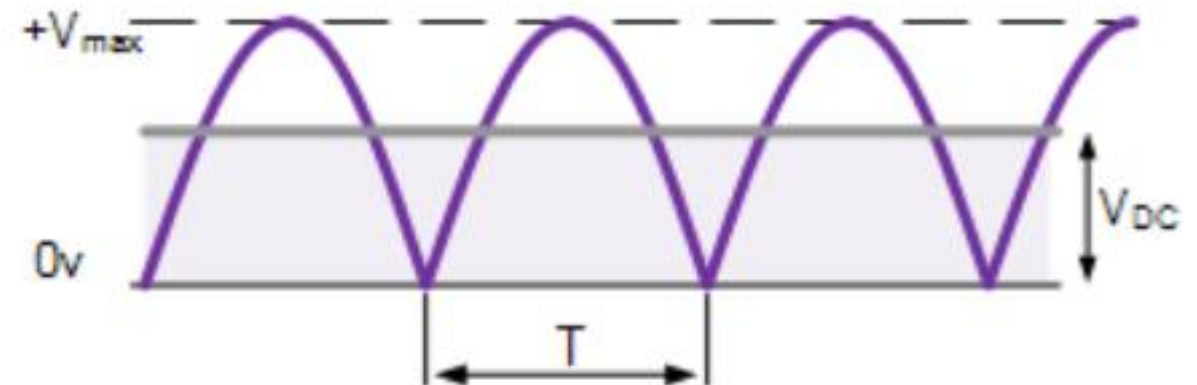
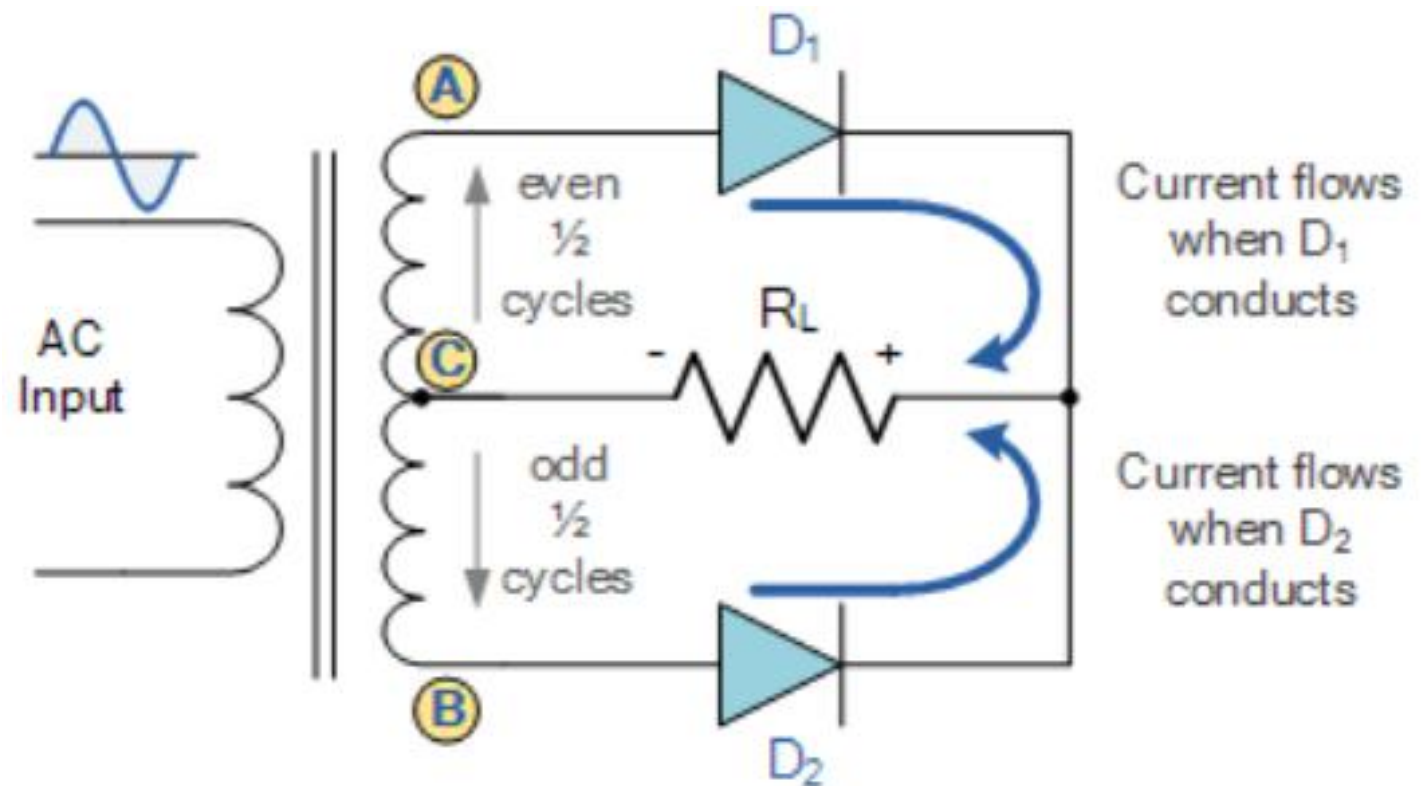
- The amount of ripple voltage can be further reduced by using larger value capacitors.
- For a given capacitor value, a greater load current (smaller load resistance) will discharge the capacitor more quickly (RC Time Constant) and so increases the ripple obtained.



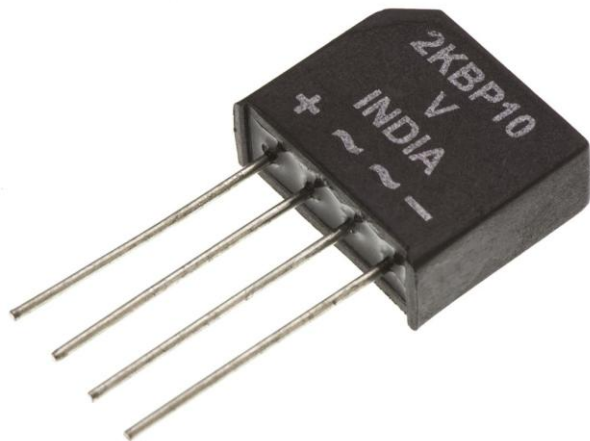
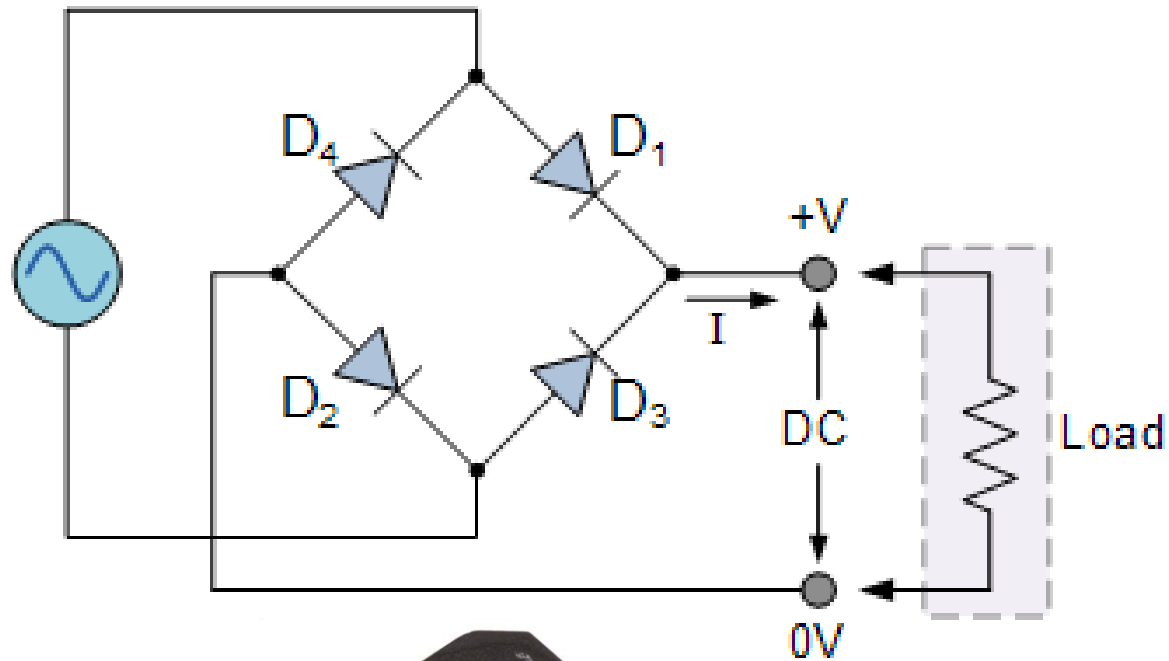
Full Wave Rectifier



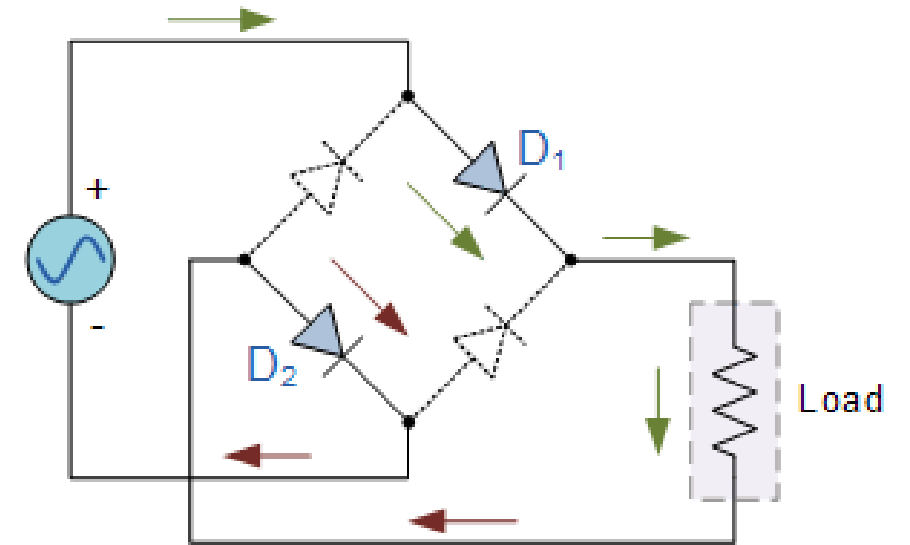
$$V_{d.c.} = \frac{2V_{max}}{\pi} = 0.637V_{max} = 0.9V_{RMS}$$



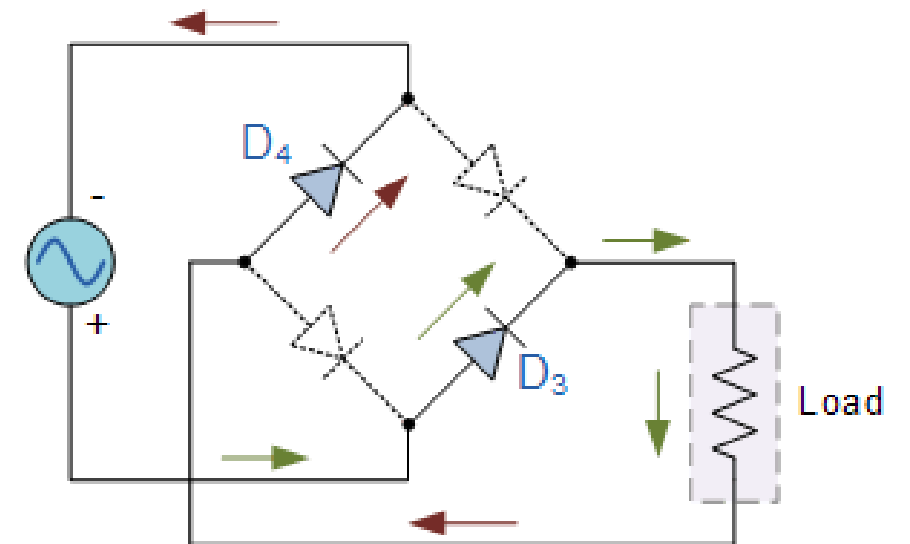
Full Wave Bridge Rectifier



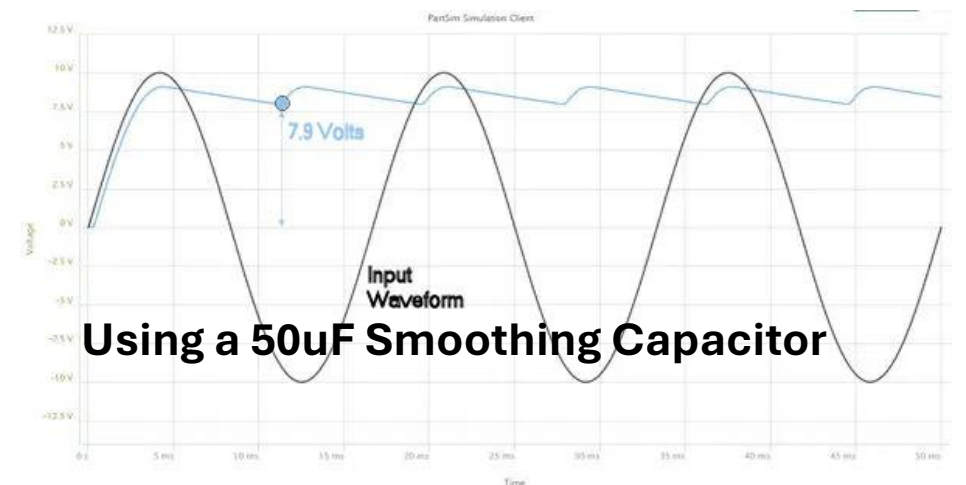
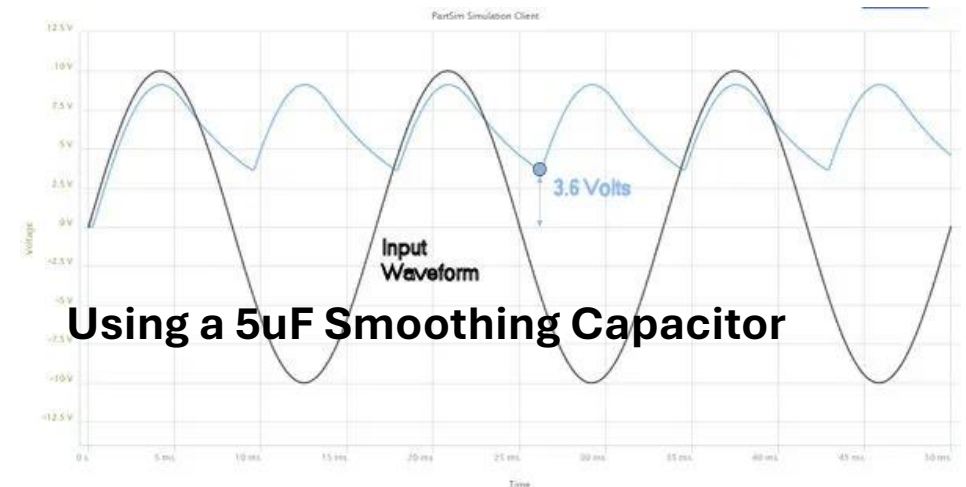
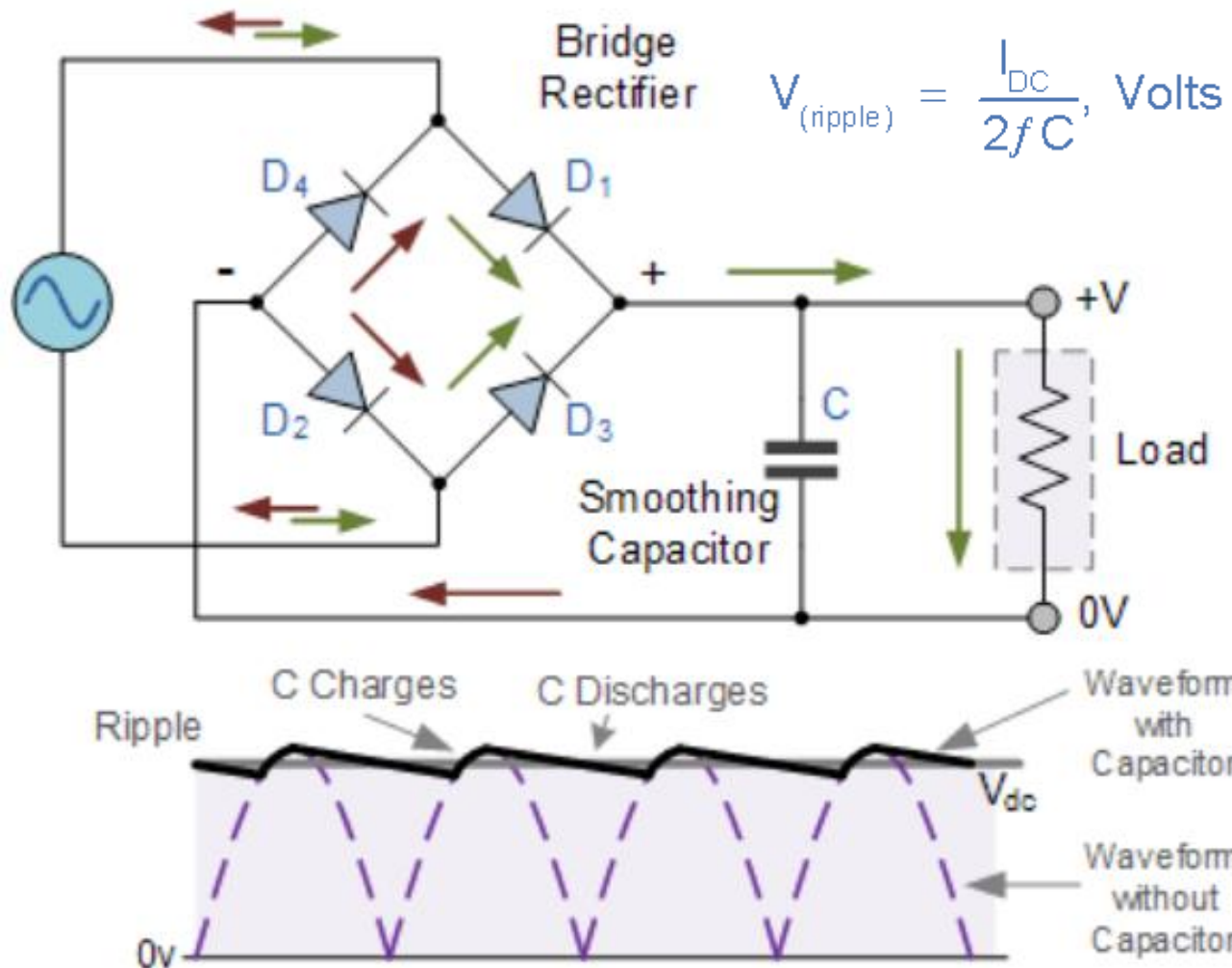
Positive Half-cycle



Negative Half-cycle



Full Wave Rectifier with Smoothing Capacitor



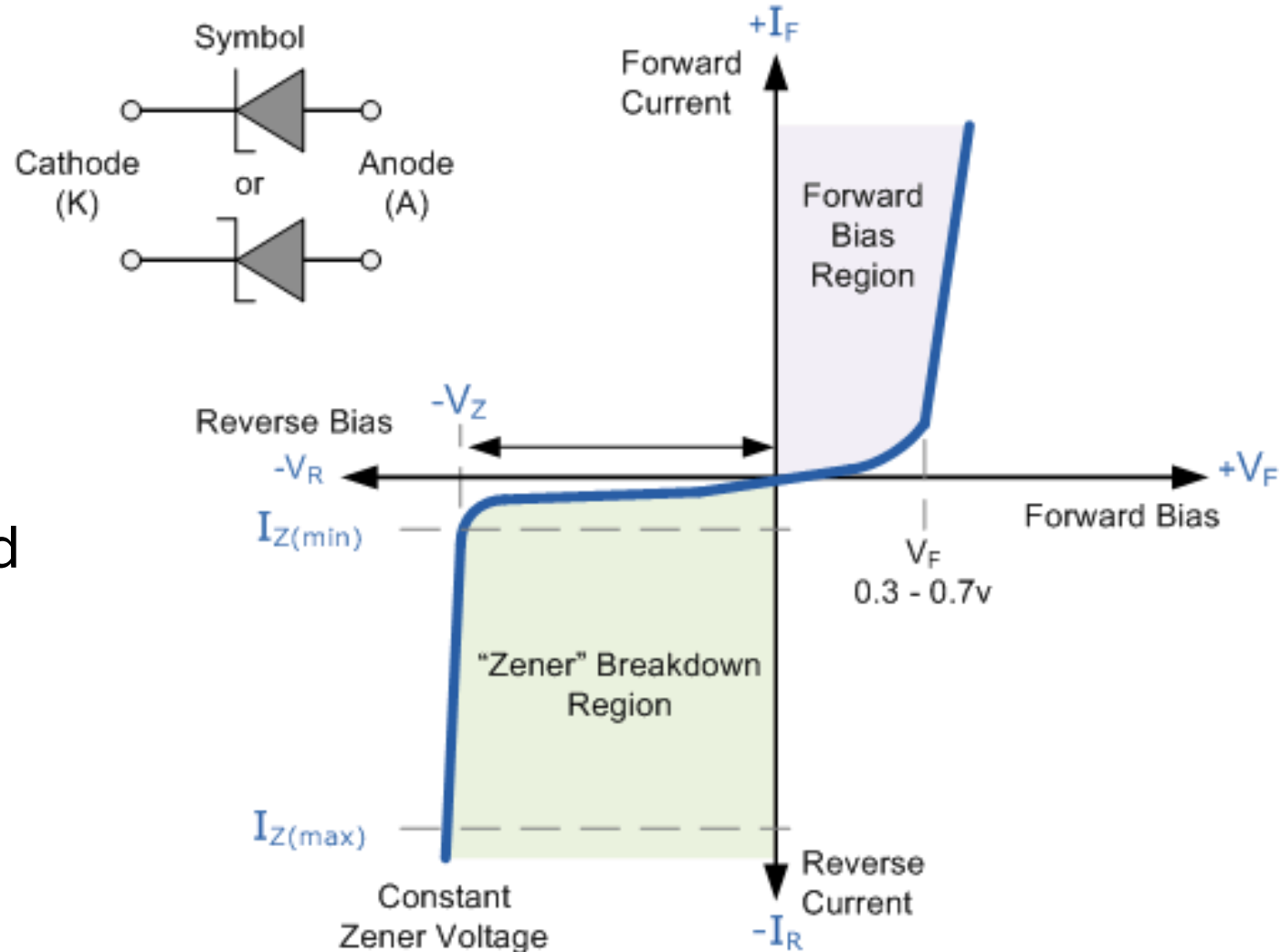
The Zener Diode



- A **Zener diode** is a specially designed semiconductor diode that is intended to operate in the **reverse breakdown region** without being damaged. Unlike ordinary diodes, which are normally damaged by excessive reverse voltage, Zener diodes are designed to maintain a nearly constant voltage across their terminals when reverse biased.
- They are widely used for voltage regulation, voltage reference circuits, overvoltage protection, and clipping and clamping circuits.

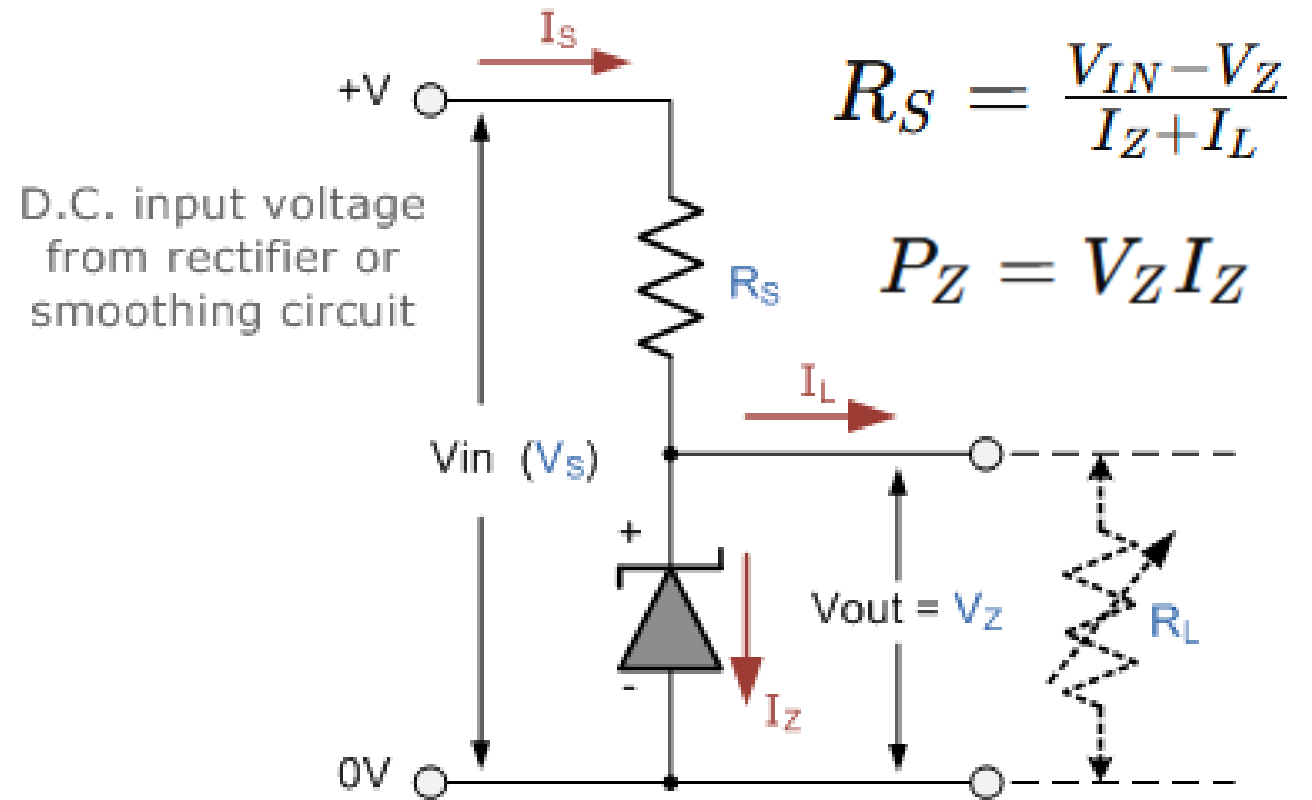
Zener Diode I-V Characteristics

- Forward Bias:
 - Acts like a normal silicon diode (~ 0.7 V drop)
- Reverse Bias:
 - Blocks current until Zener voltage is reached
- When the reverse voltage reaches a specific value called the **Zener Voltage** (V_Z), the diode enters breakdown and conducts current while maintaining approximately constant voltage.



Zener Diode as a Voltage Regulator

- Resistor, R_S is connected in series with the zener diode to limit the current flow through the diode with the voltage source, V_S being connected across the combination. The stabilised output voltage V_{out} is taken from across the zener diode.



Important Parameters of Zener Diodes

Parameter	Description
(V_Z)	Zener breakdown voltage
$(I_{Z(\min)})$	Minimum current for regulation
$(I_{Z(\max)})$	Maximum safe Zener current
(P_Z)	Maximum power dissipation
(Z_Z)	Dynamic impedance
Temperature Coefficient	Voltage change with temperature

Zener Diode Standard Zener Voltages



BZX55 Zener Diode Power Rating 500mW

2.4V	2.7V	3.0V	3.3V	3.6V	3.9V	4.3V	4.7V
5.1V	5.6V	6.2V	6.8V	7.5V	8.2V	9.1V	10V
11V	12V	13V	15V	16V	18V	20V	22V
24V	27V	30V	33V	36V	39V	43V	47V

BZX85 Zener Diode Power Rating 1.3W

3.3V	3.6V	3.9V	4.3V	4.7V	5.1V	5.6	6.2V
6.8V	7.5V	8.2V	9.1V	10V	11V	12V	13V
15V	16V	18V	20V	22V	24V	27V	30V
33V	36V	39V	43V	47V	51V	56V	62V

Exercise

- A Zener diode with a Zener voltage of **6.8 V** is connected in parallel with a load. The supply voltage is **12 V** and the series resistor is **470 Ω** . Calculate the **Zener current** when the load current is **10 mA**.

Voltage across resistor

$$V_R = V_S - V_Z = 12 - 6.8 = 5.2 \text{ V}$$

Resistor current

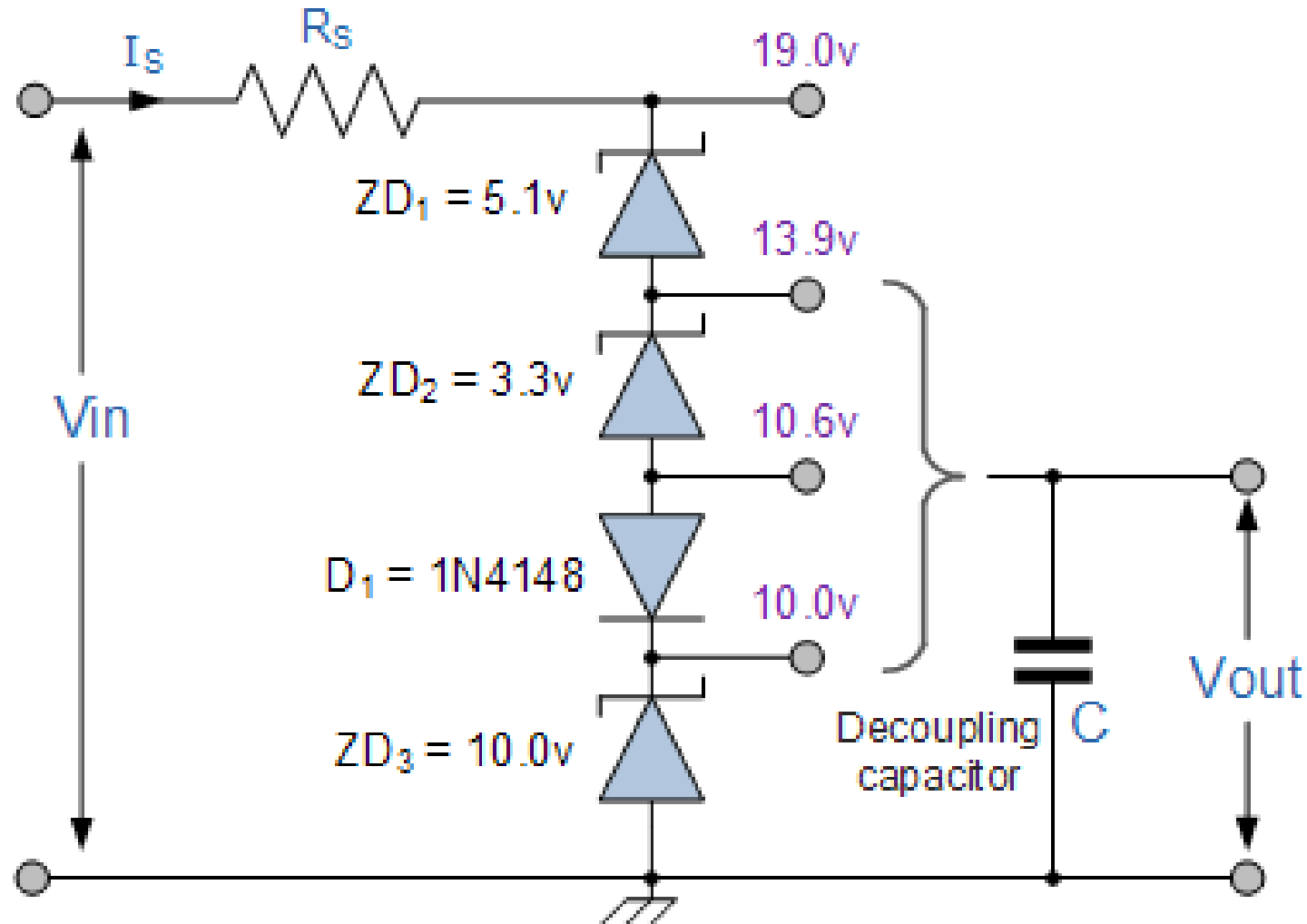
$$I_R = \frac{5.2}{470} = 11.06 \text{ mA}$$

Zener current

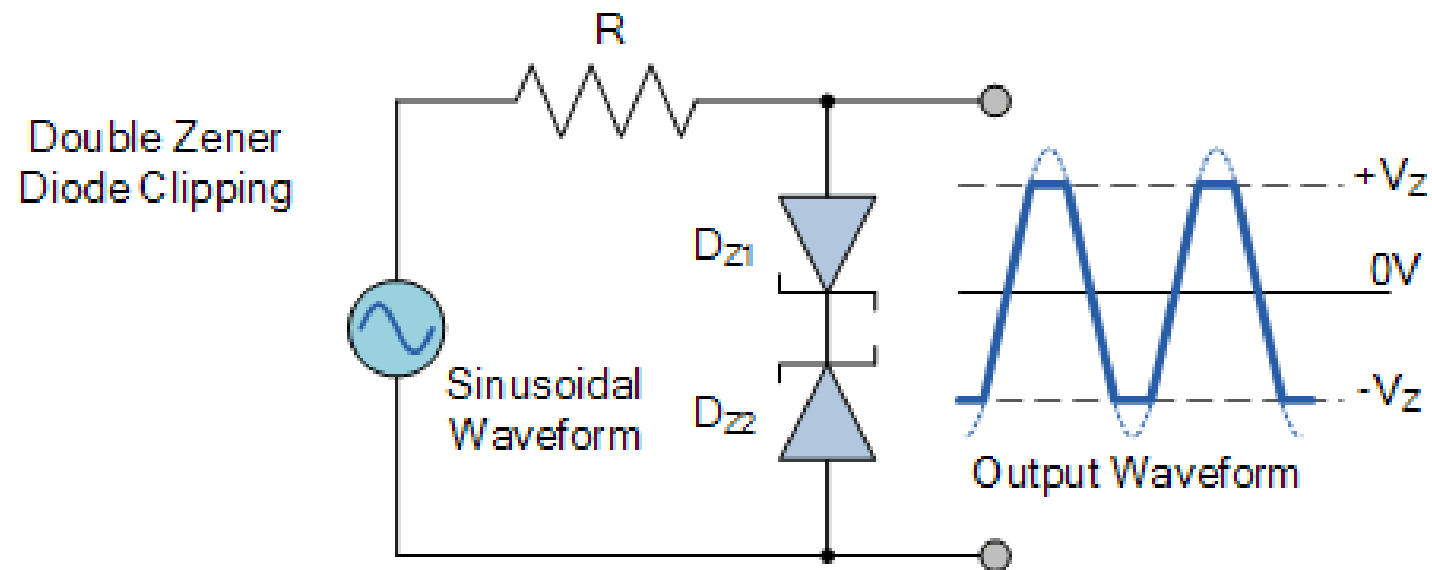
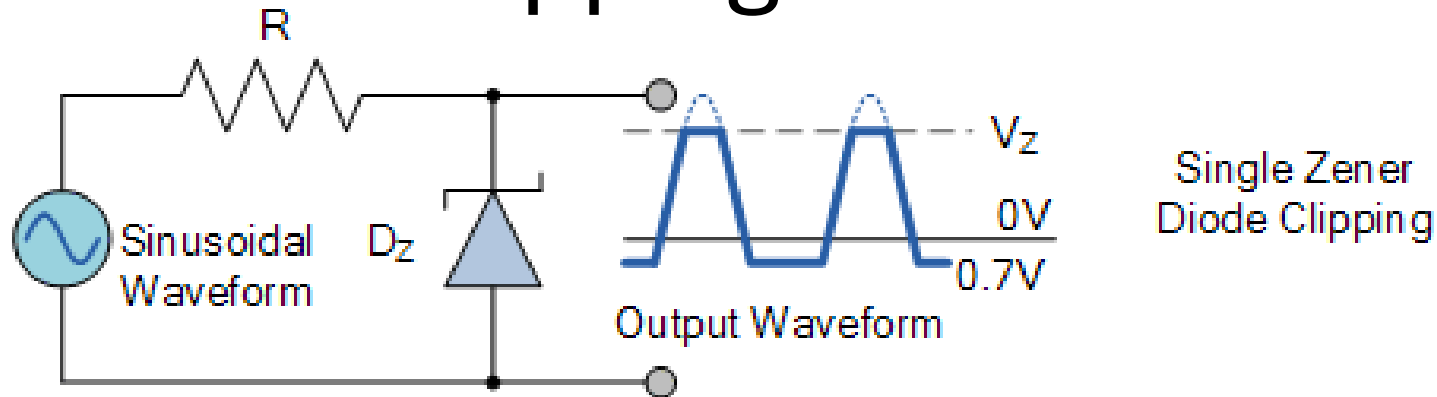
$$I_Z = I_R - I_L = 11.06 - 10 = 1.06 \text{ mA}$$

Zener current = 1.06 mA

Application: Voltage Reference

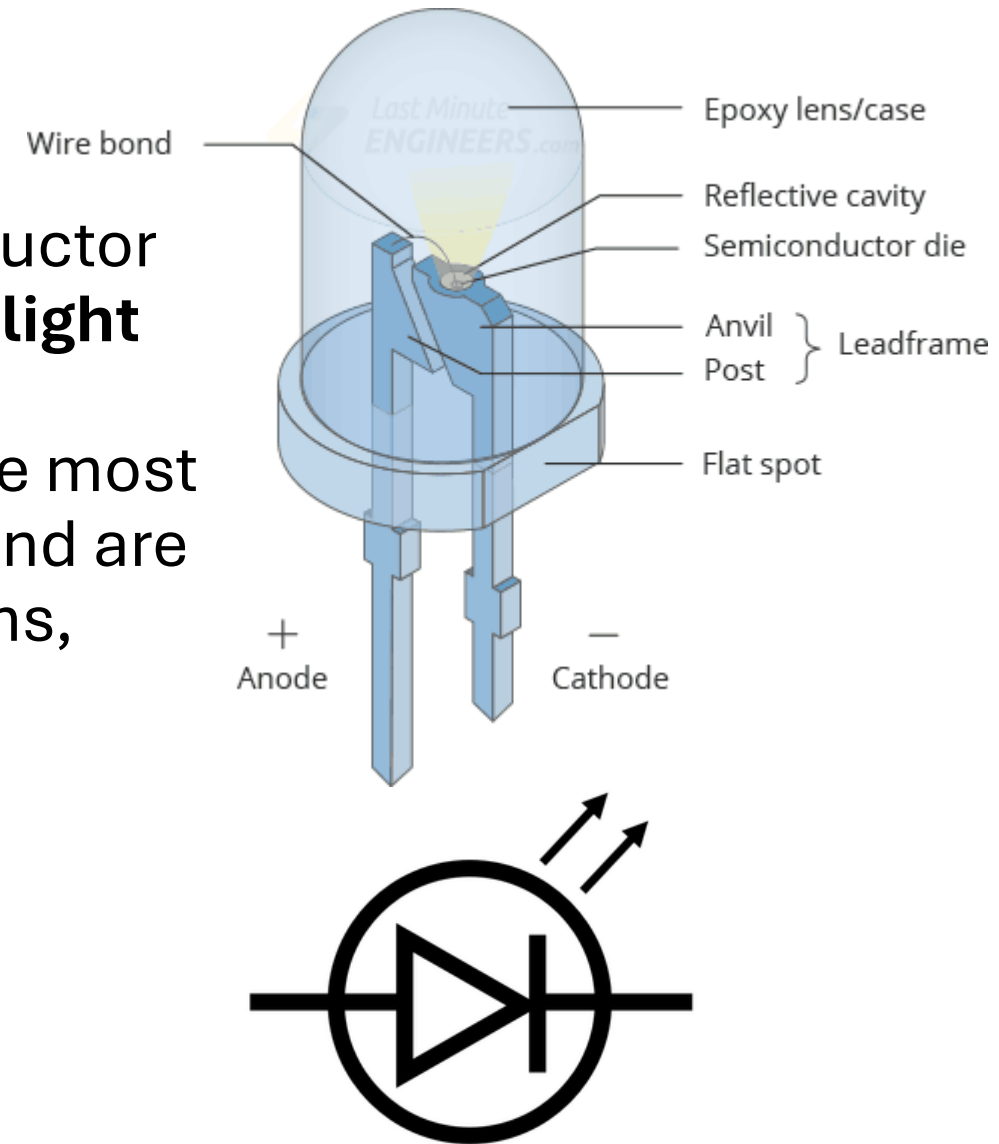


Application: Clipping Circuits



Light Emitting Diodes (LEDs)

- A **Light Emitting Diode (LED)** is a semiconductor device that converts **electrical energy into light energy** when forward biased (**electroluminescence**). LEDs are one of the most widely used electronic components today and are found in displays, indicators, lighting systems, remote controls, and modern TV screens.
- Leads:
 - **Long lead** → Anode (+)
 - **Short lead** → Cathode (-)
 - Flat side usually indicates cathode

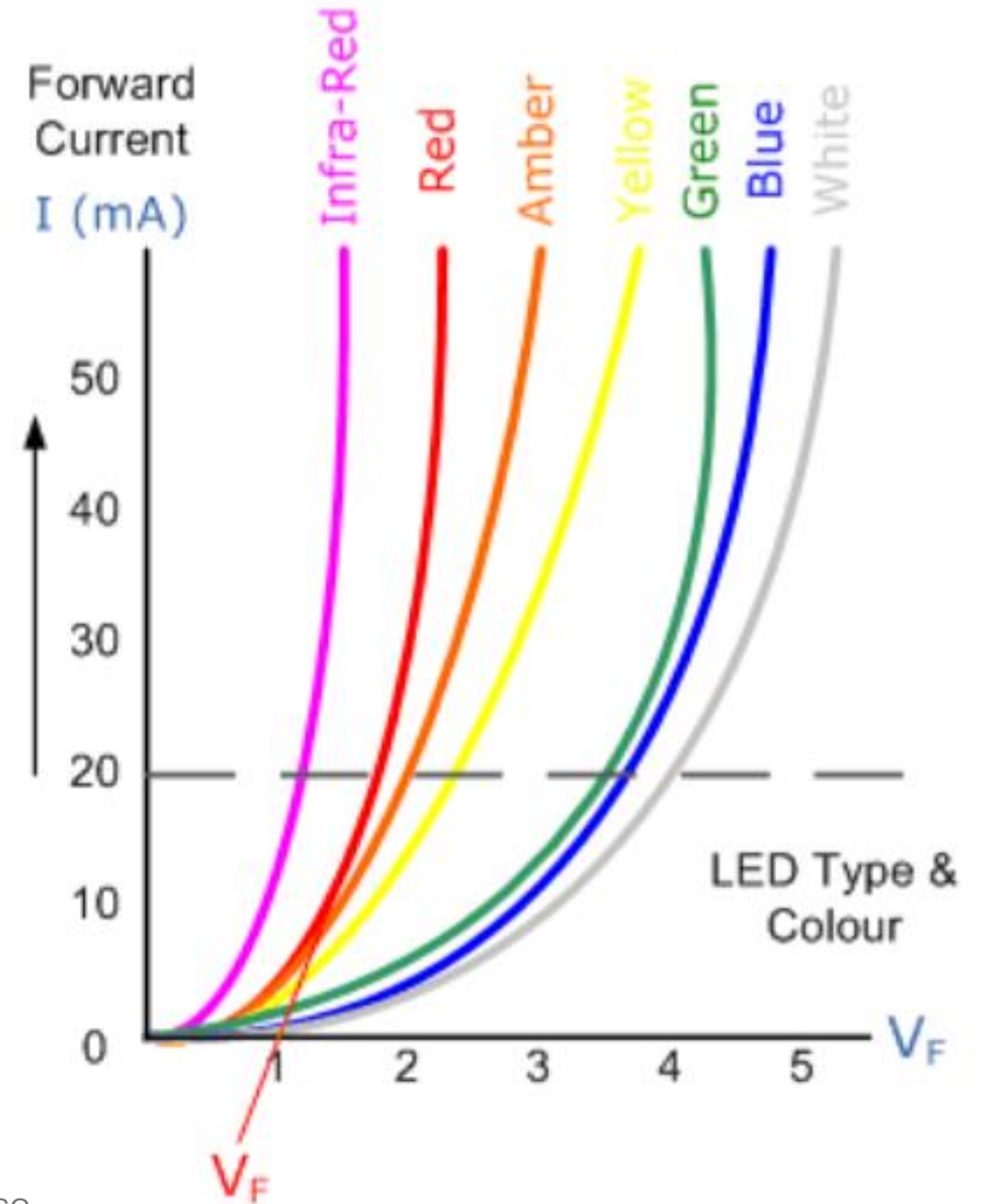
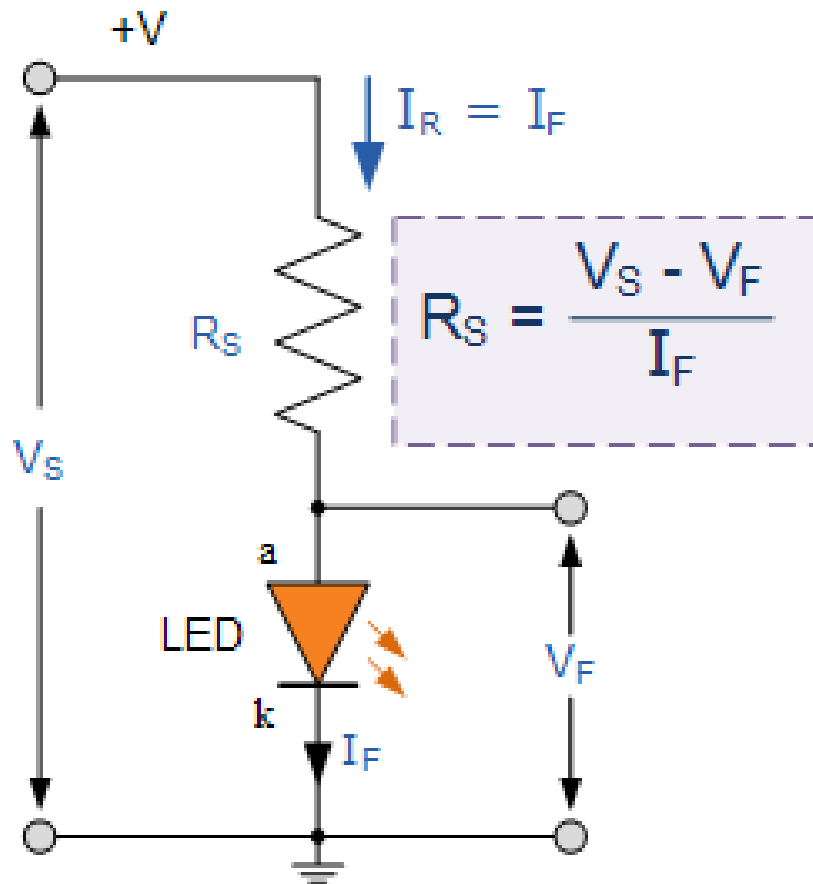


LED Materials and Colors

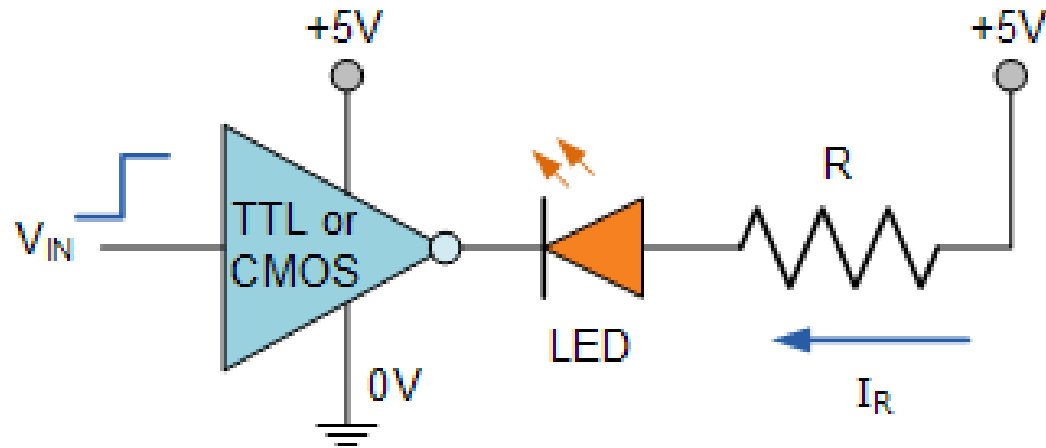
Semiconductor Material	Color Produced	Typical Forward Voltage
Gallium Arsenide (GaAs)	Infrared	1.2 V
Gallium Arsenide Phosphide (GaAsP)	Red/Amber	1.8–2.0 V
Gallium Phosphide (GaP)	Green	~2.2 V
Silicon Carbide (SiC)	Blue	~3.6 V
Gallium Indium Nitride (GaInN)	White/Blue	~4.0 V

Typical LED Characteristics			
Semiconductor Material	Wavelength	Colour	V _F @ 20mA
GaAs	850-940nm	Infra-Red	1.2v
GaAsP	630-660nm	Red	1.8v
GaAsP	605-620nm	Amber	2.0v
GaAsP:N	585-595nm	Yellow	2.2v
AlGaP	550-570nm	Green	3.5v
SiC	430-505nm	Blue	3.6v
GaInN	450nm	White	4.0v

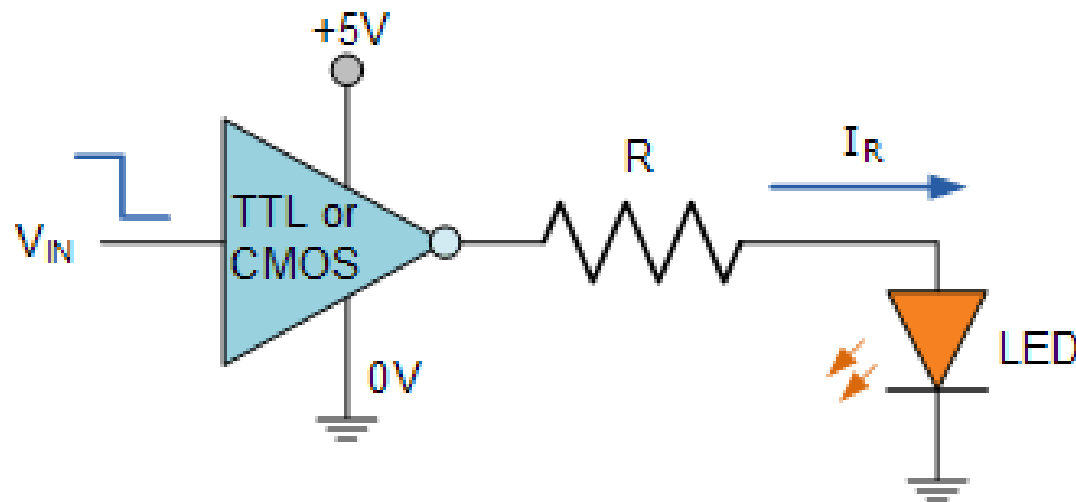
LED I-V Characteristics



IC Driver Circuit

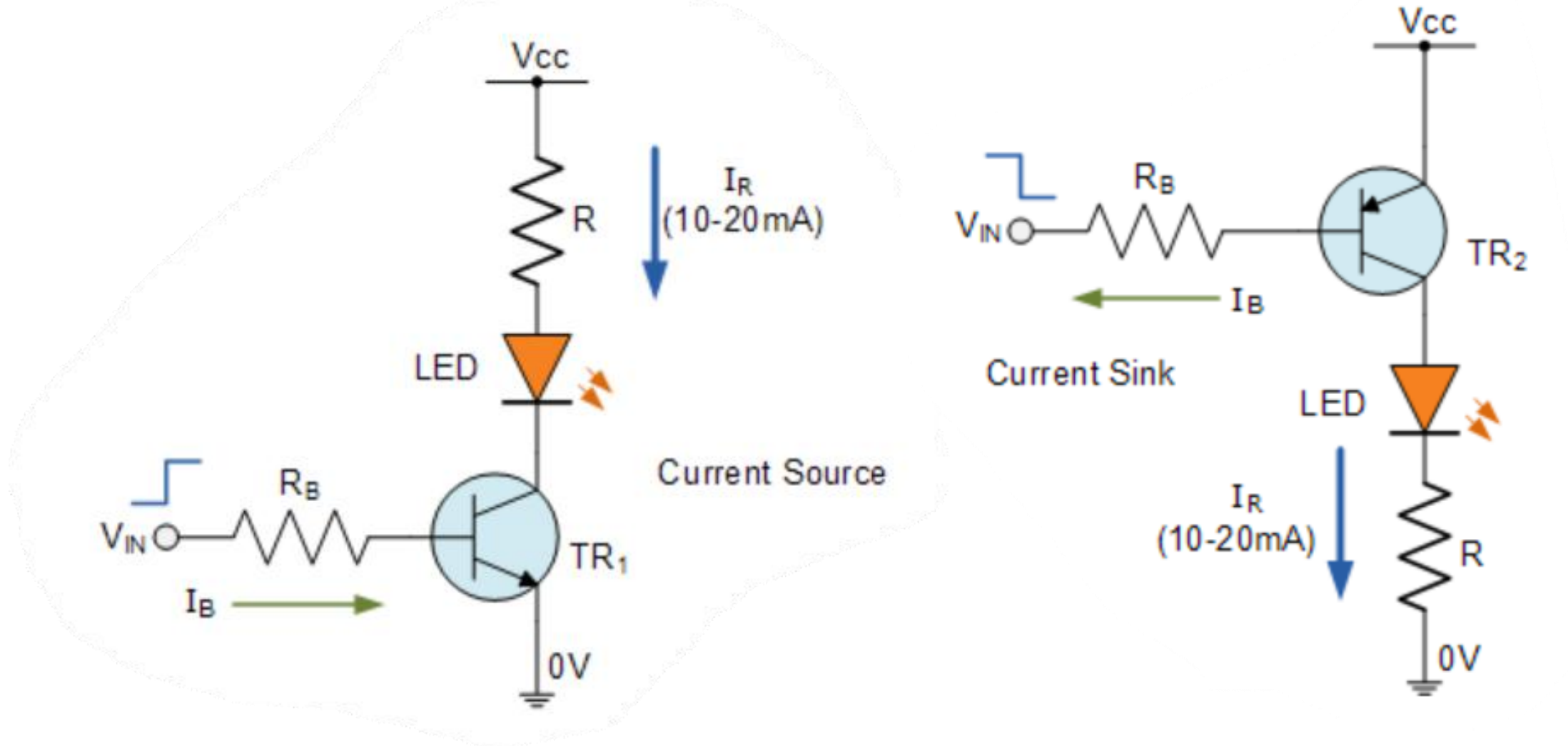


LED is "ON" when
the output is LOW
(current sink)

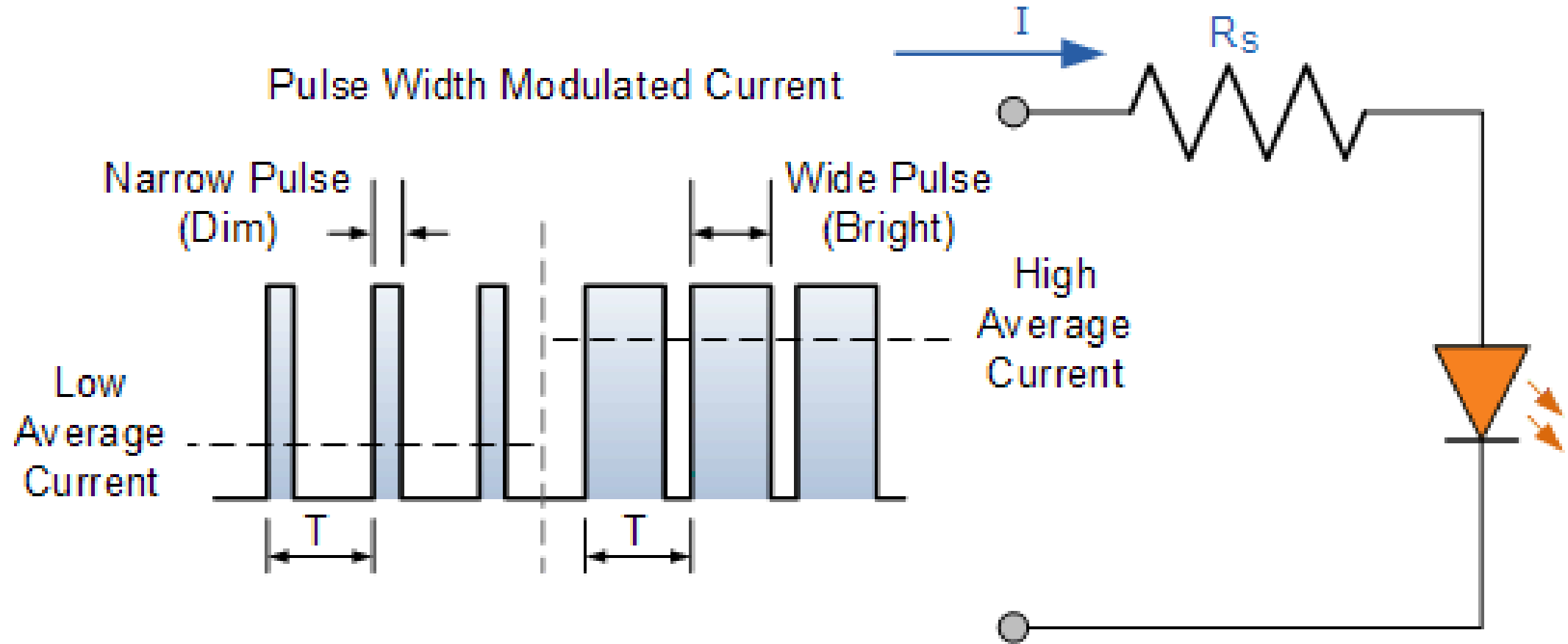


LED is "ON" when
the output is HIGH
(current source)

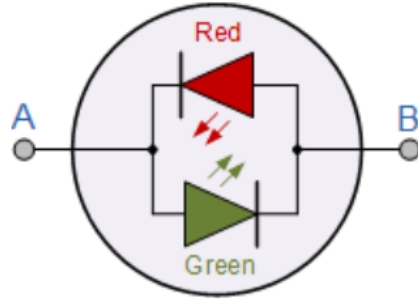
Transistor Driver Circuit



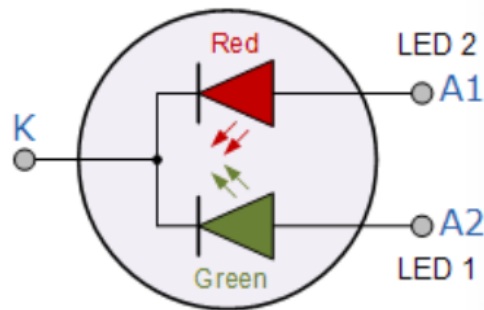
LED Light Intensity using PWM



Bi-colored, Tricolored, and Multi-colored LEDs



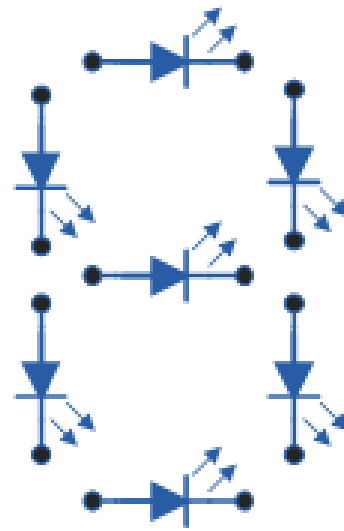
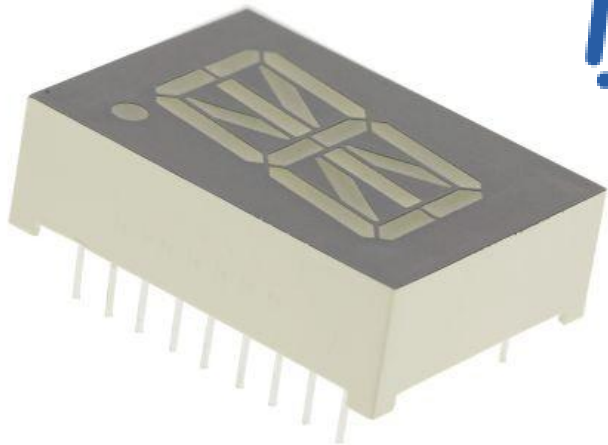
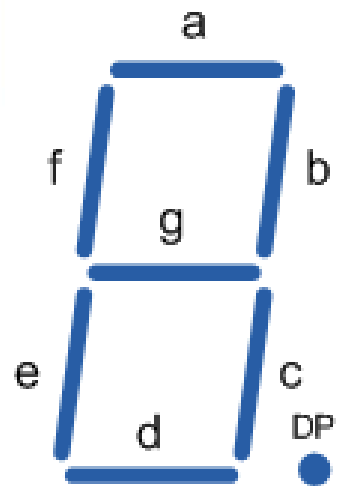
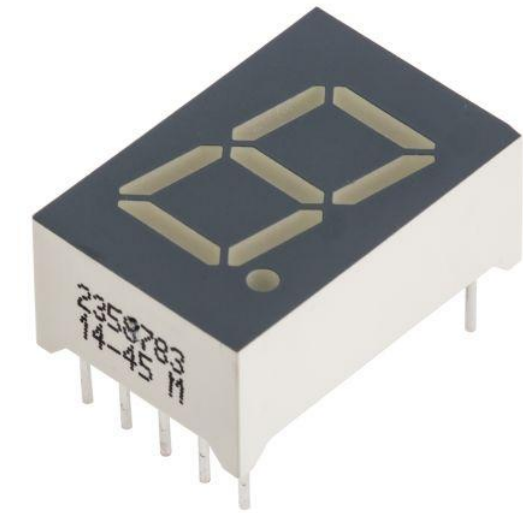
LED Selected	Terminal A		AC
	+	-	
LED 1	ON	OFF	ON
LED 2	OFF	ON	ON
Colour	Green	Red	Yellow



Output Colour	Red	Orange	Yellow	Green
LED 1 Current	0	5mA	9.5mA	15mA
LED 2 Current	10mA	6.5mA	3.5mA	0



LED Displays



ExploreEmbedded

CommonAnode

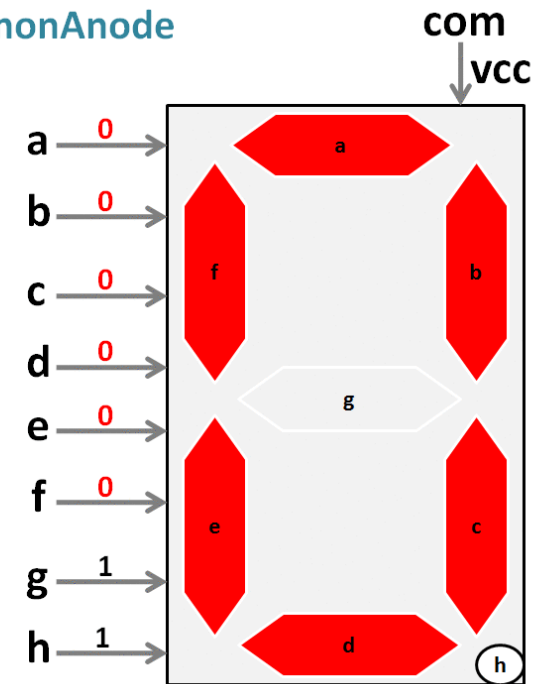
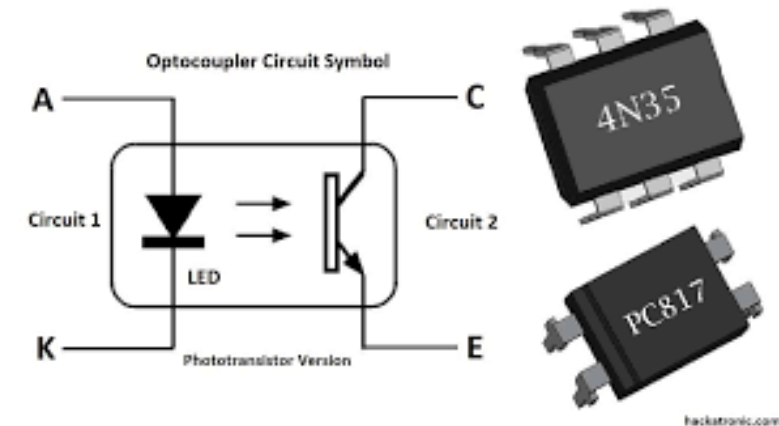
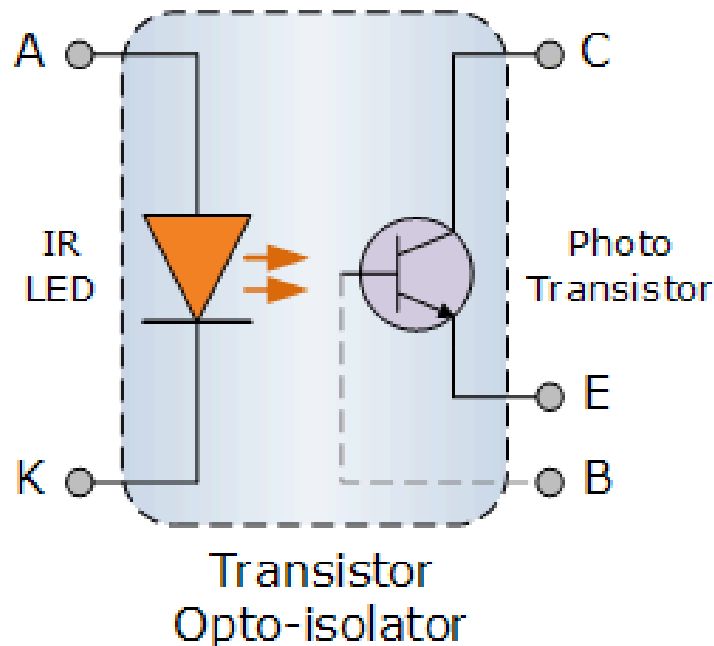
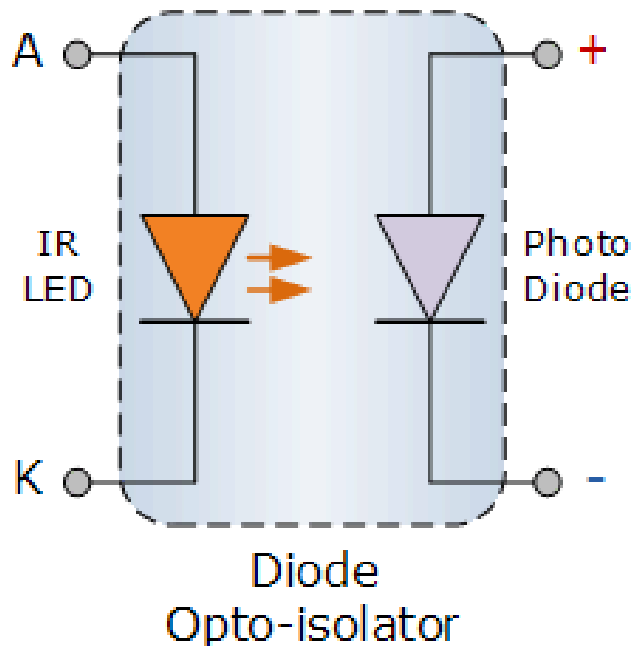


Photo-diode and Photo-transistor Optocouplers

- An optocoupler contains an LED and a photodiode or phototransistor. The LED transmits information optically while keeping circuits electrically isolated.



Transistor

Online Tutorial on Transistors:

https://www.electronics-tutorials.ws/transistor/tran_1.html

- ▶ 1. Bipolar Transistor
- 2. NPN Transistor
- 3. PNP Transistor
- 4. Transistor as a Switch
- 5. Junction Field Effect Transistor
- 6. The MOSFET
- 7. MOSFET as a Switch
- 8. Transistor Tutorial Summary
- 9. Darlington Transistors
- 10. FET Current Source
- 11. Open Collector Outputs
- 12. Phototransistor Basics
- 13. Complementary Transistors

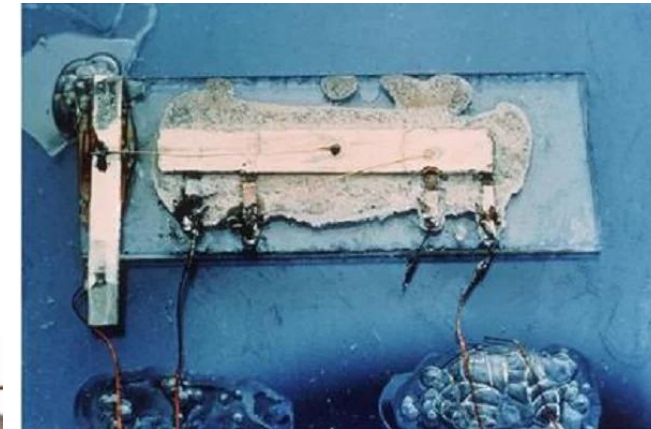
Vacuum Tube



- A **vacuum tube** (also called a **thermionic valve** or **electron tube**) is an electronic device that controls the flow of electrons through a vacuum inside a sealed glass or metal enclosure. Vacuum tubes were the foundation of early electronics before the invention of the transistor.
- A vacuum tube works by:
 - Heating a metal element called the **cathode**
 - Releasing electrons into a vacuum
 - Controlling the movement of those electrons using electric fields

Evolution Timeline from Vacuum Tube to Solid-State Electronics

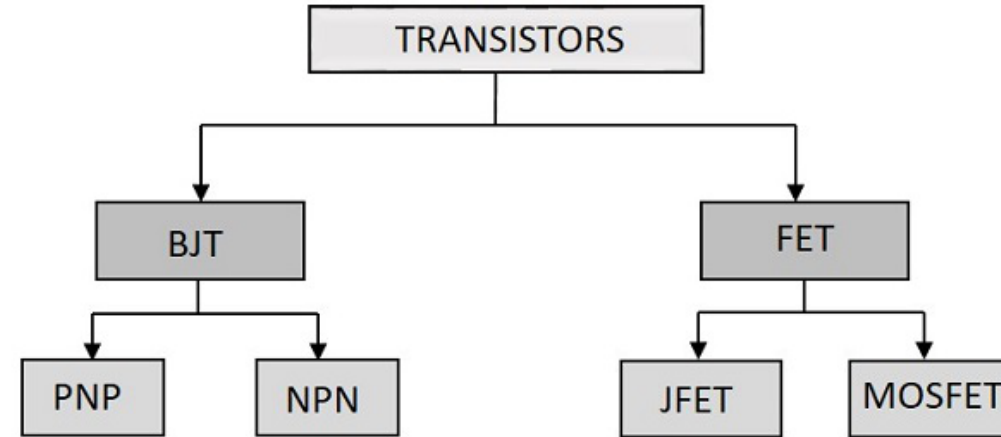
- 1904 — Vacuum Diode
 - Invented by John Ambrose Fleming
- 1906 — Triode
 - The first electronic amplifier invented by Lee de Forest. A tiny voltage on the grid could control a much larger current flowing from cathode to anode.
- 1947 — Transistor
 - Invented at Bell Labs.
- 1958 — Integrated Circuit
 - Multiple transistors placed on one chip.
- Modern Era
 - Billions of transistors on microprocessors.



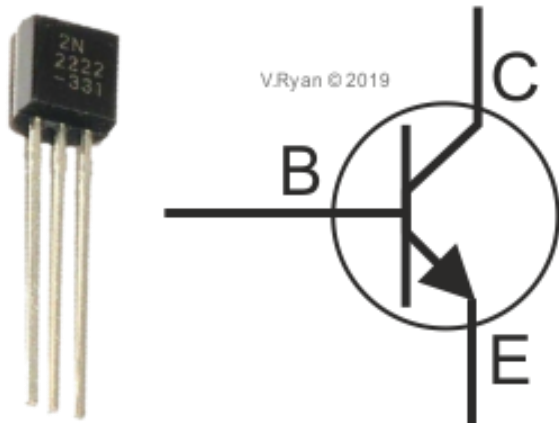
Types of Transistors

Transistors are broadly classified into two main categories:

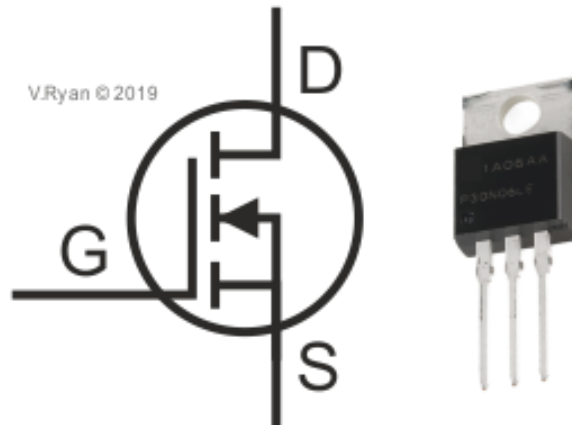
1. **Bipolar Junction Transistor (BJT)**
2. **Field-Effect Transistor (FET)**



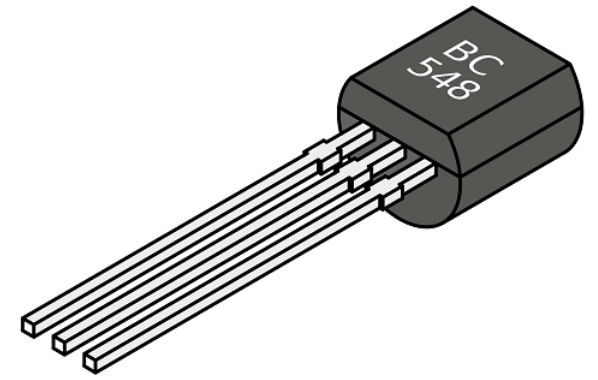
BJT
(bipolar junction transistor)



FET
(Field-Effect Transistors)



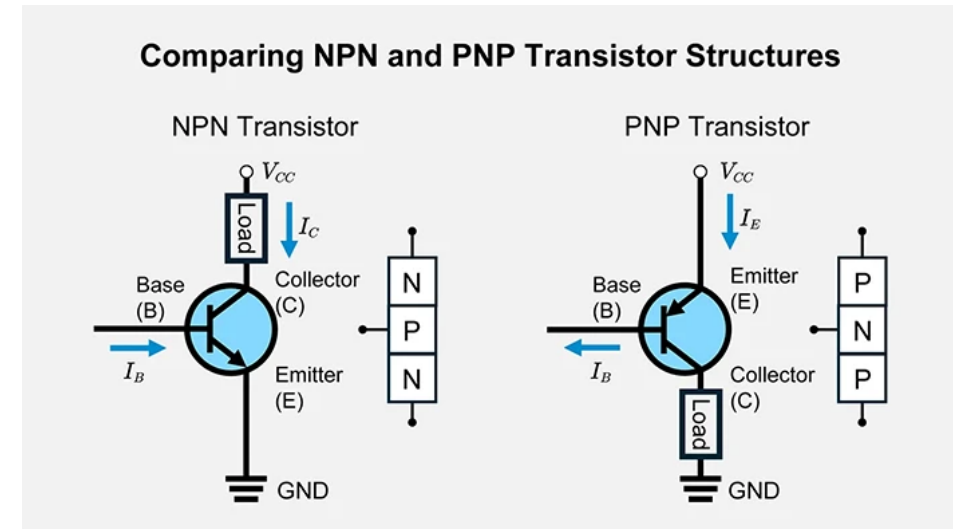
Bipolar Junction Transistors (BJTs)



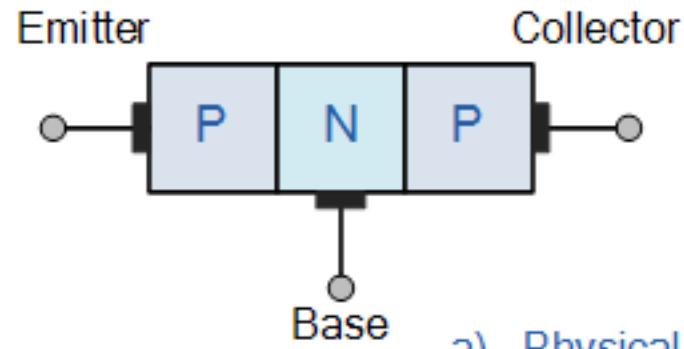
- A **Bipolar Junction Transistor (BJT)** is a three-layer semiconductor device used mainly for **amplification** of signals and electronic **switching**.
- The word **bipolar** means the device operation involves **two types of charge carriers: electrons and holes**.
- The word **junction** refers to the **two PN junctions** inside the transistor.
- A BJT therefore contains two PN junctions, three semiconductor regions, and three terminals.

Bipolar Transistor Construction

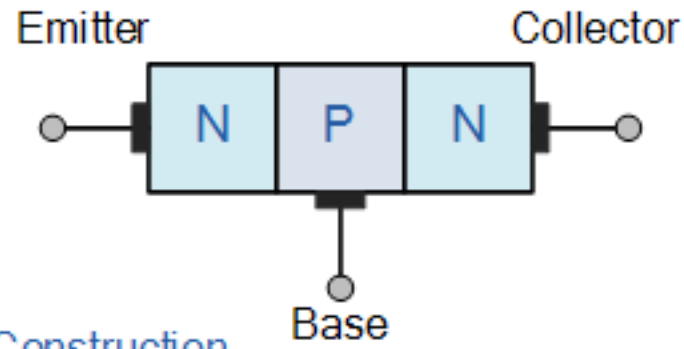
- A transistor has three terminals:
 - **Emitter (E)** – Supplies charge carriers
 - **Base (B)** – Controls transistor action
 - **Collector (C)** – Collects carriers
- There are two main transistor types:
 - **NPN** (Not Pointing iN) Transistor - Current flows from collector to emitter; Requires positive base voltage relative to emitter.
 - **PNP** (Pointing iN) Transistor - Current direction reverses; Requires negative base voltage relative to emitter.



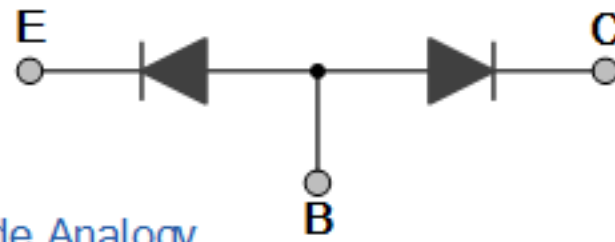
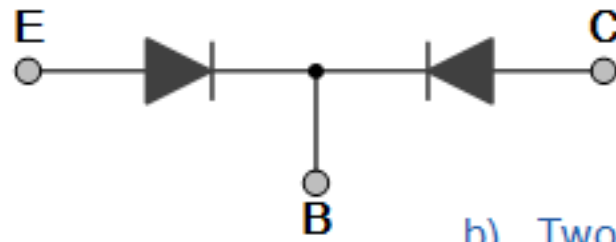
PNP Transistor



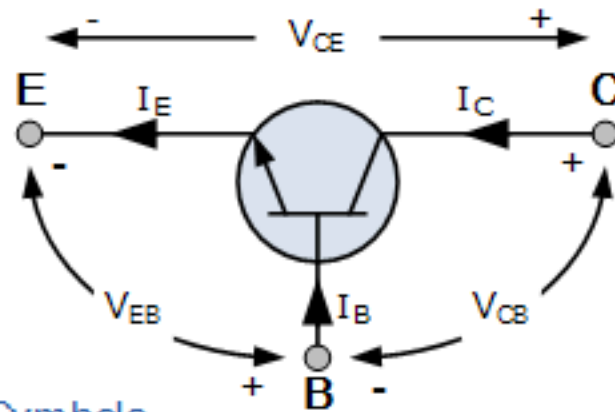
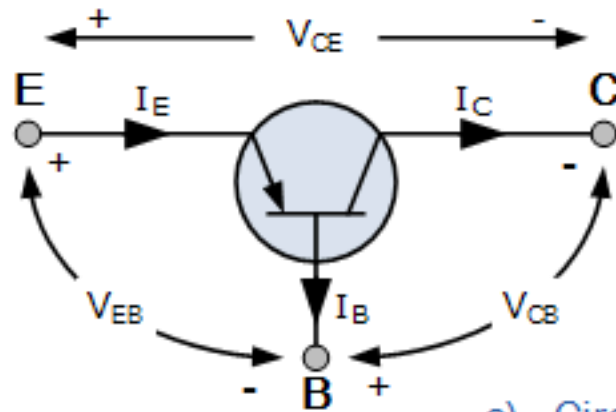
NPN Transistor



a). [Physical Construction](#)



b). [Two-diode Analogy](#)

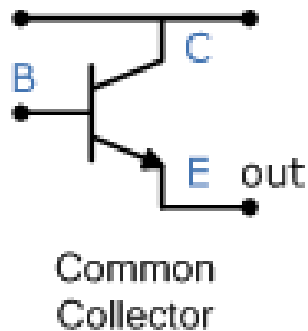
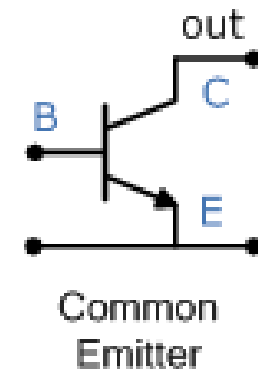
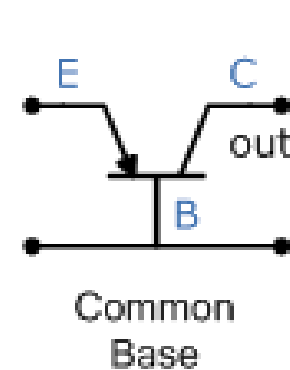
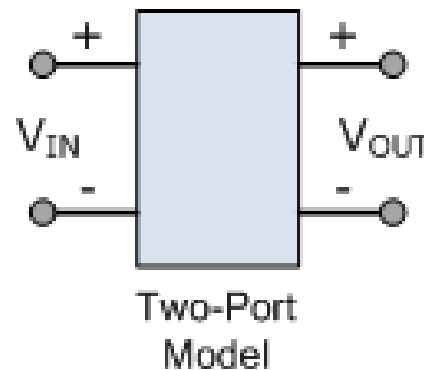


c). [Circuit Symbols](#)

$$I_E = I_B + I_C$$

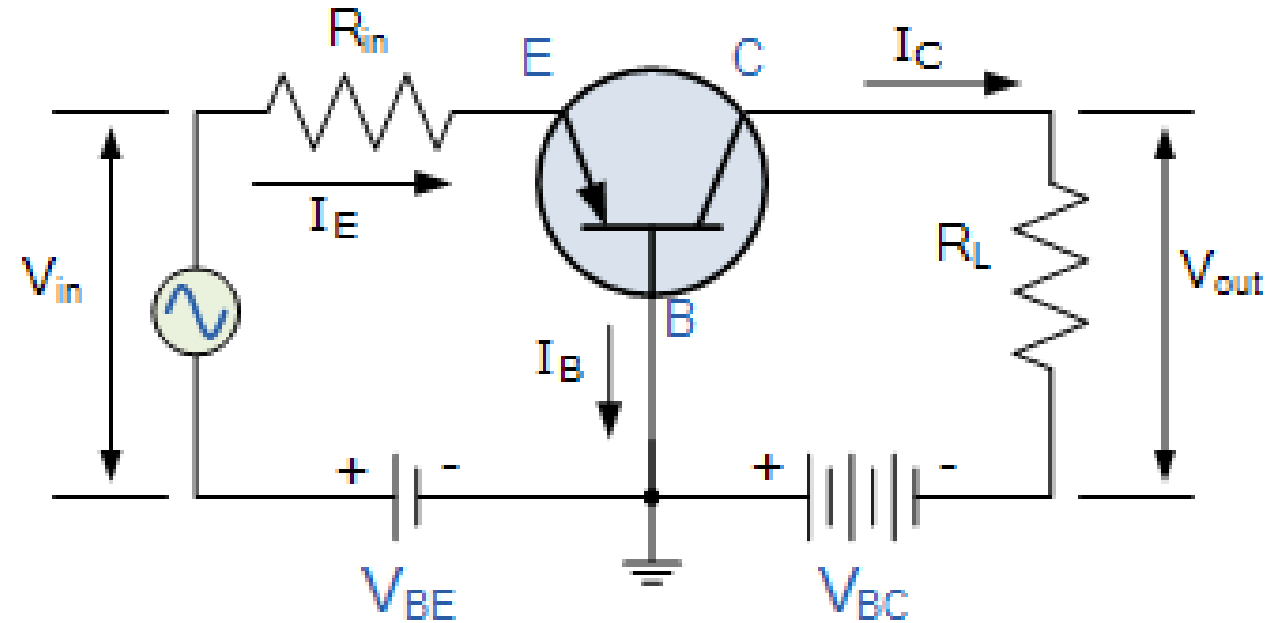
Bipolar Transistor Configurations

- A **BJT configuration** describes how the transistor terminals are connected in a circuit relative to the input and output signals.
- Since a transistor has three terminals one terminal becomes common to both input and output circuits.
- There are **three basic transistor configurations**:
 - Common Base (CB)
 - Common Emitter (CE)
 - Common Collector (CC)



Common Base (CB) Configuration

- As **non-inverting voltage amplifier**, i.e., the signal voltages V_{in} and V_{out} are “in-phase”.
- **Voltage gain (A_v) is high; current gain (α) is less than 1** (0.95 to 0.99); power gain is moderate.
- Used in single stage amplifier circuits such as microphone pre-amplifier or radio frequency (Rf) amplifiers due to its very good **high frequency response**.



$$A_v = \frac{V_{out}}{V_{in}} = \frac{I_C \times R_L}{I_E \times R_{IN}} \quad \alpha = \frac{I_C}{I_E}$$

Common Emitter (CE) Configuration

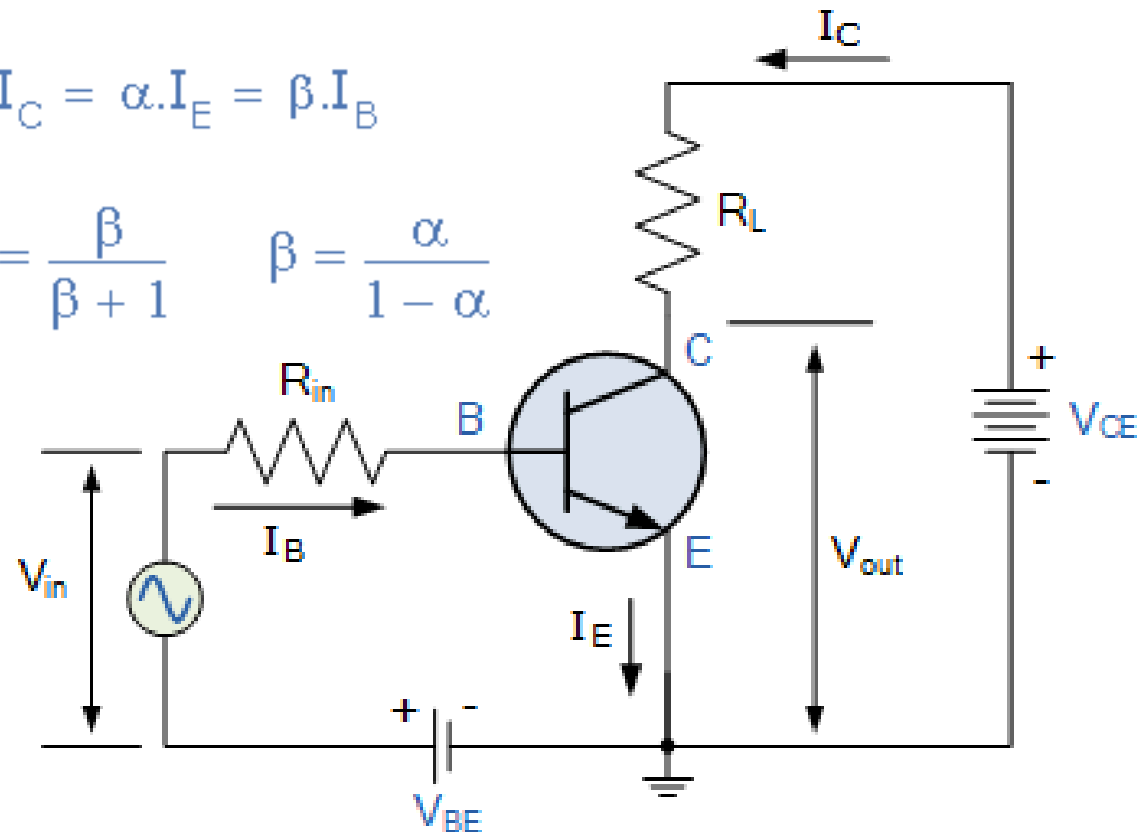
- The most used circuit for transistor-based **amplifiers**.
- Produces the **highest current gain** (β or h_{FE} ; typically, 20 to 500), lower voltage gain (A_v) and **high power gain** of all the three bipolar transistor configurations.
- Input impedance is LOW and output impedance is HIGH. Output signal has a 180° phase-shift (inverted).

$$\text{Alpha, } (\alpha) = \frac{I_C}{I_E} \quad \text{and} \quad \text{Beta, } (\beta) = \frac{I_C}{I_B}$$

$$I_E = I_C + I_B$$

$$\therefore I_C = \alpha \cdot I_E = \beta \cdot I_B$$

$$\text{as: } \alpha = \frac{\beta}{\beta + 1} \quad \beta = \frac{\alpha}{1 - \alpha}$$



Transistor Characteristics – NPN, CE Mode

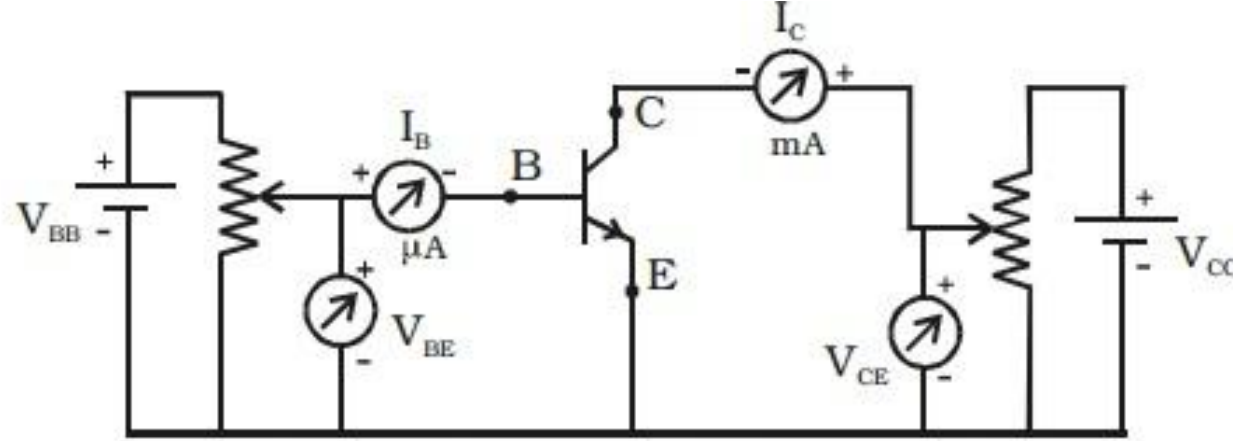


Fig Transistor circuit in CE mode.

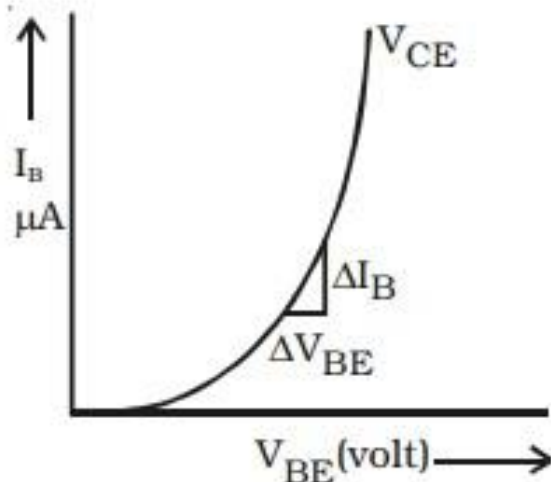


Fig Input characteristics

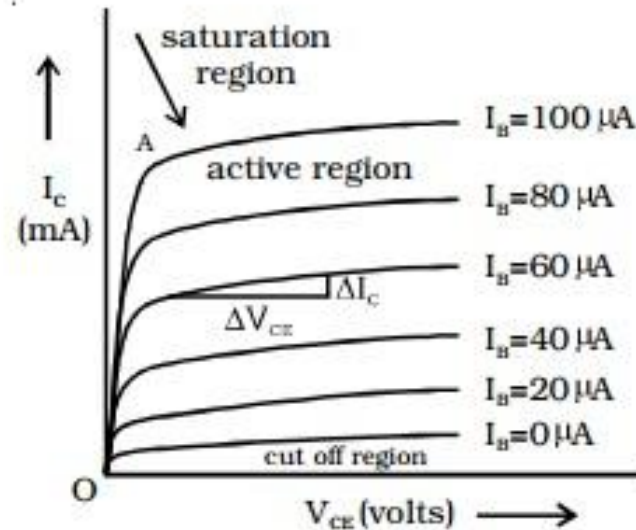


Fig Output characteristics

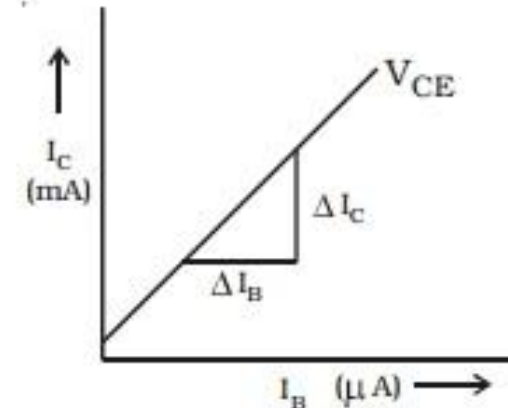


Fig Transfer characteristic curve

Input Characteristics: I_B vs V_{BE}

- Shows how **base current** (I_B) varies with **base-emitter voltage** (V_{BE}) at a constant **collector-emitter voltage** (V_{CE}).
- Similar to a **forward-biased diode curve**.
- Current starts rising sharply around $V_{BE} \approx 0.6\text{--}0.7\text{ V}$ (for silicon transistors like BC547).
- Indicates how much base current is needed to "turn on" the transistor.

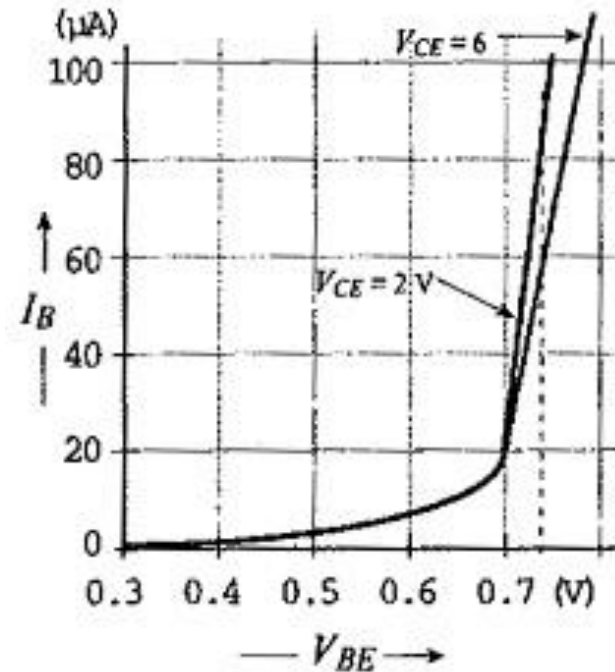
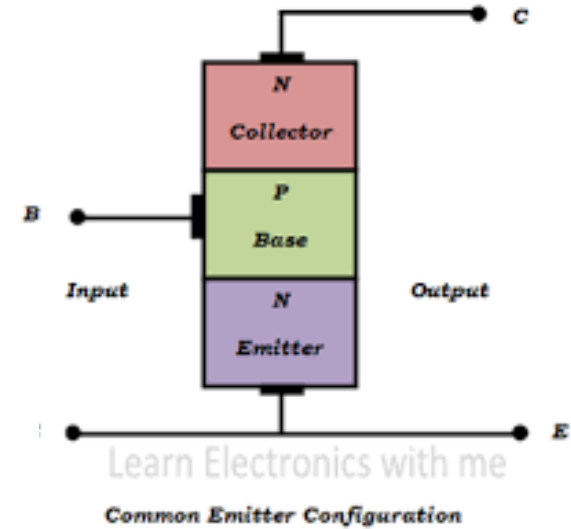


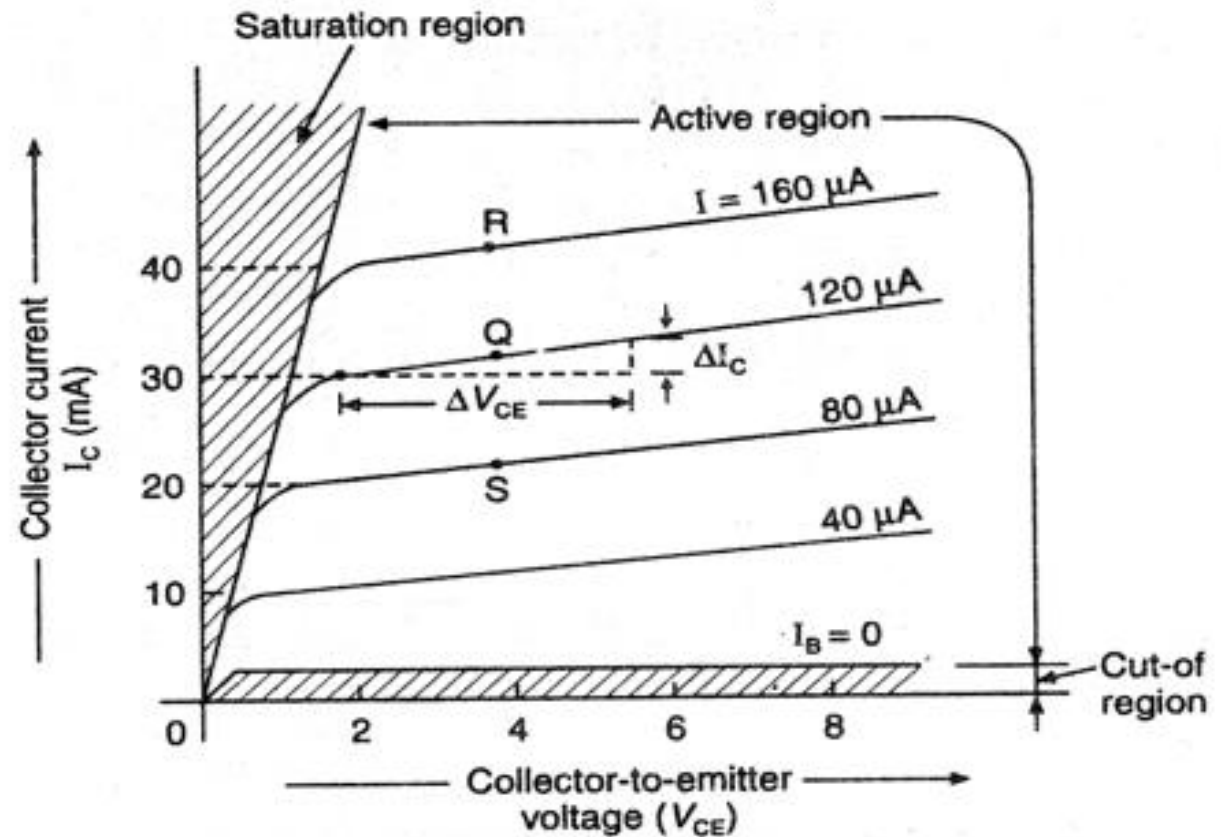
Figure 4-27

The common-emitter input characteristics for a transistor are those of a forward-biased pn-junction.



Output Characteristics: I_C vs V_{CE}

- Shows how **collector current** (I_C) varies with **collector-emitter voltage** (V_{CE}) at different fixed **base currents** (I_B).
- The graph is divided into **three regions**:
 - Cutoff Region**: $I_B = 0 \rightarrow$ transistor OFF, $I_C \approx 0$ regardless of V_{CE}
 - Active Region**: $V_{CE} > 0.2\text{ V}$, $I_C \approx \beta \times I_B$, Transistor acts as an **amplifier**
 - Saturation Region**: $V_{CE} < 0.2\text{--}0.3\text{ V}$, I_C no longer increases with I_B , $\Rightarrow$ transistor is **fully ON** like a switch



Transfer Characteristics: I_C vs I_B

- Shows how **collector current** (I_C) varies with **base current** (I_B) at **constant V_{CE}** .
- Linear relation in the **active region**: $I_C = \beta \cdot I_B$, where β (or hFE) is the current gain (e.g., 100–800 for BC547).
- As I_B increases, I_C increases proportionally **until saturation**, beyond which increasing I_B no longer increases I_C .
- Helps determine how much I_B is needed for a desired I_C , i.e., to bias the transistor correctly.

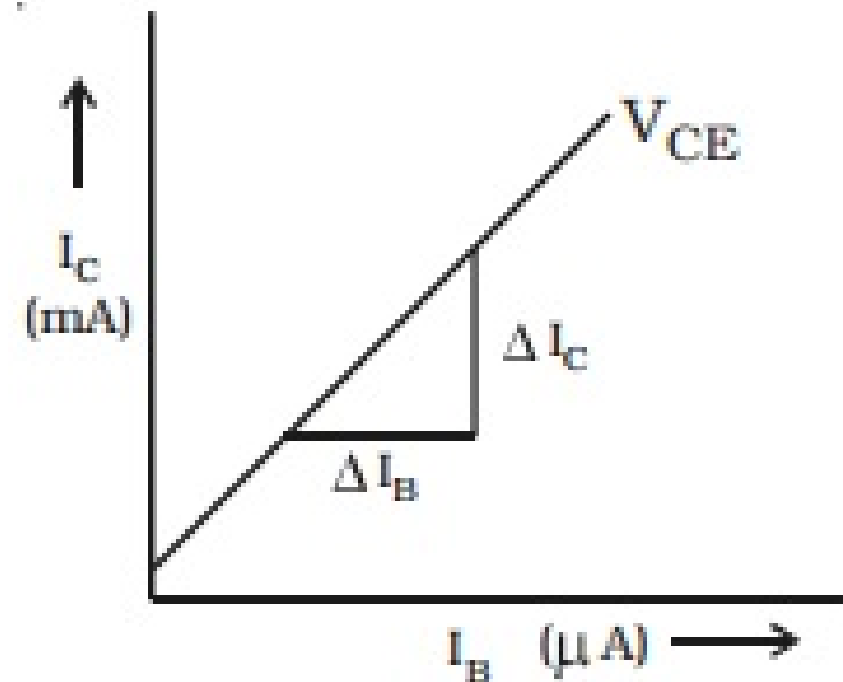
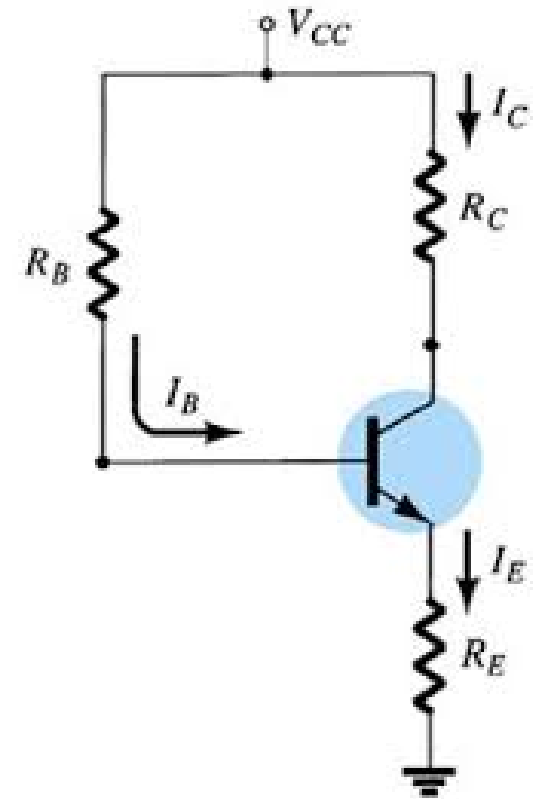


Fig Transfer characteristic curve

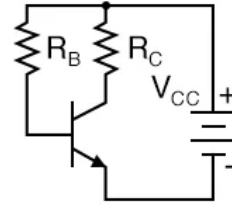
Transistor Biasing

- **Biasing** means applying **DC voltages and currents** to a transistor to set it up for proper operation in a specific region – typically the **active region** for amplification.
- **Goals of Biasing:** Ensure the transistor operates in the **desired region** (cutoff, active, or saturation); **Stabilize** the operating point against temperature and transistor variation; Allow a **faithful reproduction of input signals** (no distortion).
- For a **Common Emitter (CE) NPN Transistor**:
 - Base-Emitter junction is **forward-biased** ($V_{BE} \approx 0.6\text{--}0.7\text{ V}$)
 - Collector-Base junction is **reverse-biased**



Types of Biasing Methods

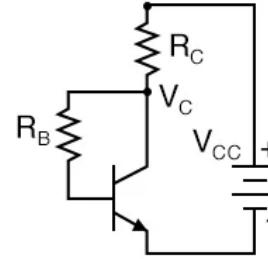
Method	Stability	Use
Fixed bias	Poor	Simple, used in switching circuits
Collector-to-base bias	Better	Improved stability
Voltage divider bias	Best	Common in amplifiers
Emitter feedback bias	High	Adds thermal stability



$$I_E = \frac{V_{BB} - V_{BE}}{R_B / \beta}$$

$$R_B = \frac{V_{BB} - V_{BE}}{I_E / \beta}$$

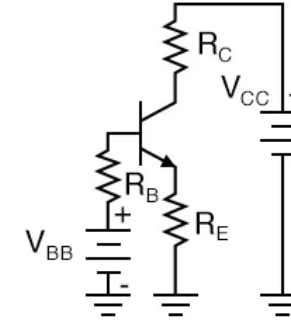
Base-bias



$$I_E = \frac{V_{CC} - V_{BE}}{R_B / \beta + R_C}$$

$$R_B = \beta \left[\frac{V_{CC} - V_{BE}}{I_E} - R_C \right]$$

Collector feedback bias



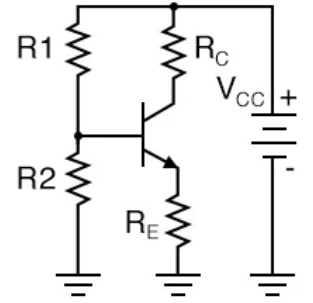
$$I_E = \frac{V_{BB} - V_{BE}}{R_B / \beta + R_E}$$

$R_E \Leftarrow R_E + r_{EE}$
to include r_{EE}

$$r_{EE} = 26\text{mV}/I_E$$

$$R_B = \beta \left[\frac{V_{BB} - V_{BE}}{I_E} - R_E \right]$$

Emitter-bias



$$V_{BB} = V_{th}$$

$$R_B = R_{th}$$

$$R1 = R_{th} \frac{V_{CC}}{V_{th}}$$

$$\frac{1}{R2} = \frac{1}{R_{th}} - \frac{1}{R1}$$

Voltage divider bias

Common Collector (CC) Configuration

- Also known as a **Voltage Follower** or **Emitter Follower** circuit - the load resistance has both the collector current and the input current of the base flowing through it.
- Very useful for impedance matching applications because of its very high input impedance, in the region of hundreds of thousands of Ohms while having a relatively low output impedance.

$$I_E = I_C + I_B$$

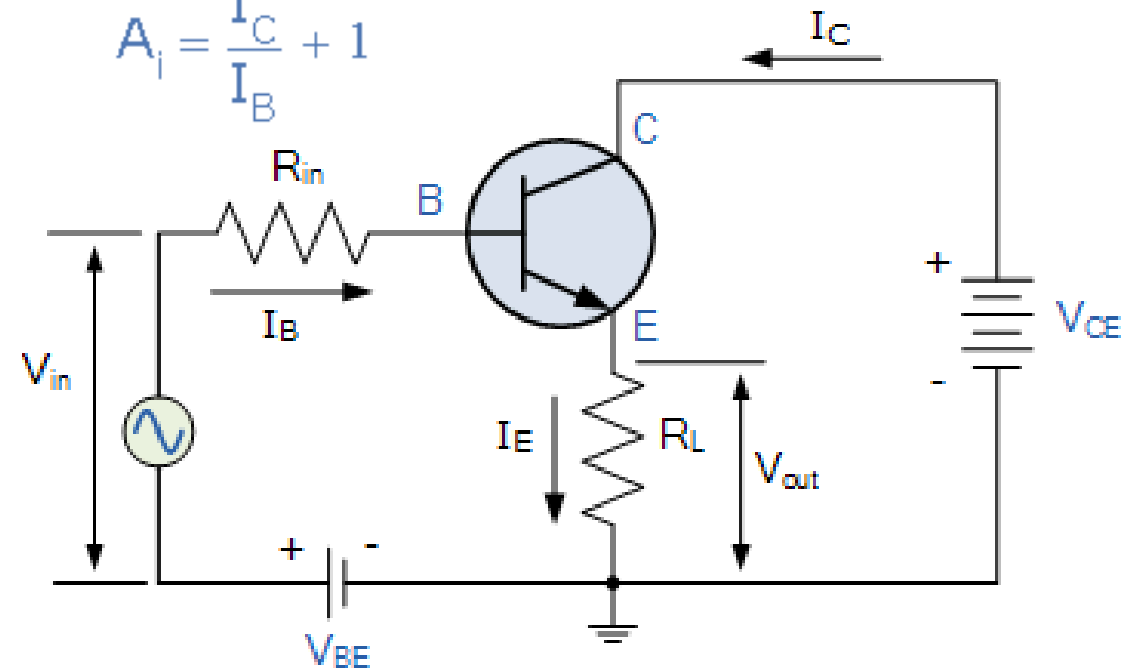
$$V_E \approx V_B - 0.7V$$

$$A_i = \frac{I_E}{I_B} = \frac{I_C + I_B}{I_B}$$

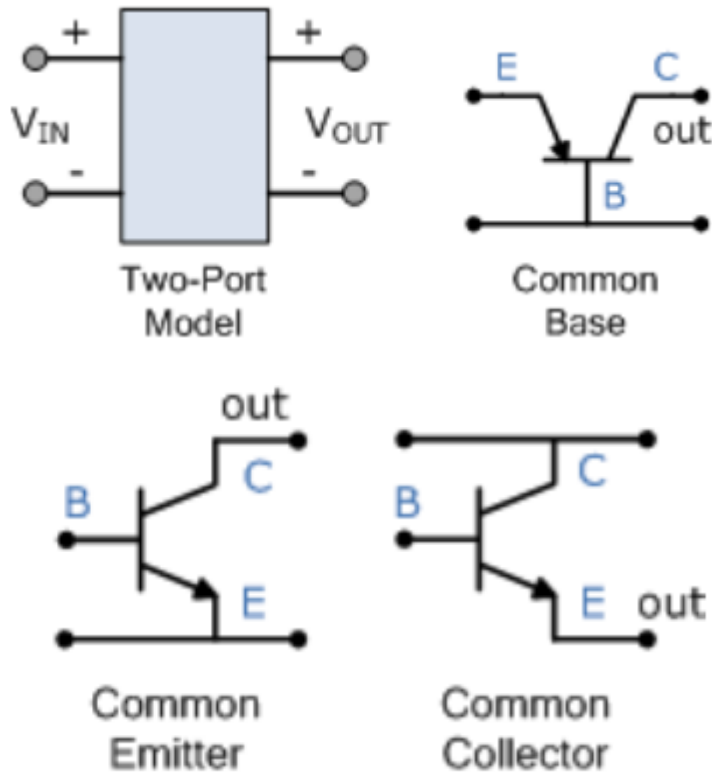
Current Gain:

$$A_i = \beta + 1$$

$$A_i = \frac{I_C}{I_B} + 1$$

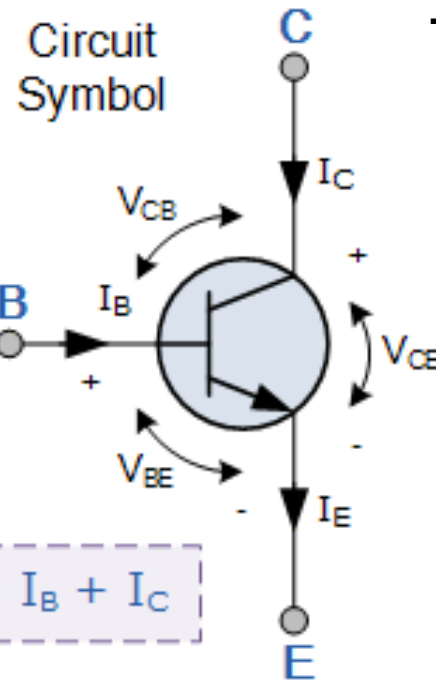
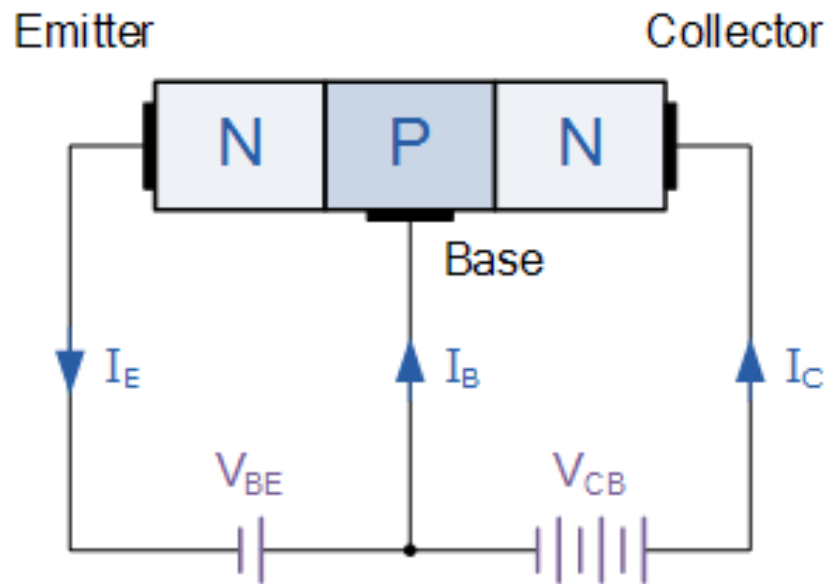


Bipolar Transistor Configurations Summary



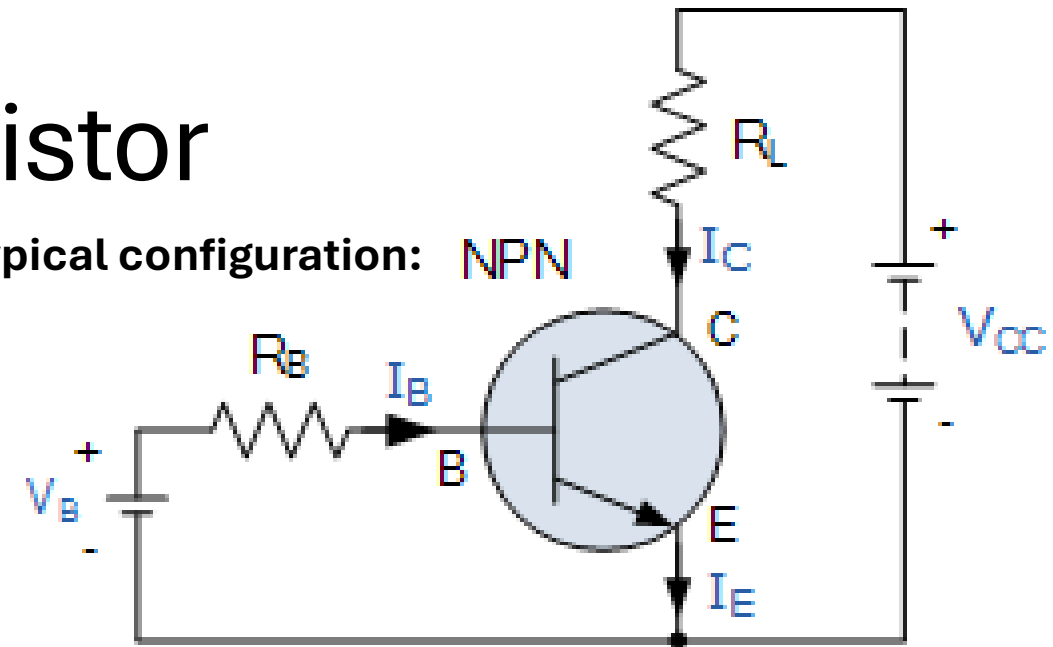
Characteristic	Common Base	Common Emitter	Common Collector
Input Impedance	Low	Medium	High
Output Impedance	Very High	High	Low
Phase Shift	0°	180°	0°
Voltage Gain	High	Medium	Low
Current Gain	Low	Medium	High
Power Gain	Low	Very High	Medium

NPN Bipolar Junction Transistor



$$I_E = I_B + I_C$$

Typical configuration: **NPN**



$$\alpha = \frac{\beta}{\beta + 1} \quad \text{or} \quad \alpha = \beta(1 - \alpha)$$

$$\beta = \frac{\alpha}{1 - \alpha} \quad \text{or} \quad \beta = \alpha(1 + \beta)$$

$$\text{If } \alpha = 0.99 \quad \beta = \frac{0.99}{0.01} = 99$$

$$\text{DC Current Gain} = \frac{\text{Output Current}}{\text{Input Current}} = \frac{I_C}{I_B}$$

$$\frac{I_C}{I_E} = \alpha$$

BC547 Transistor

Absolute Maximum Ratings $T_a=25^\circ\text{C}$ unless otherwise noted

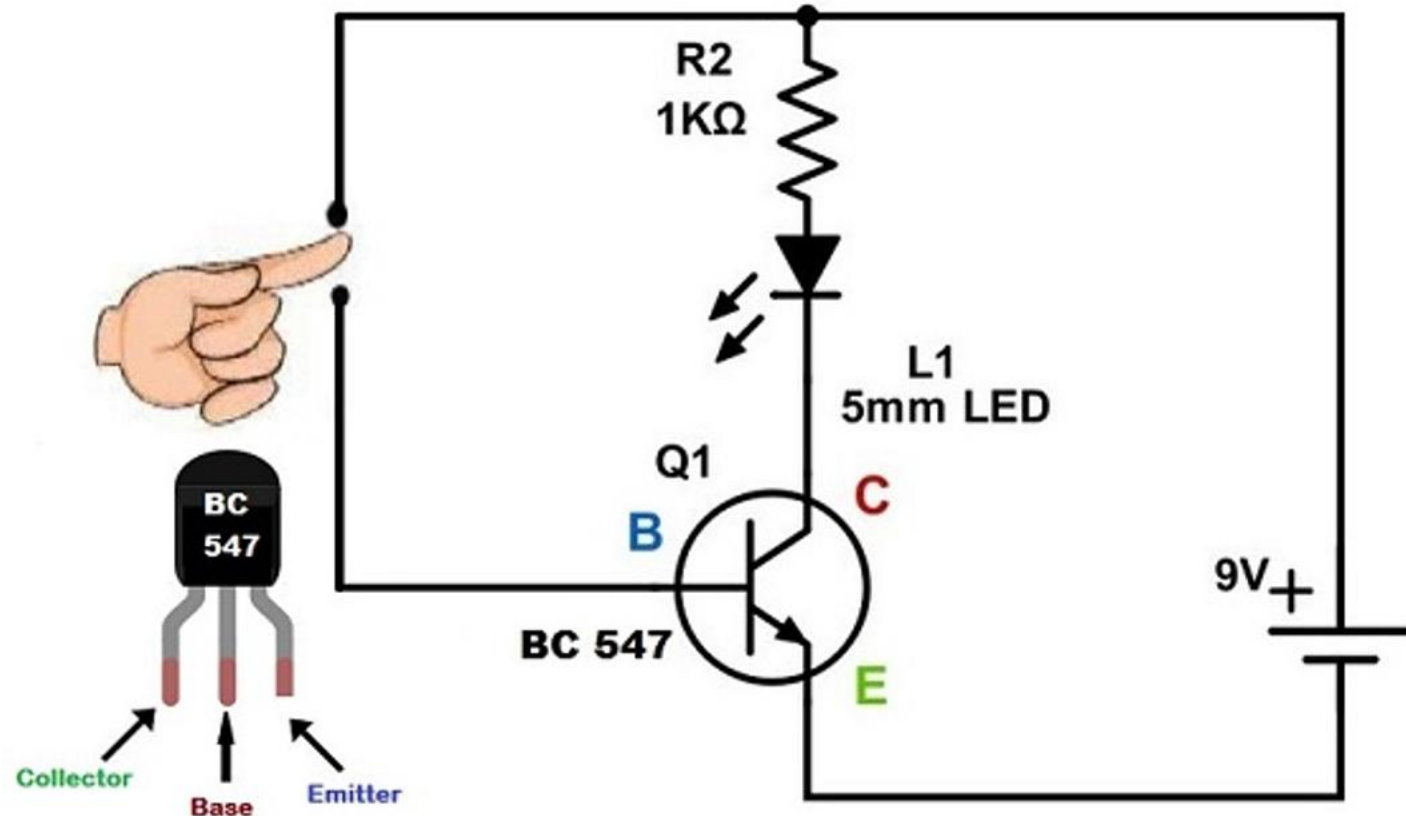
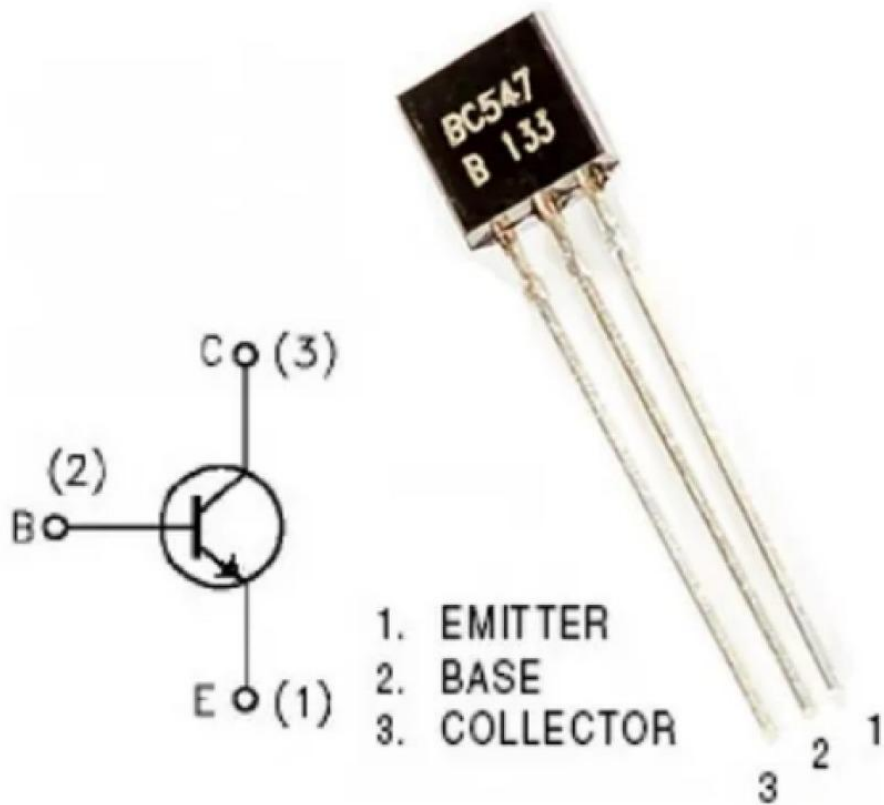
Symbol	Parameter	Value	Units
V_{CBO}	Collector-Base Voltage : BC546	80	V
	: BC547/550	50	V
	: BC548/549	30	V
V_{CEO}	Collector-Emitter Voltage : BC546	65	V
	: BC547/550	45	V
	: BC548/549	30	V
V_{EBO}	Emitter-Base Voltage : BC546/547	6	V
	: BC548/549/550	5	V
I_C	Collector Current (DC)	100	mA
P_C	Collector Power Dissipation	500	mW
T_J	Junction Temperature	150	$^\circ\text{C}$
T_{STG}	Storage Temperature	-65 ~ 150	$^\circ\text{C}$

Electrical Characteristics $T_a=25^\circ\text{C}$ unless otherwise noted

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Units
I_{CBO}	Collector Cut-off Current	$V_{CB}=30\text{V}, I_E=0$			15	nA
h_{FE}	DC Current Gain	$V_{CE}=5\text{V}, I_C=2\text{mA}$	110		800	
$V_{CE}(\text{sat})$	Collector-Emitter Saturation Voltage	$I_C=10\text{mA}, I_B=0.5\text{mA}$ $I_C=100\text{mA}, I_B=5\text{mA}$		90 200	250 600	mV mV

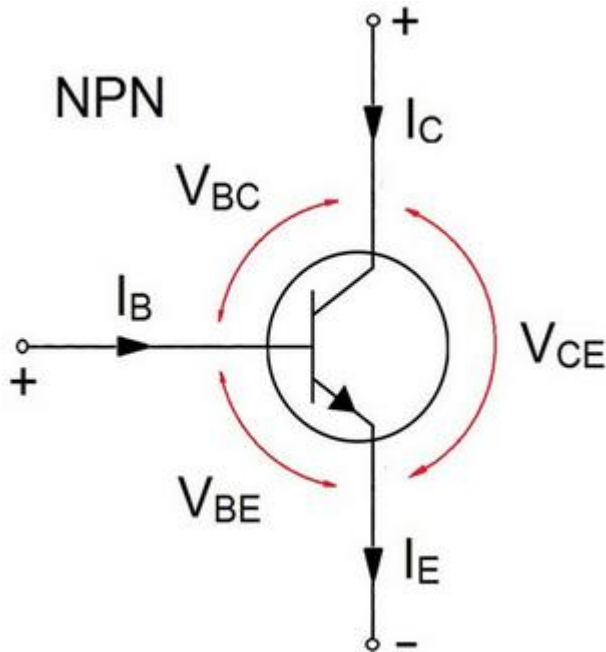
<https://www.sparkfun.com/datasheets/Components/BC546.pdf>

Simple Application



Key Parameters of the BC547

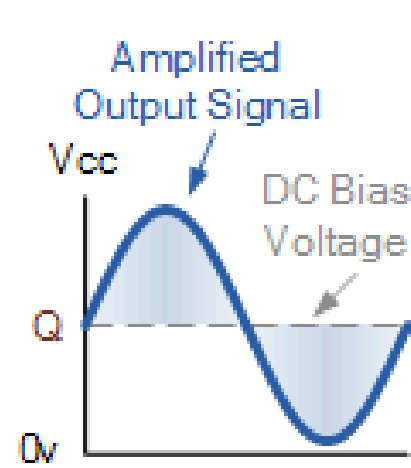
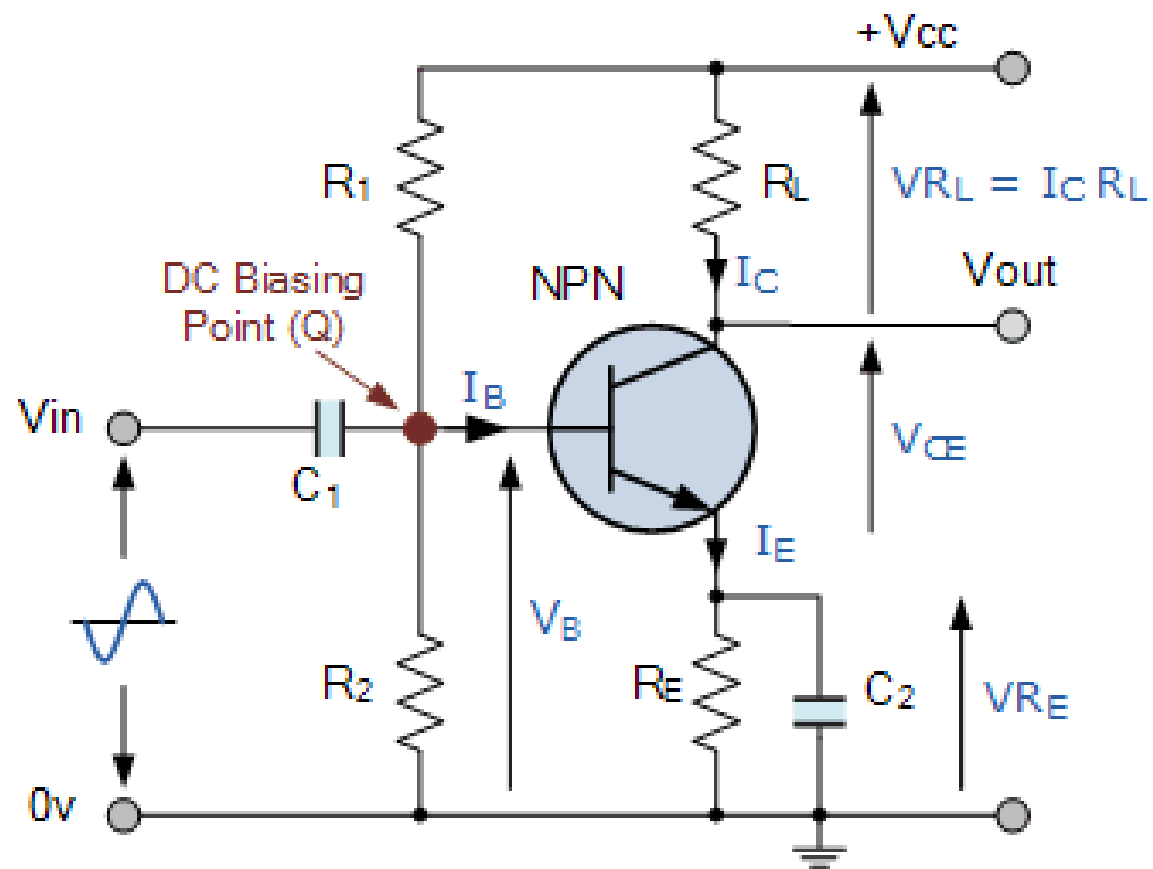
Parameter	Symbol	Typical Value for BC547	Explanation
Collector-Emitter Voltage	V _{CEO}	45 V	Maximum voltage between collector and emitter when the base is open.
Collector-Base Voltage	V _{CBO}	50 V	Max voltage between collector and base with emitter open.
Emitter-Base Voltage	V _{EBO}	6 V	Max voltage between emitter and base with collector open.
Collector Current	I _C	100 mA	Maximum DC current the collector can handle.
Total Power Dissipation	P _{tot}	500 mW (at 25°C)	Maximum power the transistor can safely dissipate.
DC Current Gain	h _{FE}	110 – 800 (depending on subtype)	Ratio of collector current to base current (I _C /I _B). A measure of amplification.
Transition Frequency	f _T	150 MHz (typical)	Frequency at which current gain falls to 1. Indicates high-speed performance.
Saturation Voltage	V _{CE(sat)}	~0.2 V (at I _C = 10 mA)	Voltage across collector-emitter when the transistor is "fully on".
Base-Emitter Voltage	V _{BE}	~0.6 – 0.7 V (active region)	Voltage required to forward-bias the base-emitter junction.



Subtype	h _{FE} Range
BC547A	110 – 220
BC547B	200 – 450
BC547C	420 – 800

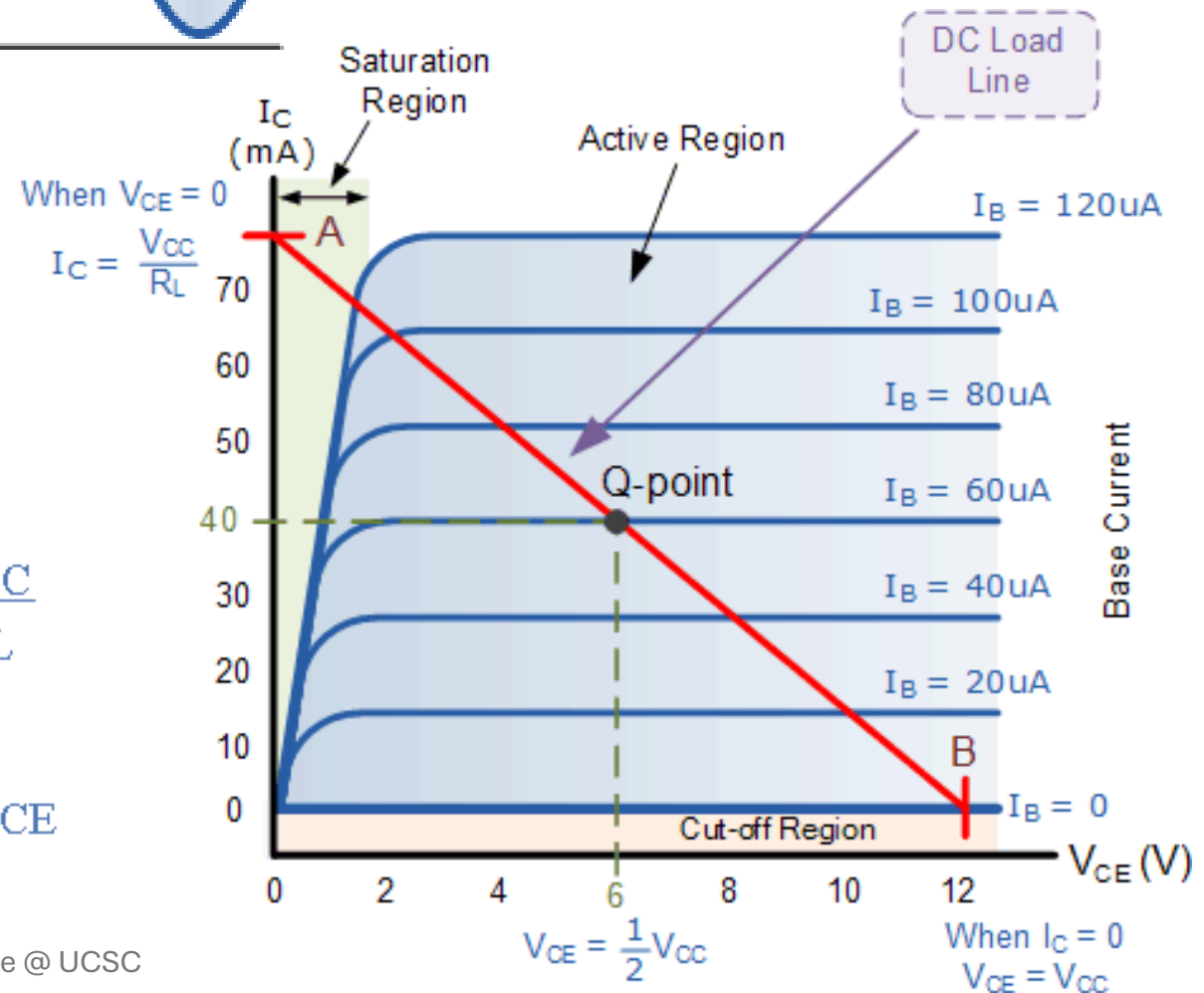
Single Stage Class A Common Emitter Amplifier Circuit

- **Class A Amplifier:** the transistors Base terminal is biased in such a way as to forward bias the Base-emitter junction.
- The transistor is always operating halfway between its cut-off and saturation regions, thereby allowing the transistor amplifier to accurately reproduce the positive and negative halves of any AC input signal superimposed upon this DC biasing voltage.
- It is necessary to set the initial value of V_{ce} correctly to allow the output voltage to vary both up and down when amplifying AC input signals and this is called setting the operating point or ***Quiescent Point***, “**Q-point**” for short.

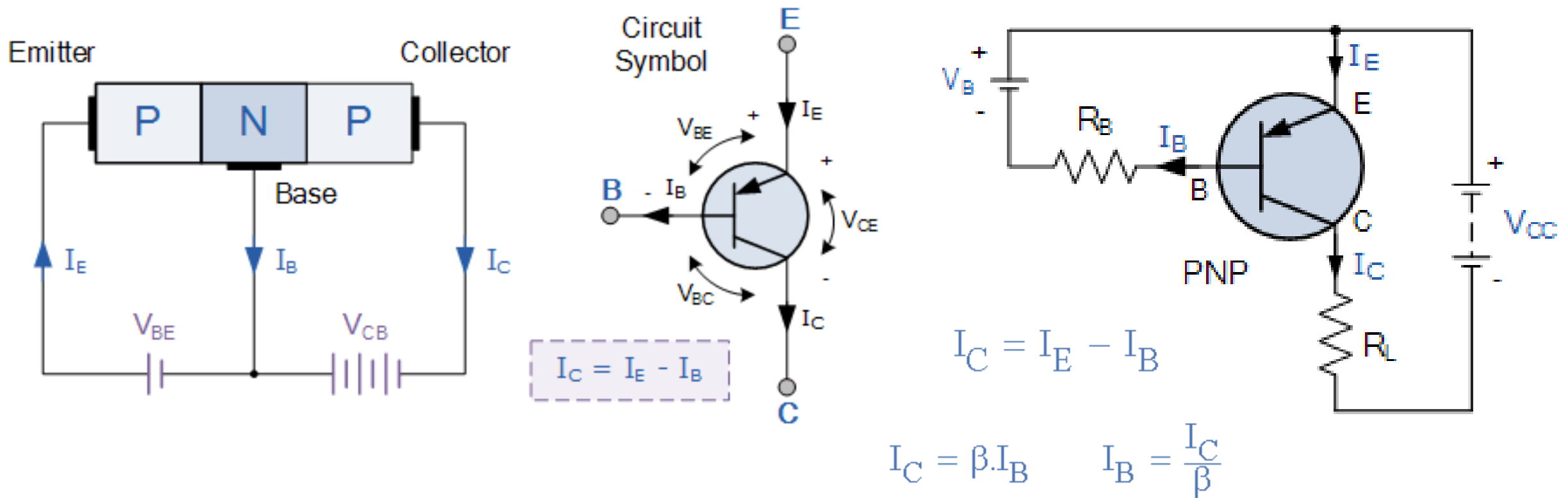


$$\text{When: } (V_{CE} = 0) \quad I_C = \frac{V_{CC} - 0}{R_L}, \quad I_C = \frac{V_{CC}}{R_L}$$

$$\text{When: } (I_C = 0) \quad 0 = \frac{V_{CC} - V_{CE}}{R_L}, \quad V_{CC} = V_{CE}$$



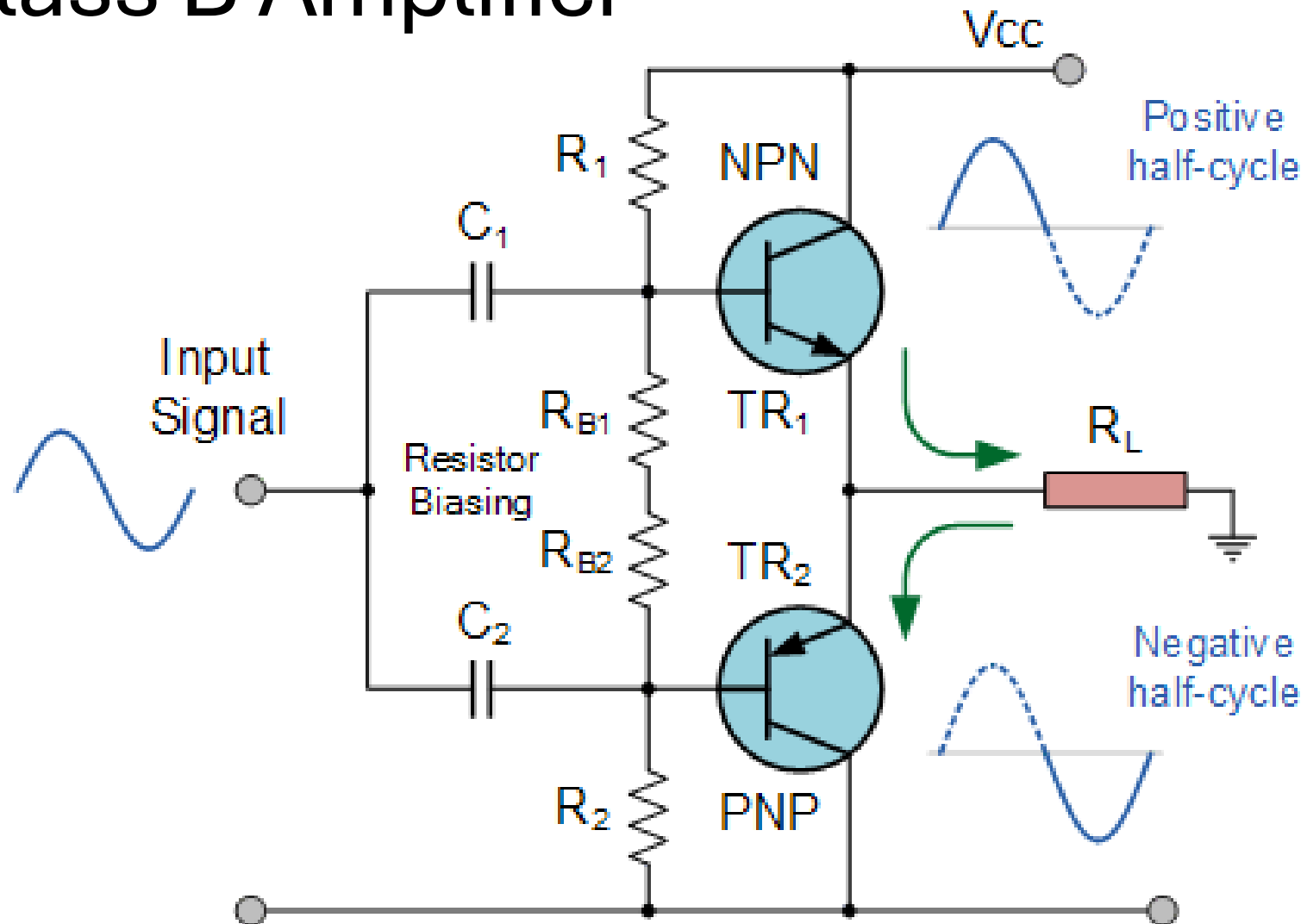
PNP Bipolar Junction Transistor



Complementary Bipolar Transistors

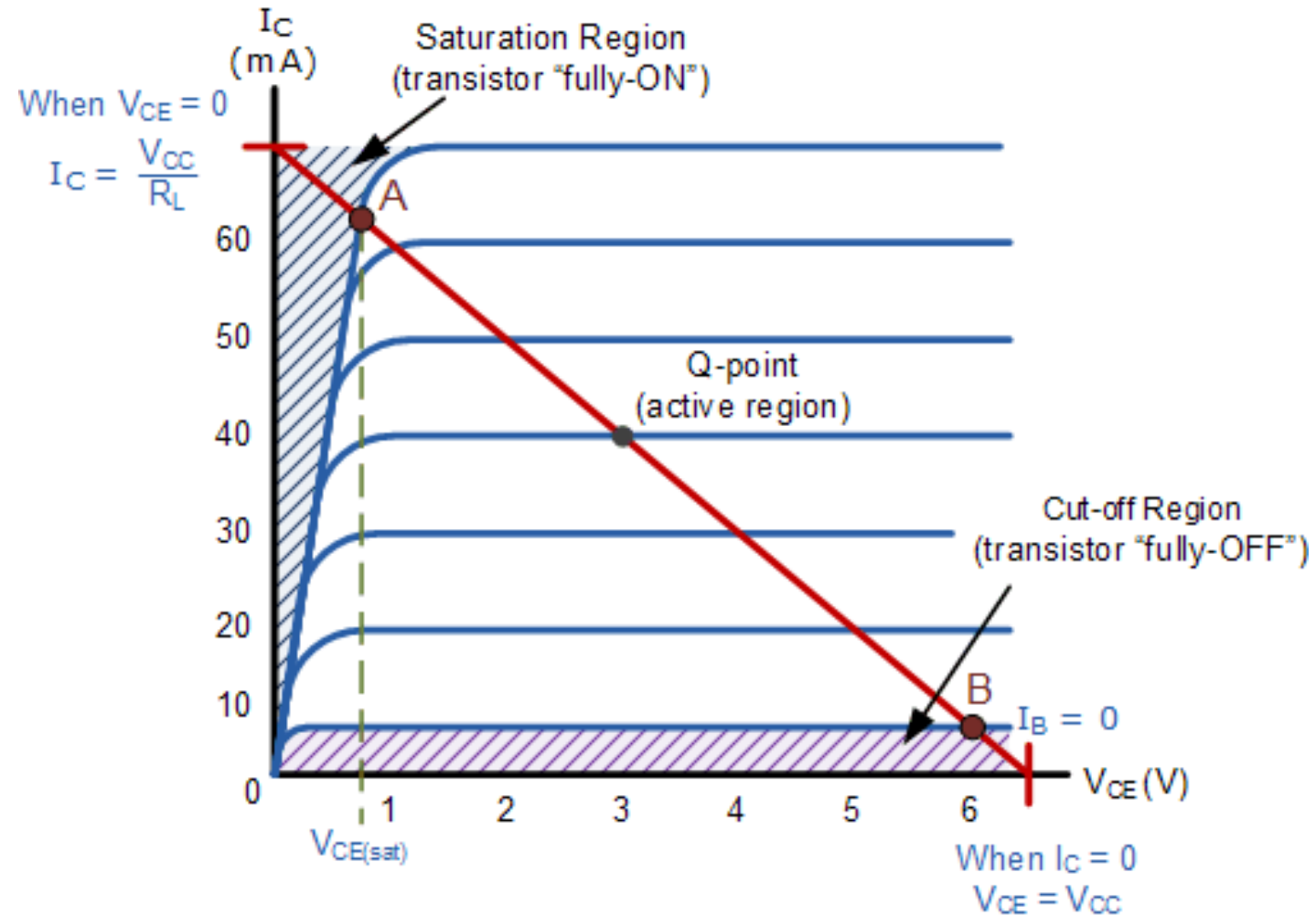
- Complementary transistors are pairs of matched NPN/PNP bipolar transistors which have nearly identical electrical characteristics and can be used together to provide symmetrical performance in many amplifier and electronic circuits.
 - E.g., 2N3904 (NPN) and 2N3906 (PNP), or the TIP3055 (NPN) and the TIP2955 (PNP)
- Complementary transistors have a DC current gain, Beta, (I_c/I_b) matched to within 10% and high Collector current of about 15A making them ideal for general motor control or robotic applications.
- Also, class B amplifiers use complementary NPN and PNP in their power output stage design. The NPN transistor conducts for only the positive half of the signal while the PNP transistor conducts for negative half of the signal.

Class B Amplifier



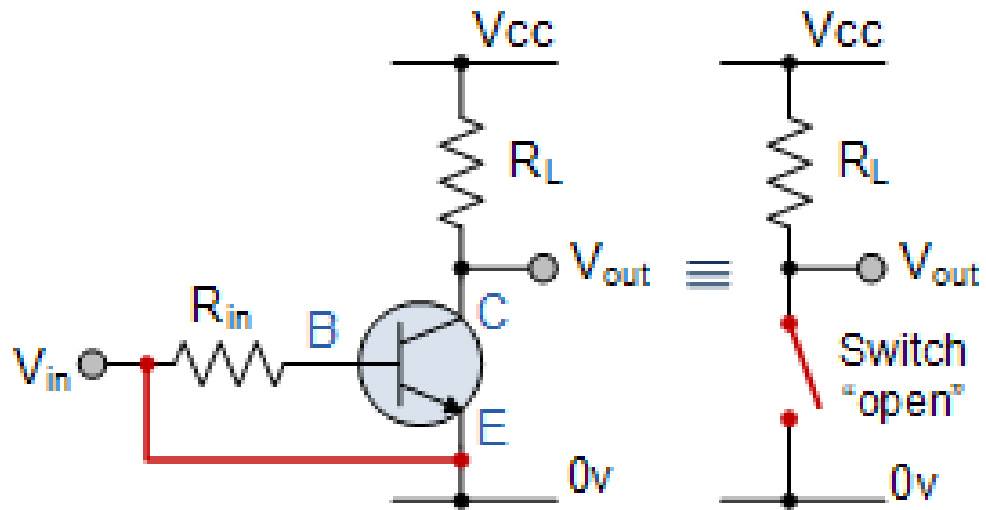
Bipolar Transistor as a Switch

- The Bipolar Junction Transistor (BJT) can be used as a switch by operating it in the saturated and cut-off regions of its characteristics curves.
- Using the **Transistor As A Switch** has many applications as a solid-state switching element to control DC power to a load.



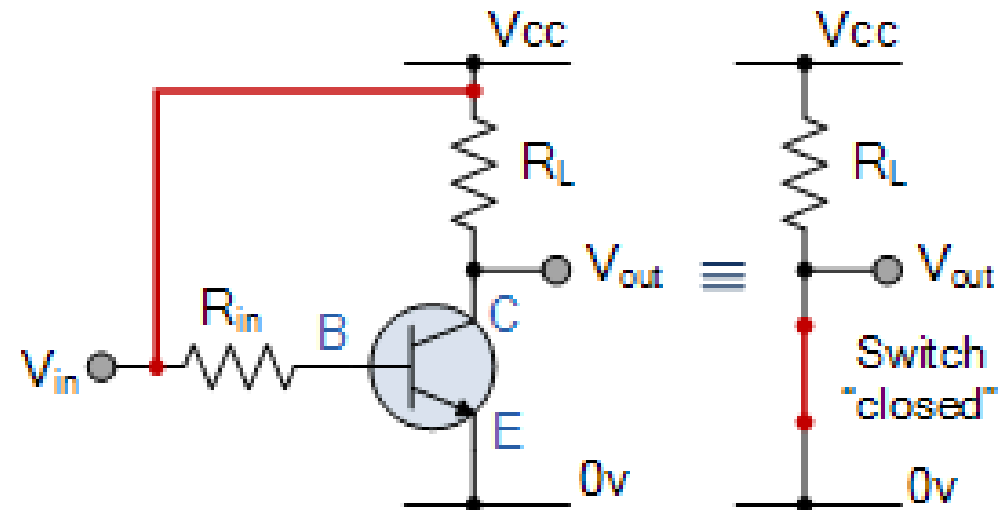
The Cut-off & Saturation Regions

The Cut-off Region



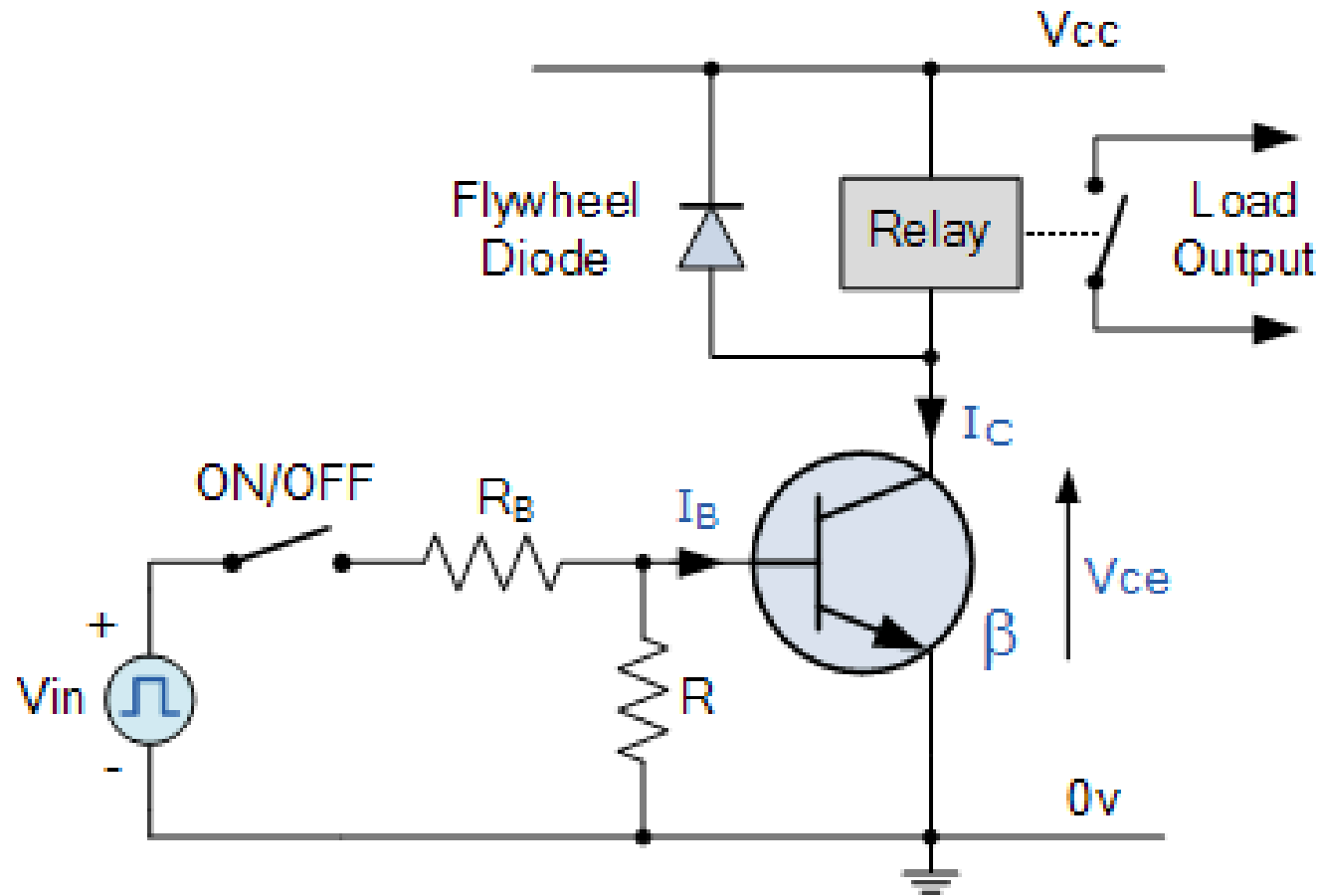
$$V_b < 0.7v \text{ and } I_c = 0$$

The Saturation Region



$$V_b > 0.7v \text{ and } I_c = \text{Maximum}$$

Basic NPN Transistor Switching Circuit



Transistor as a Switch Worked Example No1

- Using the transistor values of: $\beta = 200$, $I_c = 4\text{mA}$ and $I_b = 20\mu\text{A}$, find the value of the Base resistor (R_b) required to switch the load fully “ON” when the input terminal voltage exceeds +2.5 volts.

$$R_B = \frac{V_{in} - V_{BE}}{I_B} = \frac{2.5\text{V} - 0.7\text{V}}{20 \times 10^{-6}} = 90\text{k}\Omega$$

Preferred value is: 82k Ω

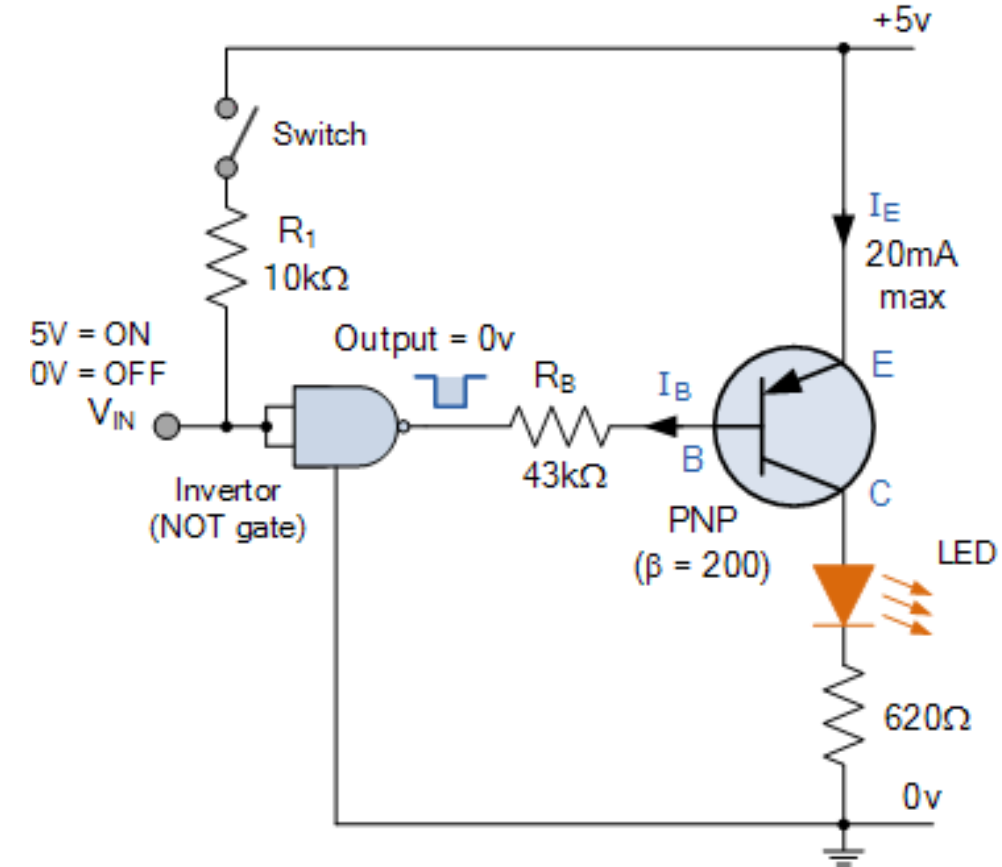
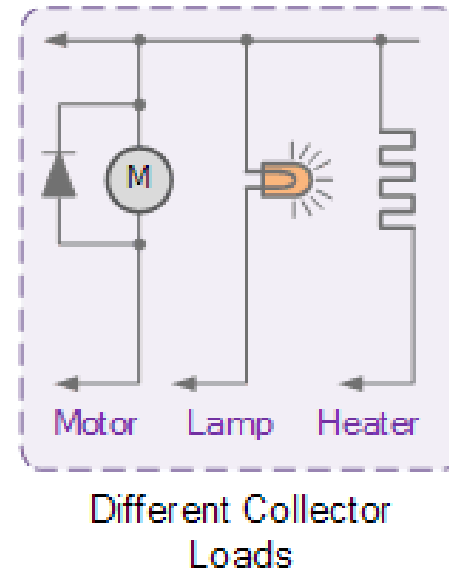
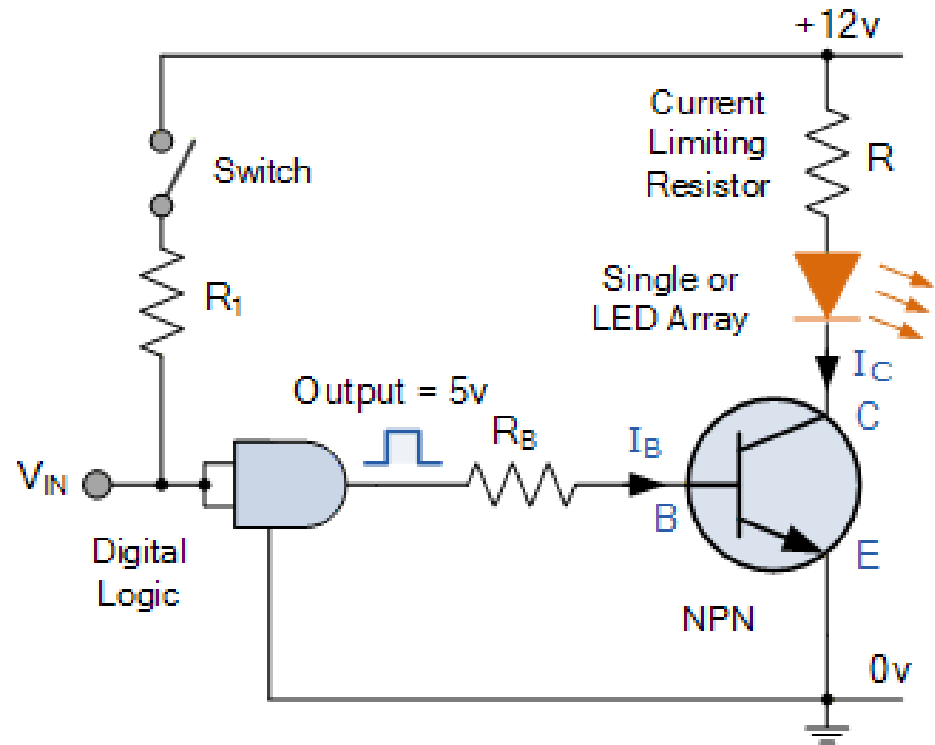
Transistor as a Switch Worked Example No2

- Using the same values ($\beta = 200$, $I_c = 4\text{mA}$ and $I_b = 20\mu\text{A}$), find the minimum Base current required to turn the transistor “fully-ON” (saturated) for a load that requires 200mA of current when the input voltage is increased to 5.0V. Also calculate the new value of R_b .

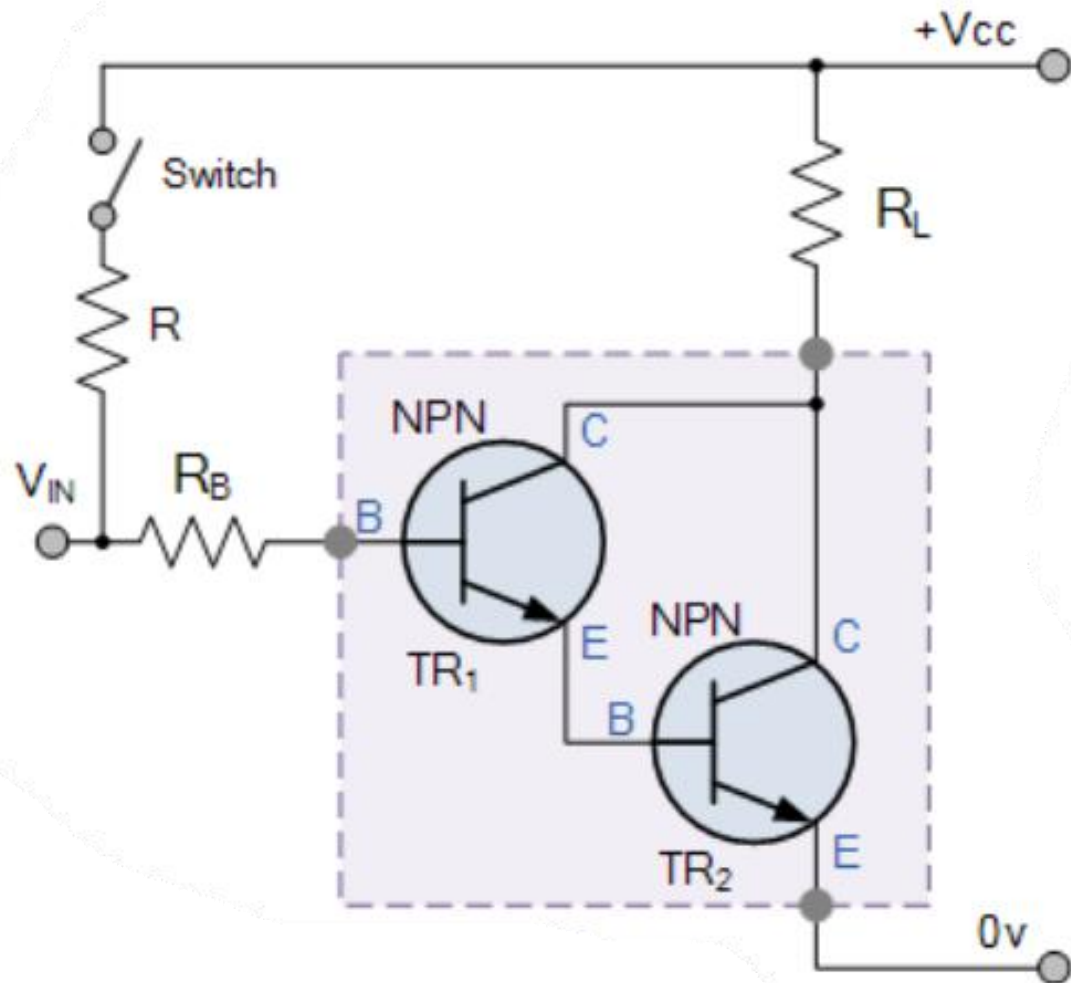
$$I_B = \frac{I_C}{\beta} = \frac{200\text{mA}}{200} = 1\text{mA}$$

$$R_B = \frac{V_{in} - V_{BE}}{I_B} = \frac{5.0\text{V} - 0.7\text{V}}{1 \times 10^{-3}} = 4.3\text{k}\Omega$$

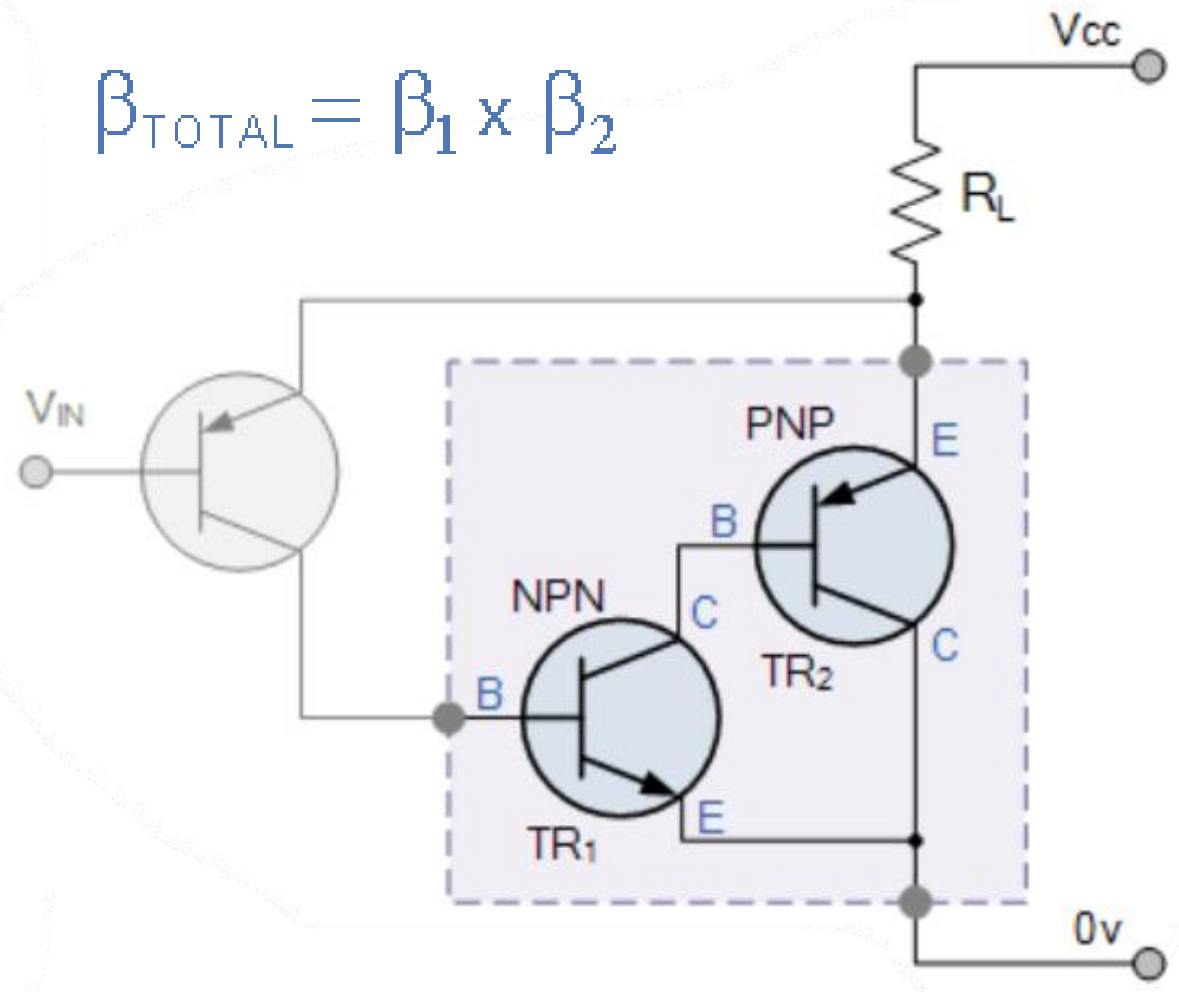
Digital Logic Transistor Switch



The Darlington Transistor Switch



$$\beta_{TOTAL} = \beta_1 \times \beta_2$$



Activity 2(a)

- Identify and describe the following electronic components: 1N4001, 1N34, 10A10, 1N4007, 1N4148, 1N5711, BR5010, BC547, BC557, 2N3055, 2N3904, D400, D313, and TIP120.
For each component:
 - State the type of component (e.g., diode, transistor, bridge rectifier, etc.)
 - Give one practical application
 - Provide the key specifications or characteristics
 - Mention the approximate local market price in Sri Lanka

Activity 2(b)

- Explain how to test and identify the condition of diodes and bipolar junction transistors (BJTs) using a digital multimeter (DMM). Your answer should include:
 - How to test a diode in forward-bias and reverse-bias modes
 - How to identify whether a transistor is NPN or PNP
 - How to determine the transistor terminals (Emitter, Base, and Collector)
 - How to measure or check the transistor gain (h_{FE}) using a DMM
 - Expected meter readings for properly functioning components

Activity 2(c)

- Design and construct an astable multivibrator circuit using BC547 transistors and other components to generate a blinking output at approximately 2 Hz. Your answer should include:
 - A complete schematic diagram of the circuit
 - Step-by-step design calculations for selecting resistor and capacitor values
 - A list of required components with approximate local prices in Sri Lanka
 - An explanation of the working principle and operation of the circuit
 - The expected output waveform and blinking behavior of the LEDs or output devices
 - Other uses of this astable multivibrator